Governing machines that act in the world: analysis report (v4)

How the machine classes in which embodied AI is deployed are governed, from 3,295 documents in eight world regions

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Key numbers

Files collected by eight regional teams (round 1)	2,557 (1,894 of 2,160 listed sources captured)
Round-2 candidates / with text / entered the collection / usable	2,075 / 1,720 / 1,641 / 1,626 (Tables 50, 51; Figure 50)
Documents annotated / usable (text readable, on topic)	3,295 / 3,165 (128 duplicate captures removed at assembly)
About robots or embodied AI ("embodied-AI set")	1,603: 509 research papers and 810 legal texts of instrument class (822 documents carry the governance flag)
About AI in general ("general-AI set")	1,015: 48 research papers, 409 governance documents
In neither set (rated not about AI, mostly round-2 sector law)	547 (Table 1), excluded from every set comparison
Principal instruments reaching robots (the headline unit: legal texts minus commencement orders, levies, exemptions, incorporation decisions, corrigenda, stubs and catalogue pages)	**505** of the 810 legal texts; 408 rated robot-relevant; 395 whose text contains autonomy or AI provisions; 153 that name AI explicitly (Table 53b)
Principal binding instruments reaching robots	**195** of 291 binding legal texts; 168 robot-relevant; 194 counted once per instrument family; 140 with autonomy or AI provisions; 15 naming AI explicitly (Table 53b); 182 of the 195 in force (Table 33b)
Precision audit of the v3 binding set (random n = 100, read by an independent auditor)	54% distinct, binding and substantive (Wilson 44 to 63), implying about 161 (132 to 190) of the 299 v3 texts; the rule-based unit keeps 61% (51 to 70) of the same sample (Table 62)
Countries with any document / with a principal embodied instrument / with a principal binding embodied instrument	116 / 56 / 38 (Table S08); 60 jurisdictions on the comparable unit of every legal text (58 countries plus the EU and the international layer; Table 53)
Share of usable documents from Europe / in English / in Chinese	42% (1,332) / 75% (2,373) / 9% (278)
Labelling	AI model under a fixed 39-field codebook; no person read or labelled any document; seven reliability checks (five on round-1 samples, two on a round-2 sample; Part F); the round-2 collectors and the reviewers were AI agents

How to read this report. Every label was produced by a language model; no person read or labelled any document. The round-2 collectors, the six adversarial reviewers of the previous version and the precision auditor were also AI agents (Claude) working in one Claude Code session, so none of the checks in Part F is a human validation. Counts of instruments are counts *in this collection*, on a stated counting unit: the headline unit is the principal instrument (Section 4), and every number that uses another unit says so. Round 1 of the collection sought research on robots and official texts on AI in general; round 2 harvested legal databases for instruments. Because the two rounds differ in genre, length and capture quality, every embodied-versus-general comparison is given pooled and then within collection round or within genre, and the text says which version survives. Percentages are given as counts wherever a group has fewer than 30 documents. The glossary defines the terms; the supplement holds the codebook, the reliability detail, the instrument lists, the version history (S10) and the precision audit (S13); the companion report describes round 2.

Five findings and one methodological result

1. **Binding law reaches robots through the law of the machine classes they belong to, and most of it says nothing about AI.** Within this collection, 505 principal instruments reach robots, 195 of them legally binding and 182 of those in force (Tables 53b, 33b). The binding layer is aviation, road-vehicle, machinery, workplace-safety and product law: on the comparable unit of every binding text, 207 of 291 are regulations or decrees and 71 statutes (Table S01); among the 195 principal binding instruments, drones are addressed by 107 and automated road vehicles by 67, industrial manipulators by 13 and delivery robots by 15 (Table 7). Only 140 of the 195 contain any provision about autonomy or automation, and 15 name artificial intelligence (Table 53b). El Salvador's Robotic Technologies Law is the only robotics statute captured; two robot-titled statutes known to exist (Korea's 2008 Intelligent Robots Act, cited by 9 documents, and Hangzhou's 2026 embodied-robot regulation, located by a collector) were not read. (Figures 4, 6; Tables 2, 7, 33, 33b, 53b.)
2. **Research and rules point at different robots, and this survives the change of counting unit.** Companion and social robots are the subject of 43% of robot research papers (219 of 509) but 3% of principal instruments (15 of 505), none binding; drones are 6% of research and 42% of instruments (214; 107 binding); self-driving vehicles 12% and 40% (201; 67 binding). The gaps are far outside their bootstrap intervals (+40 points, interval 36 to 45; −36, −41 to −31; −27, −33 to −22). Humanoids have 170 documents, 20 principal instruments and no binding instrument (gap +19, interval 15 to 23); industrial manipulators are at parity (−3, interval −7 to +1). (Figures 18, 40; Tables 7, 22.)
3. **What is distinctive about robot governance is the body; the "ethical core deficit" reported for v3 is a composition effect of round 2.** Within principal instruments (505 vs 372), physical safety (83% vs 9%), autonomy-level control (60% vs 10%), conformity assessment (66% vs 19%), human-robot proximity (33% vs 0.3%) and liability for physical damage (27% vs 3%) are far more often raised in robot instruments, and ten of fourteen core principles are raised less often (Table 21). Within the round-1

instruments alone (61 vs 245) none of the fourteen core principles differs after Holm correction, while all six embodied-specific items do; within round-2 instruments (444 vs 127) ten core principles are lower (Table 20b). With collection round added to the adjusted model (Table 20), eight core principles remain lower, two (non-maleficence 2.6, human oversight 1.7) higher and four do not differ; physical safety (odds ratio 45), liability (21), autonomy-level control (17), conformity (6) and dual use (3) remain the distinctive items, and proximity is not estimable (4 general documents). The tools follow the body: licensing and registration (adjusted odds ratio 4.1), insurance (7.0), conformity assessment (4.2) and references to standards (3.9) belong to robot instruments; oversight bodies (0.51), self-regulation (0.51), labelling (0.53) and international cooperation (0.59) to general-AI instruments. (Figures 20, 24, 38, 39; Tables 20, 20b, 21, 34.)

4. **Specification lags mention, measured on texts that can specify anything.** On content-analysed captures, physical safety is set out in detail by 62% of the 195 principal binding instruments, 33% of 207 soft-law instruments and 33% of 77 drafts; the scope pages of 62 technical standards name it in 95% and treat it in depth in 68%, but scope pages are a separate measurement and are never pooled with substantive texts. Autonomy-level control is raised by 57 to 66% of instruments of every type and set out in detail by 25% of binding instruments, 21% of guidance, 23% of standards' scope pages and 31% of drafts; liability for physical damage is absent from standards. (Figure 23; Tables 9, 35.)
5. **Regions differ most in tools; the newest shift is toward industrial policy and away from personhood; compliance machinery grows within the original collection.** On a common population of 1,576 embodied documents in six regions, 34 of 38 principles and mechanisms differ after Holm correction (Table 29), led by references to standards, licensing and beneficence; but the genre mix differs by region, and within research outputs only 2 of 14 core principles differ by region while within principal instruments 12 of 14 do (Table 29b), so regional differences on values belong to the instrument layer, and within it to the mix of standards and strategies each region supplies, not to the research literature. Between 2020 to 2024 and 2025 to 2026, after adjustment for genre and region, industrial policy rises (odds ratio 1.7, Holm p = 0.03) and legal personhood falls (0.39, Holm p = 0.04); the fall of ethics discourse does not survive adjustment. Risk-based tiering and the conformity-assessment mechanism grow by 31% and 26% a year within round-1 documents and by 10% and 5% a year with round as a covariate (Table 30); the pooled declines of proximity and liability (Table 17) coincide with the arrival of round-2 sector law and are not reported as trends. "Embodied AI" vocabulary rises from 1% of embodied documents in 2024 to 13% in 2026. (Figures 21, 25, 29 to 34, 42; Tables 16, 17, 28 to 31.)

Methodological result: the collection strategy changes instrument counts by an order of magnitude and reverses several conclusions. On the comparable unit of every legal text, round 2 raised embodied instruments from 65 to 810 and binding embodied instruments from 5 to 291 (Table 53). The rise is a property of where the teams searched, not of the world; within the round-1 documents the v2 results (no core-principle deficit, compliance machinery growing about 30% a year) still hold, and within round 2 they do not (Tables 20b, 30). Reported as a finding about method, not about governance (Section 3).

1. Scope, questions and definitions

The brief asked for a metric-heavy analysis of AI governance in the context of robotics, physical AI and embodied AI: development over time, regional differences, dominant views and principles, the newest trends, and how different robot types are governed, with a taxonomy of robots, a taxonomy of documents (including governance-instrument status) and a document dashboard. The study has gone through four versions, each after a set of independent adversarial reviews; the version history and the before-and-after comparisons are in supplement S10 and the review logs, and are referenced here only where a comparison between the two collection rounds is itself the result.

The object of study is the governance of the machine classes in which embodied AI is deployed: robots, automated road vehicles, unmanned aircraft, industrial and medical machines and autonomous weapons. Most of the binding instruments that reach these machines are machine-class rules (aviation, road vehicle, machinery, product safety) with no AI content; the AI-content tiers in Table 53b report how many instruments contain autonomy or AI provisions and how many name AI explicitly, so that the reader can choose the construct.

Definitions used throughout:

- **Usable documents** (3,165): annotated documents with readable text that are neither failed captures nor off-topic.
- **Embodied-AI set** (1,603): usable documents rated at least moderately relevant to embodied-AI governance (2 or 3 on a 0 to 3 scale) or carrying at least one robot-type label, and not rated "not about AI"; 1,374 by relevance alone (Table 32). A strict definition (relevance only) is reported as a sensitivity.
- **General-AI set** (1,015): usable documents about AI in general, rated 0 or 1 with no robot-type label. The 547 usable documents in neither set (Table 1) are mostly round-2 sector law rated not about AI; they enter only the all-document counts.
- **Legal texts** (1,696 usable; 810 in the embodied set): every usable document of instrument class (binding instrument, soft-law instrument or standard, proposal or draft). This is the comparable unit for comparisons with v2 and v3 (Table 53) and the unit of the instrument lists S01 to S03.
- **Principal instruments** (505 in the embodied set, 372 in the general-AI set; Table 21): legal texts minus commencement orders, levy and fee instruments, individual exemptions, directions, approvals and permits, definitions-only parts, tables of contents, EEA Joint Committee incorporation decisions, language-version corrigenda, byte-identical duplicate texts, catalogue or landing pages and metadata stubs under 150 words. Amendments count. This is the headline unit for every instrument count (Section 4).
- **Instrument families** group a principal instrument with its yearly renewals, successive versions and parts within one jurisdiction; family-level counts are reported as a further tier (Table 53b).
- **Robot-relevant instruments** are principal instruments rated relevance 2 or 3 (408; 168 binding); **instruments with embodied provisions** are rated 0 or 1 but carry a robot-type label (97; 27 binding) and are mostly sector or general law with an incidental robot provision (Table 33).
- **AI-content tiers**: *autonomy or AI provisions* means the full text contains at least two mentions of autonomous or automated operation, remote piloting, self-driving or AI terms; *names AI explicitly* means at least two mentions of artificial intelligence, machine learning, algorithms or their equivalents in the document's language (Table 53b).
- **Governance documents** (1,709 usable; 822 in the embodied set, 409 in the general-AI set) are the documents the annotation flagged as governing or prescribing (an annotation field, not the instrument class; it includes a few reports and web pages and misses a few legal texts). They are the base of the lifecycle, binding-nature and mechanism-share tables (Tables 2, 4, 10) and are always labelled as such.

- **Regions** follow the issuing jurisdiction (first author's country for research; Europe for EU bodies; International for ISO, IEC, IEEE, UN, OECD, UNESCO and WHO): Europe (including Russia), Asia, North America, Latin America and the Caribbean, Middle East, Africa, Oceania, International.
- **Content-analysed captures** are substantive or partial captures; catalogue, committee and landing pages are excluded from the principal unit and from every statement about what instruments say. Statistics about what instruments specify in depth (Table 9, Figure 23) use content-analysed captures of at least 150 words; technical standards are annotated on their public scope text only and are reported as a separate measurement.
- **Collection round**: round 1 is the regional teams' collection (675 embodied and 803 general-AI usable documents; Table 20b); round 2 is the legal-database and collector-agent collection (1,626 usable documents, 928 embodied and 212 general-AI; Tables 20b, 56). Round enters the adjusted models as a covariate and the stratified tables as a stratum.

2. Data and methods in brief

The full pipeline is in the supplement (S1). In round 1 the eight regional teams listed 2,160 sources and captured 1,894 as 2,557 files; files were converted to text, deduplicated (hash, URL, title, prefix and prose similarity, then a second pass on normalised title and jurisdiction), and every document was annotated by Claude Sonnet 5.5 against a 39-field codebook as a fixed system prompt with a JSON schema. Governance documents were re-annotated in v2 with the collectors' cover block stripped, minimal metadata and up to 110,000 characters of text.

Round 2 searched for instruments rather than discourse. Nine collector agents (Claude subagents working from regional, thematic and standards briefs with exclusion lists of the round-1 holdings) and scripted harvests of seven legal databases (EUR-Lex through CELLAR, legislation.gov.uk, the Japanese e-Gov law API, the Chinese State Council policy library, the Australian Federal Register of Legislation, the US eCFR and the Brazilian Câmara API) produced 2,075 de-duplicated candidates. Texts were fetched from the primary URL with documented fallbacks and a recovered-fetch pass; 1,720 yielded text (16 of them standards' scope pages through a committee mirror) and 355 did not (272 HTTP errors or thin pages, 46 blocked hosts, 37 failed fetches; Tables 51 and 52). All 1,720 were annotated under the same blind, extended-excerpt policy, model and prompt hash as the v2 governance re-annotation; 1,641 entered the collection after the assembly de-duplication and 1,626 are usable (Figure 50). The agreement between the blind labels and the collectors' catalogue metadata (binding nature 93% of 1,608, kappa 0.85; jurisdiction 98%; year within one year 94%; document class 77%, kappa 0.66; Tables 57 and 58) is a consistency check, not a validation: the collectors were AI agents and harvest scripts, not people.

Four v4 corrections apply to both rounds. First, the excerpt cleaner introduced in v2 (which drops short lines near the top of a text that carry no digit, capital letter or sentence punctuation) deleted most of the text of Chinese, Japanese and Korean documents, because those scripts have no whitespace words or ASCII full stops: a median of 24% of each Chinese document reached the model, and 97 Chinese documents naming robots or AI had none of those terms in the model input (AI researcher's review). A character-aware cleaner (lines containing CJK characters are never dropped; policy tag `cjk_clean=v4`) was written and every Chinese, Japanese and Korean document (376 round-2 and 46 round-1 documents) was re-annotated; 57 of the 422 changed embodied-set membership and the number rated not about AI fell from 205 to 160 (review log). Second, the counting unit (Section 4) and the AI-content tiers were assigned from documented title and capture rules (`scripts/07c_v4_units_and_tiers.py`). Third, status and class rules that the model got wrong at scale were added: standards still in development (AWI, WD, CD, DIS, FDIS and similar designations) are not published instruments; mandatory Chinese GB standards are binding; CCW working papers, chair's summaries and rolling texts are position documents, not draft instruments; proposals whose adopted act is in the collection are superseded. Fourth, the governance expert's audit of 159 v3 rows (108 with at least one error, 68%; 98 with a material one, 62%; the released row table has 165 rows, 119 with an error code and 107 material; a risk-stratified sample, so an upper bound for the whole set) was converted into 157 documented doc-level corrections on 91 documents (`data/manual_overrides_v4.csv`: 50 counting-unit, 32 relevance, 32 robot-type, 17 status, 14 class, 7 year and 5 binding-nature corrections). The 79 field corrections on 27 round-1 documents and the 128 duplicate captures removed in earlier versions are unchanged.

Two measurements were added. A stratified sample of 160 round-2 documents (116 English, 29 Chinese, 6 Japanese; 63 binding, 52 soft-law, 23 drafts; 27 landing pages; median 848 words) was re-annotated by a second model (Claude Opus 5.5) and by a same-model re-run (Part F, supplement S4.7). An independent auditor read a simple random sample of 100 of the 299 v3 binding embodied texts against five yes-or-no criteria and the result is compared with the rule-based unit on the same documents (Table 62; supplement S13).

Statistics use Wilson intervals; chi-square tests with a Monte-Carlo p-value for sparse tables and Fisher's exact test for sparse 2 by 2 tables; bias-corrected Cramér's V with bootstrap intervals; logistic regression adjusting for document class, region, collection round and year, with a sparse-cell flag where one set contributes fewer than five events (such odds ratios are reported as not estimable); Mantel-Haenszel tests stratified by class; bootstrap intervals for the governance-gap index and for regional distances, with split-half noise floors. Holm correction is applied within one family per table (Tables 8, 10, 15, 16, 17, 20, 28, 29 and 34) and within each stratum for the stratified tables (20b, 21); no correction spans tables. Model inputs for round 2, the CJK re-annotation, the governance re-annotation and the blind reliability run are stored under `annotations/<run>/_inputs/`; `data/run_manifest.json` records models, prompt hashes, excerpt policies, counts and cost per annotation folder (Claude Code CLI 2.1.314; about 334 USD of model calls in all).

3. What collection round 2 changed, on the comparable unit

Table 53. Before/after on the comparable unit (every text of instrument class): v2 (original collection) versus v3/v4 (original plus round 2), content-analysed documents only

measure	v2	v3	added	ratio
Documents (annotated)	1656	3295	1639	1.99
Usable documents	1538	3165	1627	2.06
Embodied-AI set	675	1603	928	2.37
General-AI set	806	1015	209	1.26
Governance documents (incl. reports)	343	1709	1366	4.98
Governance instruments, all	345	1696	1351	4.92
Governance instruments, embodied-AI set	65	810	745	12.46

measure	v2	v3	added	ratio
Robot-specific instruments (relevance 2+)	34	699	665	20.56
Binding instruments, all	34	702	668	20.65
Binding instruments, embodied-AI set	5	291	286	58.2
Binding instruments, robot-specific	4	254	250	63.5
Soft-law instruments, embodied-AI set	50	411	361	8.22
Proposals and drafts, embodied-AI set	10	108	98	10.8
Government reports, embodied-AI set	24	97	73	4.04
Jurisdictions with an embodied instrument	30	60	30	2

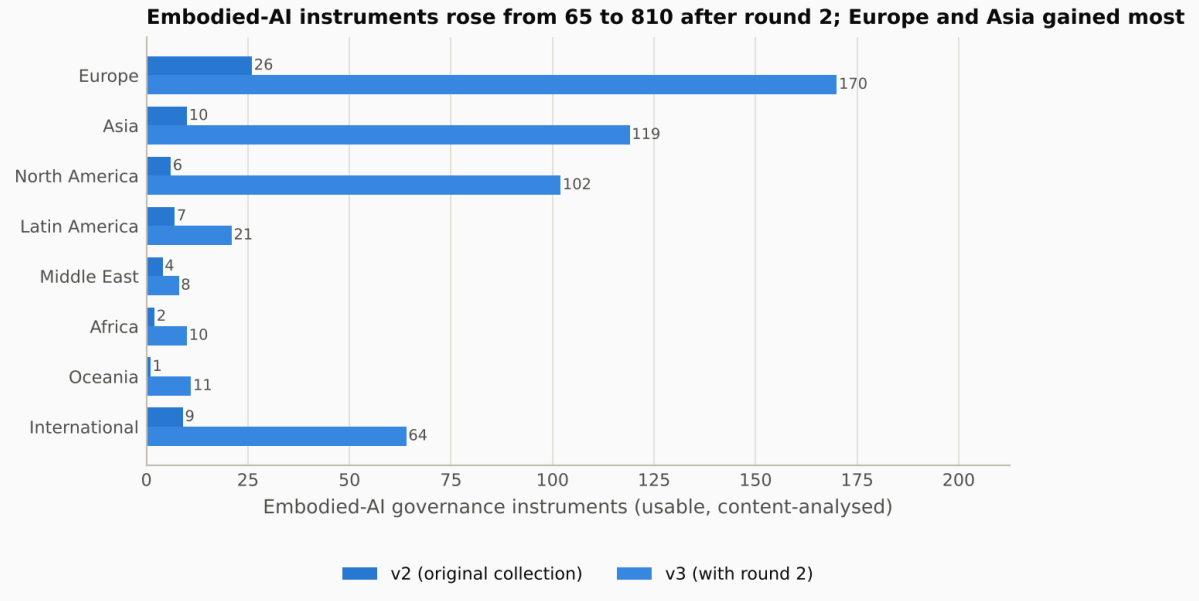


Figure 51. Embodied-AI governance instruments by jurisdiction region, v2 (original collection) versus v3 (with round 2).

Table 53 compares the collection before and after round 2 on the unit the two versions share, every legal text of instrument class. Embodied-AI instruments went from 65 to 810, robot-specific instruments from 34 to 699, binding embodied instruments from 5 to 291 (robot-specific and binding from 4 to 254), soft-law instruments from 50 to 411, drafts from 10 to 108 and government reports from 24 to 97; jurisdictions with an embodied instrument went from 30 to 60. The general-AI set grew from 806 to 1,015 and the research literature by one paper. Table 54 and Figure 51 give the additions by region and Table 55 by robot group, with the v4 column on the principal unit (505 principal instruments; 195 binding): Europe 26 to 170 (binding 1 to 63), Asia 10 to 119 (0 to 32), North America 6 to 102 (1 to 65), the international layer 9 to 64 (0 to 5), Oceania 1 to 11 (0 to 7); vehicles and field machines absorbed most of it (48 to 389 instruments, binding 5 to 174) and humanoids none of the binding layer (9 to 20 instruments, 0 binding).

These ratios are consequences of collection design, not findings about the world. Round 1 asked regional teams to search the web for material about embodied AI; round 2 queried legal databases for instruments that name the machines. The binding share among settled embodied instruments (286 legally binding, 236 voluntary, 254 standards against 79, 258 and 20 in the general set; Cramér's V 0.41) and the doubling of jurisdictions follow from that choice and are reported here rather than in the findings. The order-of-magnitude change is nevertheless the study's clearest methodological result: within the instrument genre, the embodied side is now 444 of 505 round-2 (88%) and the general side 127 of 372 (34%; Table 20b); round-2 texts are shorter (median about 850 words against about 2,650 for round 1, master table), 254 of the 1,626 usable round-2 documents are landing pages and 379 are in Chinese, Japanese or Korean, so every pooled embodied-versus-general contrast is also a round-1-versus-round-2 contrast. Parts C and E therefore place each pooled result next to its within-round or within-genre version and say which survives.

4. Counting units and constructs

Table 53b. How many instruments reach robots depends on the counting unit and the construct: the same collection under successively stricter definitions (v4)

tier	count
All legal texts of instrument class, embodied-AI set	810
Principal instruments (amendments kept; commencement, levy, exemption, incorporation, corrigenda, stubs and catalogue pages excluded)	505
Principal instruments rated robot-relevant (relevance 2 or 3)	408
Principal embodied instruments whose text contains autonomy or AI provisions	395
Principal embodied instruments whose text names AI explicitly	153
Binding legal texts, embodied-AI set	291
Principal binding instruments, embodied-AI set	195
Principal binding instruments rated robot-relevant (relevance 2 or 3)	168
Principal binding instruments counted once per instrument family	194
Principal binding instruments with autonomy or AI provisions in the text	140
Principal binding instruments naming AI explicitly	15

How many instruments reach robots depends on what is counted. Table 53b gives the same collection under successively stricter units. Every legal text in the embodied set: 810. Principal instruments (amendments kept; commencement orders, levies, exemptions, incorporation decisions, corrigenda, stubs and catalogue pages excluded): 505. Principal instruments rated robot-relevant: 408. Principal instruments whose text contains autonomy or AI provisions: 395; naming AI explicitly: 153. For binding law: 291 texts, 195 principal instruments, 168 robot-relevant, 194 once per instrument family, 140 with autonomy or AI provisions and 15 naming AI explicitly. The excluded texts are not errors of the collection but of the unit (master table, `instrument_unit`): among the 810 embodied legal texts, 222 are catalogue or landing pages (191 of them technical standards captured as catalogue entries), 42 are amendments (kept), 35 are stubs under 150 words, 30 are ancillary instruments (commencement orders, levies, exemptions, approvals, directions), 10 dependent parts, 6 EEA incorporation decisions and 2 language corrigenda; among the 291 binding texts, 28 are ancillary, 25 stubs, 28 catalogue pages, 7 dependent parts, 6 incorporation decisions and 2 corrigenda. Australia is the clearest case: 47 embodied legal texts on the comparable unit, 6 principal instruments, because the Federal Register harvest returned register entries for individual CASA exemptions, directions, area approvals and levy instruments (Table 4).

Table 62. Independent precision audit of 100 randomly sampled v3 binding embodied instruments versus the v4 rule-based counting unit, on the same documents (Wilson 95% intervals)

criterion	audit_share	rule_share	agreement	kappa	rule_passes_audit_fails	audit_passes_rule_fails
distinct instrument (audit a)	67% (57 to 75)	74% (65 to 82)	0.79	0.5	14	7
distinct, binding and embodied-substantive (audit a+b+c)	54% (44 to 63)	61% (51 to 70)	0.77	0.53	15	8
a+b+c and autonomy/AI provision (audit a+b+c+d)	53% (43 to 62)	46% (37 to 56)	0.71	0.42	11	18

The rule-based unit was checked against an independent reading. The auditor's random sample of 100 of the 299 v3 binding embodied texts found 67% to be distinct instruments (Wilson 57 to 75), 97% binding, 56% to contain substantive embodied provisions (46 to 65), 67% to contain autonomy or AI content and 14% to name AI explicitly (9 to 22); 54% passed the three criteria of distinct, binding and substantive together (44 to 63), which implies about 161 such instruments (132 to 190) in the v3 set of 299. On the same 100 documents the v4 rules keep 61% (51 to 70) as principal binding instruments; agreement with the auditor is 77% (kappa 0.53), with 15 documents the rules keep and the auditor rejects and 8 the auditor keeps and the rules exclude (Table 62). The rules therefore remove most but not all of the inflation: the v4 counts of 195 and 168 should be read as upper bounds on the number of distinct, binding, substantive instruments, with the audit's 161 (132 to 190) as the independent estimate for the v3 population. The eight counting-unit rules the auditor proposed, and which of them v4 implements, are in supplement S13.

Table 33. Embodied-AI governance instruments by tier and capture quality (n=505; 'fraction seen' is the share of the text inside the v1 excerpt)

tier	doc_class	n	substantive	partial	landing	median_fraction_seen
general-AI instrument with embodied provisions (relevance 0-1)	Binding instrument	27	22	5	0	0.09
general-AI instrument with embodied provisions (relevance 0-1)	Soft-law instrument	64	63	1	0	0.51
general-AI instrument with embodied provisions (relevance 0-1)	Proposal or draft	6	6	0	0	0.33
robot-specific instrument (relevance 2-3)	Binding instrument	168	153	15	0	0.56
robot-specific instrument (relevance 2-3)	Soft-law instrument	176	112	64	0	1
robot-specific instrument (relevance 2-3)	Proposal or draft	64	48	16	0	1

Table 33 splits the 505 principal instruments by relevance tier and capture quality. The 408 robot-relevant instruments are 168 binding (153 substantive captures, 15 partial), 176 soft-law instruments and standards (112 substantive, 64 partial) and 64 drafts; the 97 instruments with embodied provisions (27 binding, 64 soft-law, 6 drafts) are mostly sector or general law in which robots appear as an example, a list entry or a scope clause, and only a few are AI instruments in the ordinary sense (the AI Act, its Digital Omnibus amendment, the two US executive orders). No landing page remains in the principal unit. The median share of a text that the annotation excerpt covered is 0.56 for robot-relevant binding instruments and 0.09 for binding instruments with embodied provisions, which are long consolidated statutes: the labels of the longest texts rest on the first 100,000 characters plus two short slices, and a full-text pass is listed as future work.

5. Part A. The document landscape

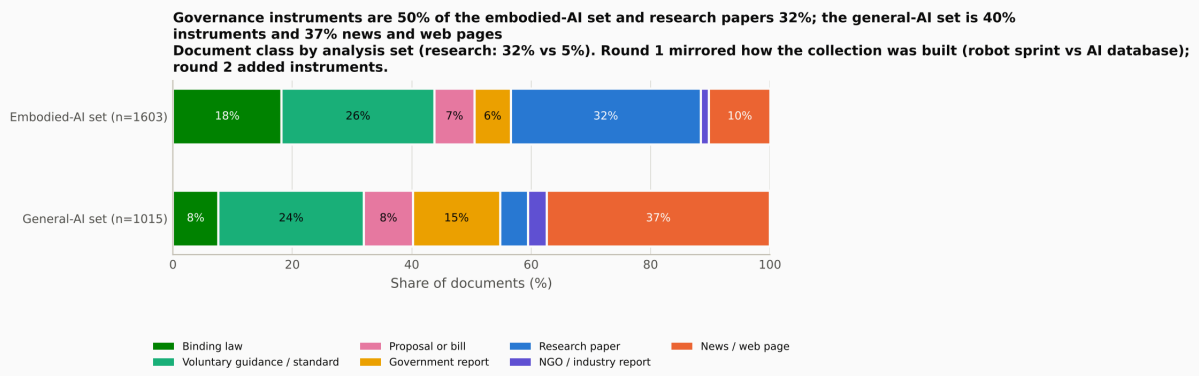


Figure 01. Document classes in the embodied-AI set and the general-AI set, usable documents only (n in the axis labels).

The embodied-AI set is half legal texts (810 of 1,603) and a third research papers (509), with news and web pages (165), drafts (108), government reports (97) and grey literature (22) making up the rest; its issuers are academic (516) but also national executives (276), standards bodies (228), intergovernmental bodies (217), regulators (179) and legislatures (91). The general-AI set is 37% news, web

pages and press releases (380 of 1,015), 24% soft-law instruments (248) and 15% government reports (148), with 48 research papers; its issuers are intergovernmental bodies (382), national executives (323), regulators (121) and legislatures (62). The split mirrors the collection design: round 1's robot sprint drew on academic literature and its database on official sources, and round 2 added legal texts almost exclusively (745 of the 810 embodied legal texts and 444 of the 505 principal instruments are round-2 captures), which is why every comparison in Parts C and D is made within genre, within round or adjusted for both.

Table 01. Document types by analysis set (usable documents)

doc_type	Embodied-AI set	General-AI set	Other	total
Regulation or decree	209	60	299	568
Web page informational	116	234	27	377
Journal article	246	32	10	288
Technical standard	254	21	8	283
Policy guideline or code of practice	84	83	70	237
National strategy or action plan	89	85	22	196
Government report or review	57	116	9	182
Press release or government news	32	126	2	160
Statute or act	71	12	34	117
Conference paper	105	4	3	112
Bill or draft law	36	40	27	103
Intergovernmental recommendation or resolution	27	39	14	80
Consultation or white paper	38	38	0	76
Parliamentary material	30	33	4	67
Preprint or working paper	47	3	3	53
Industry or ngo report	19	30	1	50
Book or chapter	42	1	2	45
Editorial or commentary	35	4	6	45
Ethical framework or principles	8	15	0	23
Review or survey article	21	1	0	22
Blog or opinion	10	12	0	22
Thesis or dissertation	12	3	0	15
News article	8	4	0	12
Procurement or internal policy	0	8	4	12
Other	3	6	1	10
Court decision or judicial notice	3	4	1	8
International treaty or convention	1	1	0	2

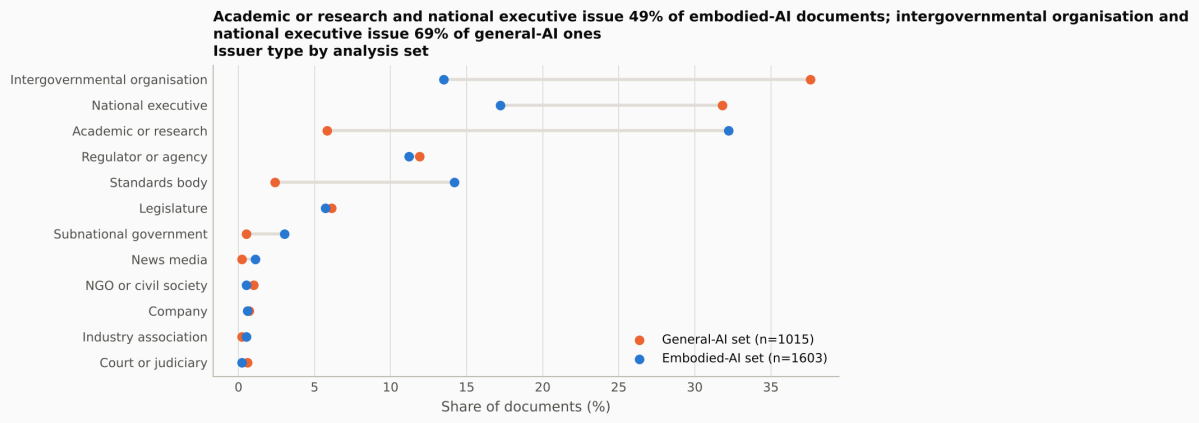


Figure 03. Issuer types by set.

Regulators and agencies issue 179 embodied documents and 121 general-AI ones; legislatures 91 and 62; standards bodies 228 and 24. In the original collection the institutions that apply robot rules were present mainly as authors of general-AI rules; after round 2 they are present as authors of robot rules.

5.1 Instruments: lifecycle and binding nature

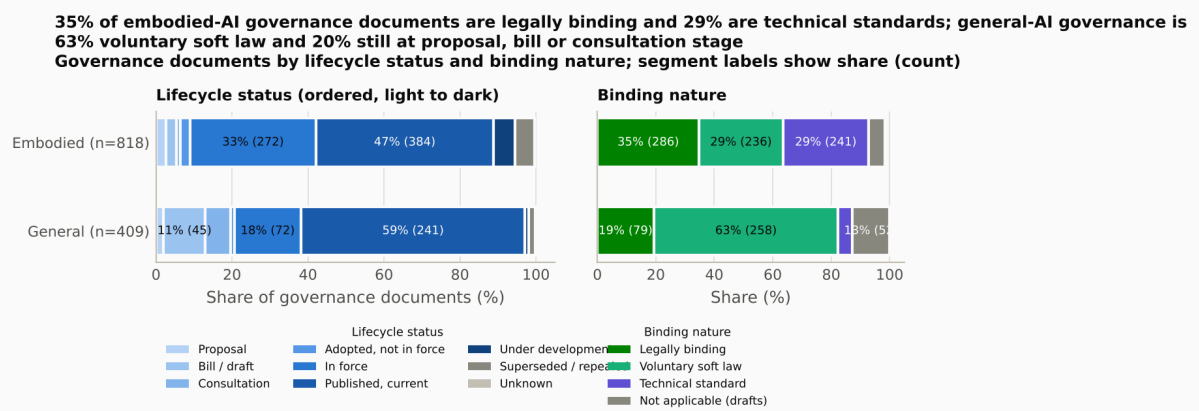


Figure 04. Governance documents: lifecycle status (left) and binding nature (right), embodied-AI set vs general-AI set (n in the axis labels).

Table 02. Governance documents: lifecycle status by binding nature and set

setname	instrument_status	legally_binding	not_applicable	technical_standard_mandated	technical_standard_voluntary	voluntary_soft_law
Embodied-AI set	adopted_not_yet_in_force	5	0	8	7	1
Embodied-AI set	consultation_open	0	3	0	0	5
Embodied-AI set	draft_or_bill_under_consideration	0	22	0	0	1
Embodied-AI set	in_force	260	0	5	0	7
Embodied-AI set	proposal_or_concept	0	5	0	0	16
Embodied-AI set	published_current	1	1	0	179	203
Embodied-AI set	superseded_or_repealed	20	11	0	8	3
Embodied-AI set	under_development	0	0	0	47	0
Embodied-AI set	withdrawn_or_lapsed	0	4	0	0	0
General-AI set	adopted_not_yet_in_force	4	0	0	0	0
General-AI set	consultation_open	0	9	0	0	18
General-AI set	draft_or_bill_under_consideration	0	38	0	0	7
General-AI set	in_force	69	0	0	0	3
General-AI set	proposal_or_concept	0	1	0	0	7
General-AI set	published_current	1	1	0	16	223
General-AI set	superseded_or_repealed	4	3	0	0	0
General-AI set	under_development	0	0	0	4	0
General-AI set	unknown	1	0	0	0	0
Other	adopted_not_yet_in_force	19	0	0	0	0
Other	draft_or_bill_under_consideration	0	21	0	0	0
Other	in_force	253	0	0	0	3
Other	published_current	0	2	0	8	102
Other	superseded_or_repealed	63	5	0	0	1
Other	withdrawn_or_lapsed	0	1	0	0	0

Of the 822 documents in the embodied set that carry the governance flag, 384 are published and current, 272 in force, 42 superseded, 47 under development (standards in preparation), 23 bills, 21 proposals, 21 adopted but not yet in force, 8 open consultations and 4 lapsed; by binding nature 286 are legally binding, 241 voluntary technical standards, 236 voluntary soft law, 13 mandatory standards and 46 have no binding nature yet. The general-AI governance documents (409) are more legislative in their pending items (45 bills, 27 open consultations) and far less binding in their settled ones (72 in force; 258 voluntary, 79 binding). The embodied set's higher binding share (Cramér's V 0.41 on the 2 by 3 table of binding, voluntary and standard; Section 3) is what round 2 was designed to find and is not a finding about anyone's propensity to legislate.

Table 33b. Binding instruments in the embodied-AI set after corrections, with relevance tier and capture quality

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
D0005	Robotic Technologies Law (Decree No. 340)	SV	2025	in_force	3	substantive_document	professional_service_robots; care_assistive_robots; mobile_industrial_logistics; autonomous_road_vehicles; collaborative_robots; industrial_manipulators; educational_robots
D0016	Resolution 710/2024 - Creation of the Unit of Artificial Intelligence Applied to Security (UIAAS)	AR	2024	in_force	2	substantive_document	drones_uas; security_policing_robots
D0228	Law No. (9) of 2023 Regulating the Operation of Autonomous Vehicles in the Emirate of Dubai	AE	2023	in_force	3	substantive_document	autonomous_road_vehicles
D0915	Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence	US	2023	superseded_or_repealed	1	substantive_document	autonomous_road_vehicles; surgical_medical_robots
D1291	Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (Artificial Intell	EU	2024	in_force	2	substantive_document	autonomous_road_vehicles; drones_uas; surgical_medical_robots; care_assistive_robots; industrial_manipulators
R2-82f8813a	Civil Aviation Regulations 2016	MY	2016	in_force	1	substantive_document	drones_uas
R2-d2567eac	Civil Aviation Directive 6011 Part (I) – Unmanned Aircraft System Remote Pilot Training Organisation	MY	2022	in_force	2	substantive_document	drones_uas; agricultural_robots
R2-723e3d80	Civil Aviation Directive 6011 Part (II) – Unmanned Aircraft System Agricultural UAS Operations	MY	2022	in_force	3	substantive_document	drones_uas; agricultural_robots
R2-1617d407	Civil Aviation Directive 6011 Part (V) – Unmanned Aircraft System Special UAS Project	MY	2022	in_force	3	substantive_document	drones_uas
R2-0ea46a4c	Decree No. 288/2025/ND-CP providing the management of unmanned aircraft and other aerial vehicles	VN	2025	in_force	2	substantive_document	drones_uas
R2-d0d19353	Law on People's Air Defense No. 49/2024/QH15	VN	2024	in_force	2	substantive_document	drones_uas; military_autonomous_weapons
R2-f67080c2	Standards Document: Operation of Aircraft – Part IV – International Operations (Remotely Piloted Air	FJ	2025	in_force	3	substantive_document	drones_uas
R2-5de2dafe	Civil Aviation Rules Part 102 — Unmanned Aircraft and Remotely Piloted Aircraft Systems	PG	2025	in_force	2	substantive_document	drones_uas
R2-c5eb5314	Civil Aviation Safety Regulations 1998	AU	1998	in_force	2	partial_or_excerpt	drones_uas
R2-873de2ab	Part 101 (Unmanned Aircraft and Rockets) Manual of Standards 2019	AU	2019	in_force	2	partial_or_excerpt	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-ab6b4f47	Civil Aviation Safety Amendment (Remotely Piloted Aircraft and Model Aircraft—Registration and Accre	AU	2019	in_force	2	partial_or_excerpt	drones_uas
R2-295bf91b	Air Navigation (Aircraft Noise) Regulations 2018	AU	2018	in_force	1	partial_or_excerpt	drones_uas
R2-35fea8b3	Part 101 Gyrogliders and Parasails, Unmanned Aircraft (including Balloons), Kites, and Rockets - Ope	NZ	2025	in_force	2	partial_or_excerpt	drones_uas; other_embodied
R2-2b0fd47f	Part 774—The Commerce Control List (15 CFR Part 774, including Supplement No. 1)	US	2026	in_force	1	partial_or_excerpt	military_autonomous_weapons
R2-675a9e05	15 CFR Part 742—Control Policy—CCL Based Controls	US	2026	in_force	1	substantive_document	drones_uas; military_autonomous_weapons; space_or_extreme_environment
R2-8a885ff8	Part 596—Automatic Emergency Braking Test Devices	US	2024	in_force	2	substantive_document	autonomous_road_vehicles
R2-e589d498	14 CFR Part 47—Aircraft Registration	US	2025	in_force	1	substantive_document	drones_uas
R2-c5427e07	Part 107—Small Unmanned Aircraft Systems (14 CFR)	US	2021	in_force	3	substantive_document	drones_uas
R2-459e8712	Part 89—Remote Identification of Unmanned Aircraft	US	2021	in_force	2	substantive_document	drones_uas
R2-f3b175ef	Part 48—Registration and Marking Requirements for Small Unmanned Aircraft	US	2021	in_force	2	substantive_document	drones_uas
R2-0d0ca0f5	PART 124—COUNTER-UNMANNED AIRCRAFT SYSTEM AUTHORITY FOR STATE, LOCAL, TRIBAL, AND TERRITORIAL LAW EN	US	2026	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-022998ee	FAR Part 40—Information Security and Supply Chain Security (Subpart 40.2 Security Prohibitions and E	US	2024	in_force	2	substantive_document	drones_uas
R2-12e32782	Part 102-33—Management of Government Aircraft	US	2025	in_force	1	substantive_document	drones_uas
R2-9c282a09	49 CFR Part 830—Notification and Reporting of Aircraft Accidents or Incidents and Overdue Aircraft,	US	2022	in_force	2	substantive_document	drones_uas
R2-f9c66a43	15 CFR Part 791—Securing the Information and Communications Technology and Services Supply Chain (in	US	2025	in_force	2	substantive_document	autonomous_road_vehicles; drones_uas
R2-ddd7d9c2	Commission Implementing Regulation (EU) 2019/947 of 24 May 2019 on the rules and procedures for the	EU	2019	in_force	3	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-6c14c9c2	Commission Delegated Regulation (EU) 2019/945 on unmanned aircraft systems and on third-country oper	EU	2019	in_force	3	substantive_document	drones_uas
R2-b0d4346b	Commission Delegated Regulation (EU) 2024/1107 laying down detailed rules for the continuing airwort	EU	2024	in_force	2	substantive_document	drones_uas
R2-0383c9ff	Commission Delegated Regulation (EU) 2024/1108 of 13 March 2024 amending Regulation (EU) No 748/2012	EU	2024	in_force	2	substantive_document	drones_uas
R2-d85d64e1	Commission Implementing Regulation (EU) 2024/1109 on competent authority requirements and administra	EU	2024	in_force	2	substantive_document	drones_uas
R2-b4af497c	Commission Implementing Regulation (EU) 2022/425 of 14 March 2022 amending Implementing Regulation (	EU	2022	in_force	2	substantive_document	drones_uas
R2-a283b42e	Commission Implementing Regulation (EU) 2024/1110 amending Regulation (EU) No 748/2012 as regards th	EU	2024	in_force	3	substantive_document	drones_uas
R2-77acbca3	Commission Implementing Decision (EU) 2024/2103 of 30 July 2024 on harmonised standard for direct re	EU	2024	in_force	2	substantive_document	drones_uas
R2-c5c7a0a6	Commission Implementing Regulation (EU) 2026/1821 amending Implementing Regulation (EU) 2015/1018 as	EU	2026	adopted_not_yet_in_force	2	substantive_document	drones_uas
R2-e039001f	Commission Implementing Regulation (EU) 2022/1426 of 5 August 2022 laying down rules for the applica	EU	2022	in_force	3	substantive_document	autonomous_road_vehicles
R2-c47d3176	Commission Implementing Regulation (EU) 2026/481 of 3 March 2026 amending Implementing Regulation (E	EU	2026	in_force	3	substantive_document	autonomous_road_vehicles
R2-984ee7c4	Commission Delegated Regulation (EU) 2022/2236 amending Annexes I, II, IV and V to Regulation (EU) 2	EU	2022	in_force	2	substantive_document	autonomous_road_vehicles

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-80171de4	Directive 2006/42/EC of the European Parliament and of the Council of 17 May 2006 on machinery, and	EU	2006	in_force	2	substantive_document	industrial_manipulators; collaborative_robots; mobile_industrial_logistics
R2-6db84260	Commission Implementing Decision (EU) 2020/480 of 1 April 2020 amending Implementing Decision (EU) 2	EU	2020	in_force	2	substantive_document	agricultural_robots; mobile_industrial_logistics
R2-885d8000	Commission Implementing Decision (EU) 2024/1329 of 13 May 2024 amending and correcting Implementing	EU	2024	in_force	2	substantive_document	mobile_industrial_logistics; agricultural_robots; industrial_manipulators
R2-7bb2229f	Regulation (EU) 2023/1230 on machinery and repealing Directive 2006/42/EC and Council Directive 73/3	EU	2023	in_force	3	substantive_document	industrial_manipulators; collaborative_robots; mobile_industrial_logistics; professional_service_robots; other_embodied
R2-98aa0eab	Regulation (EU) 2025/14 on the approval and market surveillance of non-road mobile machinery circula	EU	2025	in_force	2	substantive_document	other_embodied
R2-710c5344	Commission Implementing Decision (EU) 2026/2015 of 4 September 2026 amending and repealing Commissio	EU	2026	in_force	2	substantive_document	industrial_manipulators; mobile_industrial_logistics; agricultural_robots
R2-833ba7e9	Council Directive 93/42/EEC of 14 June 1993 concerning medical devices	EU	1993	superseded_or_repealed	1	substantive_document	surgical_medical_robots
R2-87faf76a	Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical de	EU	2017	in_force	1	substantive_document	surgical_medical_robots; rehabilitation_exoskeleton_prosthetic
R2-80ec293b	Commission Regulation (EU) 2015/562 of 8 April 2015 amending Regulation (EU) No 347/2012 with regard	EU	2015	in_force	2	substantive_document	autonomous_road_vehicles
R2-d37cc122	Commission Implementing Regulation (EU) 2021/666 of 22 April 2021 amending Regulation (EU) No 923/20	EU	2021	in_force	2	substantive_document	drones_uas
R2-14ad2059	Commission Implementing Regulation (EU) 2021/665 amending Implementing Regulation (EU) 2017/373 as r	EU	2021	in_force	2	substantive_document	drones_uas
R2-210faac2	Commission Implementing Regulation (EU) 2021/664 of 22 April 2021 on a regulatory framework for the	EU	2021	in_force	2	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-34d7721c	Commission Regulation (EU) 2019/543 of 3 April 2019 amending Annex IV to Regulation (EC) No 661/2009	EU	2019	in_force	1	substantive_document	autonomous_road_vehicles
R2-2b891013	Commission Delegated Regulation (EU) 2021/1341 on type-approval of motor vehicles with regard to dri	EU	2021	in_force	2	substantive_document	autonomous_road_vehicles
R2-11555522	Commission Delegated Regulation (EU) 2022/545 on type-approval of motor vehicles with regard to thei	EU	2022	in_force	1	substantive_document	autonomous_road_vehicles
R2-457d18eb	Commission Implementing Regulation (EU) 2021/646 on type-approval of motor vehicles with regard to t	EU	2021	in_force	2	substantive_document	autonomous_road_vehicles
R2-b6c5f44e	Commission Delegated Regulation (EU) 2021/1958 laying down test procedures and technical requirement	EU	2021	in_force	2	substantive_document	autonomous_road_vehicles
R2-ada72e0e	Commission Implementing Regulation (EU) 2020/746 of 4 June 2020 amending Implementing Regulation (EU	EU	2020	in_force	2	substantive_document	drones_uas
R2-3e221bc0	Commission Implementing Regulation (EU) 2021/1166 of 15 July 2021 amending Implementing Regulation (	EU	2021	in_force	2	substantive_document	drones_uas
R2-9c6e8103	Commission Implementing Regulation (EU) 2020/639 of 12 May 2020 amending Implementing Regulation (EU	EU	2020	in_force	2	substantive_document	drones_uas
R2-0c29bfda	Regulation (EU) 2026/1744 amending Regulations (EU) 2024/1689, (EU) 2018/1139 and (EU) 2023/1230 as	EU	2026	in_force	2	substantive_document	industrial_manipulators; collaborative_robots; surgical_medical_robots; drones_uas; autonomous_road_vehicles
R2-2e97da82	Interim Regulations on the Administration of Unmanned Aircraft Flights	CN	2023	in_force	3	substantive_document	drones_uas; agricultural_robots
R2-118bce75	Road Traffic Act	JP	1960	in_force	2	substantive_document	autonomous_road_vehicles; delivery_robots
R2-2f6acdc5	Road Traffic Act Enforcement Order	JP	1960	in_force	2	partial_or_excerpt	autonomous_road_vehicles; delivery_robots
R2-47723d4c	Civil Aeronautics Act (Aviation Act) of Japan	JP	1952	in_force	2	substantive_document	drones_uas
R2-16343b99	Road Traffic Act Enforcement Regulations	JP	1960	in_force	2	substantive_document	autonomous_road_vehicles; delivery_robots
R2-dd7f8d70	Order on Road Signs, Lane Markings and Road Markings	JP	2026	in_force	1	substantive_document	delivery_robots
R2-f57251ed	Ordinance on Industrial Safety and Health	JP	1972	in_force	2	partial_or_excerpt	industrial_manipulators; mobile_industrial_logistics; collaborative_robots

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-31fbdf75	Ordinance for Enforcement of the Act on Efficiency of Distribution of Goods	JP	2005	in_force	1	substantive_document	mobile_industrial_logistics; industrial_manipulators
R2-15dba33a	Road Act Enforcement Regulations	JP	2026	in_force	1	substantive_document	autonomous_road_vehicles
R2-431b7f3a	Ministerial Ordinance Establishing Technical Standards for Railways	JP	2001	in_force	1	substantive_document	autonomous_maritime_or_rail
R2-d3b0fc1c	Act on Special Measures for the Reconstruction and Revitalization of Fukushima	JP	2012	in_force	1	partial_or_excerpt	space_or_extreme_environment
R2-7d33c29c	National Strategic Special Zones Act	JP	2013	in_force	2	partial_or_excerpt	autonomous_road_vehicles; drones_uas
R2-03e6e0ff	Cabinet Office Ordinance Specifying Demonstration Projects etc. under Article 25-2 of the National S	JP	2020	in_force	2	substantive_document	autonomous_road_vehicles; drones_uas
R2-ee6d92ca	Remote Islands Development Act	JP	1953	in_force	1	substantive_document	care_assistive_robots; drones_uas
R2-59a29a12	Ordinance for Enforcement of the National Strategic Special Zones Act	JP	2014	in_force	1	substantive_document	surgical_medical_robots; care_assistive_robots
R2-3ce1939d	Road Vehicles Act (Act on Road Transport Vehicles)	JP	1951	in_force	2	substantive_document	autonomous_road_vehicles
R2-fad4ccaf	Enforcement Regulation of the Road Vehicles Act	JP	1951	in_force	2	partial_or_excerpt	autonomous_road_vehicles
R2-46f20dd6	Passenger Motor Vehicle Transport Business Transport Regulations	JP	2026	in_force	2	substantive_document	autonomous_road_vehicles
R2-f47247d3	Enforcement Regulations of the Freight Forwarding Business Act	JP	1990	in_force	1	substantive_document	autonomous_road_vehicles
R2-fe102091	Enforcement Regulations of the Motor Truck Transportation Business Act	JP	1990	in_force	1	substantive_document	autonomous_road_vehicles
R2-aa8d88b9	Ordinance on Transport Safety in the Freight Motor Vehicle Transportation Business	JP	1990	in_force	2	substantive_document	autonomous_road_vehicles
R2-2b6aea0b	Ordinance on Type Designation of Devices (Road Vehicle Equipment)	JP	1998	in_force	2	substantive_document	autonomous_road_vehicles
R2-ce2171ab	Ordinance for Enforcement of the National Strategic Special Zones Act (MLIT-related)	JP	2014	in_force	1	substantive_document	autonomous_road_vehicles
R2-d97f5421	Automated Vehicles Act 2024	GB	2024	in_force	3	substantive_document	autonomous_road_vehicles
R2-5a3159c0	The Automated Vehicles (Marketing Restrictions) Regulations 2026	GB	2026	adopted_not_yet_in_force	3	substantive_document	autonomous_road_vehicles
R2-6140944d	The Road Vehicles (Construction and Use) (Automated Vehicles) Order 2022	GB	2022	in_force	3	substantive_document	autonomous_road_vehicles
R2-4ef7d8d3	Air Traffic Management and Unmanned Aircraft Act 2021	GB	2021	in_force	2	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-fbfa4dd7	The Unmanned Aircraft (Market Surveillance Authority) Regulations 2025	GB	2025	in_force	2	substantive_document	drones_uas
R2-2abf00a7	The Unmanned Aircraft (Amendment) (EU Exit) Regulations 2020	GB	2020	in_force	2	substantive_document	drones_uas
R2-0322d77f	The Air Navigation (Minimum Age for Operators of Small Unmanned Aircraft) Regulations 2019	GB	2019	in_force	2	substantive_document	drones_uas
R2-5e3b65c9	Beijing Municipal Measures for Safety Assessment of Autonomous Vehicles (Trial)	CN	2025	in_force	3	substantive_document	autonomous_road_vehicles
R2-e53e2440	Beijing Measures for Safety Inspection of Pilot Vehicles in Autonomous Vehicle Road Application (Tri	CN	2025	in_force	3	substantive_document	autonomous_road_vehicles
R2-2344fe88	Notice of the General Office of the Hunan Provincial People's Government on Issuing Several Policy M	CN	2024	in_force	2	substantive_document	drones_uas; delivery_robots; autonomous_road_vehicles
R2-eac3219d	§ 21a Procedures and competent authorities in the "open" operational category under Implementing Reg	DE	nan	in_force	2	substantive_document	drones_uas
R2-02ef2598	Ministerial Decree of 28 February 2018 - Implementing Rules and Operational Tools for Road Testing o	IT	2018	in_force	3	substantive_document	autonomous_road_vehicles
R2-cc3a603e	Instruction VEH 2022/07: Authorisation of research tests or trials with automated driving vehicles o	ES	2022	in_force	3	substantive_document	autonomous_road_vehicles
R2-e957136a	Royal Decree 1036/2017 of 15 December regulating the civil use of remotely piloted aircraft	ES	2017	superseded_or_repealed	2	substantive_document	drones_uas
R2-c11308f4	Act of 26 September 2018 amending the Road Traffic Act 1994 to enable experiments with automated sys	NL	2018	in_force	3	substantive_document	autonomous_road_vehicles
R2-b73cce0b	NFM Decree 11/2017 (IV. 12.) amending KöHÉM Decrees 5/1990 and 6/1990 on technical inspection and te	HU	2017	in_force	3	substantive_document	autonomous_road_vehicles
R2-eadb3869	Regulation on Type Approval of Motor Vehicles with Regard to the Automated Driving System of Fully A	TR	2024	in_force	3	substantive_document	autonomous_road_vehicles
R2-4cac38c9	Council of State, Summary Judge Order of 18 May 2020, No. 440442 – Drone Surveillance (La Quadrature	FR	2020	in_force	2	substantive_document	drones_uas; security_policing_robots

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-6ecde273	Decision No. 2021-817 DC of 20 May 2021 - Law for Comprehensive Security Preserving Freedoms	FR	2021	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-fcdcfae6	Decision No. 2021-834 DC of 20 January 2022 - Law on criminal responsibility and internal security	FR	2022	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-7fa23c8d	Russian Government Resolution No. 1415 of 26 Nov 2018 on the Experiment on Pilot Operation of Highly	RU	2018	superseded_or_repealed	3	substantive_document	autonomous_road_vehicles
R2-6da0ddec	Royal Decree 517/2024 of 4 June developing the legal regime for the civil use of unmanned aircraft s	ES	2024	in_force	2	substantive_document	drones_uas
R2-f16a72e1	UN Regulation No 157 – Uniform provisions concerning the approval of vehicles with regards to Automa	INT	2021	in_force	3	substantive_document	autonomous_road_vehicles
R2-66142fe4	Automated and Electric Vehicles Act 2018	GB	2018	in_force	3	substantive_document	autonomous_road_vehicles
R2-f8592e32	The Air Navigation Order 2016	GB	2016	in_force	1	substantive_document	drones_uas
R2-e8e3e681	Commission Implementing Regulation (EU) 2019/947 of 24 May 2019 on the rules and procedures for the	GB	2019	in_force	2	substantive_document	drones_uas
R2-08a1ad4b	Commission Delegated Regulation (EU) 2019/945 of 12 March 2019 on unmanned aircraft systems and on t	GB	2019	in_force	2	substantive_document	drones_uas
R2-3811ea49	S.I. No. 563/2015 - Irish Aviation Authority (Small Unmanned Aircraft (Drones) and Rockets) Order, 2	IE	2015	superseded_or_repealed	2	substantive_document	drones_uas
R2-4fa6065a	Ordinance (2017:309) on trial activities with automated vehicles	SE	2017	superseded_or_repealed	3	substantive_document	autonomous_road_vehicles
R2-4300b76f	Act amending the Road Traffic Act (Revision of trial schemes for self-driving motor vehicles and sel	DK	2025	in_force	3	substantive_document	autonomous_road_vehicles; delivery_robots
R2-e74e320b	Executive Order on trials with self-driving motor vehicles	DK	2025	in_force	3	substantive_document	autonomous_road_vehicles
R2-18066a4a	Executive Order on supplementary provisions to EU Regulation 2019/947 on rules and procedures for th	DK	2020	in_force	2	substantive_document	drones_uas
R2-36f79aa1	Regulation on the Operation of Remotely Piloted Aircraft No. 990/2017	IS	2017	superseded_or_repealed	2	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-11e0efaf	Regulation on the operation of unmanned aircraft and unmanned aircraft systems	IS	2024	in_force	2	substantive_document	drones_uas
R2-be225994	Regulations Amending the Canadian Aviation Regulations (Remotely Piloted Aircraft Systems): SOR/2019	CA	2018	in_force	3	substantive_document	drones_uas
R2-3a3314f7	Regulations Amending the Canadian Aviation Regulations (RPAS – Beyond Visual Line-of-Sight and Other	CA	2025	in_force	3	substantive_document	drones_uas
R2-ab778283	O. Reg. 306/15: Pilot Project - Automated Vehicles	CA	2025	in_force	3	substantive_document	autonomous_road_vehicles
R2-2a34bcd4	O. Reg. 161/25: Pilot Project - Automated Commercial Motor Vehicles	CA	2025	in_force	3	substantive_document	autonomous_road_vehicles
R2-1cc83318	Order No. 2018-16 of the Minister of Transport — Pilot Project on Autonomous Buses and Minibuses	CA	2018	superseded_or_repealed	3	substantive_document	autonomous_road_vehicles
R2-4a23d4bc	B.C. Reg. 63/2024 – Order in Council 147/2024 (Motor Vehicle Amendment Act, 2023 brought into force;	CA	2024	in_force	2	substantive_document	autonomous_road_vehicles
R2-28459b8b	CAR-UAR Unmanned Aircraft (UA) Recreational Flights, Issue 04	AE	2025	in_force	2	substantive_document	drones_uas
R2-0d86786b	Law No. (4) of 2020 Regulating Unmanned Aircraft in the Emirate of Dubai	AE	2020	in_force	3	substantive_document	drones_uas
R2-6adfafc8	Decision No. 1412/2023 Issuing the Regulation Governing Unmanned Aircraft Systems (Drones)	OM	2023	in_force	3	substantive_document	drones_uas
R2-b72982f8	Part 21 - Remotely Piloted Aircraft System (RPAS)	NG	2023	in_force	2	partial_or_excerpt	drones_uas
R2-351926f5	Rwanda Civil Aviation Regulations Part 27: Unmanned Aircraft Systems	RW	2026	in_force	3	substantive_document	drones_uas
R2-9a89dd96	Ghana Civil Aviation (Safety) Directives, 2016 – Part 28: Remotely Piloted Aircraft Systems	GH	2016	in_force	3	substantive_document	drones_uas
R2-b068f0c0	The Civil Aviation (Remotely Piloted Aircraft Systems) (Amendment) Regulations, 2020	TZ	2020	in_force	2	substantive_document	drones_uas
R2-93cc2a83	The Civil Aviation (Remotely Piloted Aircraft Systems) (Amendment) Regulations, 2023	TZ	2023	in_force	2	partial_or_excerpt	drones_uas
R2-66e4d71c	The Civil Aviation (Unmanned Aircraft Systems) Regulations, 2022	UG	2022	in_force	2	substantive_document	drones_uas
R2-83a4b662	DECEA Ordinance No. 2,094/DNOR8 of 18 March 2026 – ICA 100-40 Unmanned Aircraft and Access to Brazil	BR	2026	adopted_not_yet_in_force	3	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-e4e60299	Resolution 880/2019 - Regulation of Unmanned Aerial Vehicles (UAV) and Unmanned Aerial Vehicle Syste	AR	2019	superseded_or_repealed	2	partial_or_excerpt	drones_uas
R2-66d1186f	Law No. 30740, Law regulating the use and operations of remotely piloted aircraft systems (RPAS)	PE	2018	in_force	3	substantive_document	drones_uas
R2-c3888d81	Supreme Decree No. 012-2026-MTC approving the Regulation of Law No. 30740 regulating the use and ope	PE	2026	adopted_not_yet_in_force	3	substantive_document	drones_uas
R2-8991668e	Official Mexican Standard NOM-107-SCT3-2019, establishing requirements to operate a remotely piloted	MX	2019	in_force	3	substantive_document	drones_uas
R2-9c4dc434	Civil Aviation Law	MX	2025	in_force	1	substantive_document	drones_uas
R2-d012a8fb	Presidential Resolution approving DINACIA Resolutions No. 595-2024 and 663-2024 (authorization of DR	UY	2026	in_force	2	substantive_document	drones_uas; delivery_robots
R2-bcb83a37	The Civil Aviation [(No. 19) Unmanned Aircraft Systems] Regulations, 2016	TT	2016	in_force	2	substantive_document	drones_uas
R2-2a896e72	RDAC 101 Unmanned Aircraft Systems (UAS) - Resolution No. DGAC-DGAC-2024-0100-R	EC	2024	in_force	3	substantive_document	drones_uas
R2-9afd6f5a	De Novo Classification Order for ReWalk Powered Exoskeleton (K131798)	US	2014	in_force	2	substantive_document	rehabilitation_exoskeleton_prosthetic
R2-322c6200	Workplace Safety and Health Regulation, Manitoba Regulation 217/2006	CA	2006	in_force	2	substantive_document	industrial_manipulators
R2-06573313	Part 12: Tools, Machinery and Equipment (BC Occupational Health and Safety Regulation)	CA	2022	in_force	1	substantive_document	industrial_manipulators
R2-fd1469ee	Occupational Health and Safety Code (Alberta Regulation 191/2021)	CA	2021	in_force	2	substantive_document	industrial_manipulators; mobile_industrial_logistics
R2-c1a0fce3	Regulatory Standard NR 12 - Occupational Safety in Machinery and Equipment	BR	2022	in_force	2	substantive_document	industrial_manipulators; collaborative_robots; agricultural_robots
R2-7c6034a5	UN Regulation No 155 – Uniform provisions concerning the approval of vehicles with regards to cybers	INT	2021	in_force	2	substantive_document	autonomous_road_vehicles
R2-77e55f86	UN Regulation No 156 – Uniform provisions concerning the approval of vehicles with regards to softwa	INT	2021	in_force	2	substantive_document	autonomous_road_vehicles

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-bc159f08	UN Regulation No 152 – Uniform provisions concerning the approval of motor vehicles with regard to t	INT	2020	in_force	2	substantive_document	autonomous_road_vehicles
R2-a586cca8	UN/ECE Regulation No 79 — Uniform provisions concerning the approval of vehicles with regard to stee	INT	2018	in_force	2	substantive_document	autonomous_road_vehicles
R2-98d55e2b	FAA Reauthorization Act of 2024 (Public Law 118-63)	US	2024	in_force	2	partial_or_excerpt	drones_uas
R2-03dffb43	Public Law 115-254 (FAA Reauthorization Act of 2018), H.R. 302	US	2018	in_force	1	partial_or_excerpt	drones_uas
R2-7e501c83	FAA Modernization and Reform Act of 2012 (Public Law 112-95)	US	2012	in_force	1	partial_or_excerpt	drones_uas
R2-1b00f0cb	6 U.S. Code § 124n - Protection of certain facilities and assets from unmanned aircraft	US	2025	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-fb1d4b44	49 U.S. Code § 44809 - Exception for limited recreational operations of unmanned aircraft	US	2024	in_force	2	substantive_document	drones_uas
R2-44b2fa61	18 U.S. Code § 39B - Unsafe operation of unmanned aircraft	US	2018	in_force	2	substantive_document	drones_uas
R2-e0b61aab	49 U.S. Code § 44810 - Airport safety and airspace hazard mitigation and enforcement	US	2024	in_force	2	substantive_document	drones_uas
R2-ac756a44	Public Safety and Homeland Security Bureau Announces Addition of Uncrewed Aircraft Systems (UAS) and	US	2025	in_force	2	substantive_document	drones_uas
R2-69068222	Alabama SB226 Enrolled Act: Automated Driving System-Equipped Motor Vehicles	US	2024	in_force	3	substantive_document	autonomous_road_vehicles
R2-803015db	HB 2422 – Personal Delivery Devices	US	2018	in_force	3	substantive_document	delivery_robots
R2-91089001	Arizona Revised Statutes 13-3729. Unlawful operation of model or unmanned aircraft; state preemption	US	nan	in_force	2	substantive_document	drones_uas
R2-2ccae4d0	Act 468 of 2019 (HB 1561): An Act to Authorize the Operation of Autonomous Vehicles or Fully Autonom	US	2019	in_force	3	substantive_document	autonomous_road_vehicles
R2-18be461c	Personal Delivery Device Act (Arkansas HB 1767, Act 926 of 2021)	US	2021	in_force	3	substantive_document	delivery_robots
R2-5a3ba5cd	An Act Concerning the Regulation of Aeronautics; To Restrict the Use of Unmanned Aircraft Systems Un	US	2015	in_force	2	substantive_document	drones_uas
R2-0f7421dd	Code of the District of Columbia § 50–2351. Definitions	US	nan	in_force	3	substantive_document	autonomous_road_vehicles

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-f60a2aeb	Florida Statutes s. 316.85 – Autonomous vehicles; operation; compliance with traffic and motor vehic	US	2026	in_force	3	substantive_document	autonomous_road_vehicles
R2-0e639229	Florida Statutes 316.2071 Personal delivery devices and mobile carriers	US	2018	in_force	3	substantive_document	delivery_robots
R2-84e63b91	Searches and seizure using a drone (F.S. 934.50, Freedom from Unwarranted Surveillance Act)	US	2026	in_force	3	substantive_document	drones_uas; security_policing_robots
R2-f51b01be	Unmanned Aircraft Systems Act (Florida Statutes s. 330.41)	US	2026	in_force	2	substantive_document	drones_uas; delivery_robots
R2-a846a9fc	Iowa Code §321.514 Definitions (automated driving systems)	US	2019	in_force	3	substantive_document	autonomous_road_vehicles
R2-ed0d2da3	House Bill 7 (2024 Regular Session): AN ACT relating to autonomous vehicles	US	2024	in_force	3	substantive_document	autonomous_road_vehicles
R2-4c85870b	Maine Revised Statutes Title 25 §4501. Regulation of unmanned aerial vehicles	US	2015	in_force	3	substantive_document	drones_uas; security_policing_robots
R2-f5e63b92	Executive Order 18-04: Establishing the Governor's Advisory Council on Connected and Automated Vehic	US	2018	in_force	3	substantive_document	autonomous_road_vehicles
R2-8e251575	626.19 Use of Unmanned Aerial Vehicles (2026 Minnesota Statutes)	US	2025	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-b53750a0	Missouri Revised Statutes Section 304.900 – Personal delivery devices, authority to operate	US	2021	in_force	3	substantive_document	delivery_robots
R2-baba510e	46-5-109 Limitations on unmanned aerial vehicles	US	2019	in_force	2	substantive_document	drones_uas
R2-b78e9f87	Nevada Revised Statutes Chapter 482A - Autonomous Vehicles	US	2017	in_force	3	substantive_document	autonomous_road_vehicles
R2-7229d3e9	Nevada Administrative Code Chapter 482A - Autonomous Vehicles	US	2026	in_force	3	substantive_document	autonomous_road_vehicles
R2-ba5d81a5	Nevada Revised Statutes Chapter 493 - General Provisions (Aeronautics, incl. Unmanned Aerial Vehicle	US	2025	in_force	2	substantive_document	drones_uas
R2-785c9ea7	Article 18. Regulation of Fully Autonomous Vehicles (N.C. Gen. Stat. § 20-400 Definitions)	US	2017	in_force	3	partial_or_excerpt	autonomous_road_vehicles
R2-1e959e91	N.C. Gen. Stat. § 15A-300.1 Restrictions on use of unmanned aircraft systems (Article 16B)	US	2017	in_force	2	substantive_document	drones_uas

doc_id	title_en	jurisdiction_country	year_final	instrument_status	relevance	page_quality	embodied_categories
R2-b58572ff	N.C. Gen. Stat. § 14-401.24. Unlawful possession and use of unmanned aircraft systems	US	2023	in_force	2	substantive_document	drones_uas
R2-63dcc910	Section 4511.513 Operation of personal delivery device on sidewalks and crosswalks	US	2021	in_force	3	substantive_document	delivery_robots
R2-4f2fca64	Chapter 837 — Aircraft Operation (Oregon Revised Statutes, 2025 Edition)	US	2025	in_force	2	substantive_document	drones_uas
R2-4faa992d	H.B. 101 Autonomous Vehicle Regulations	US	2019	in_force	3	partial_or_excerpt	autonomous_road_vehicles
R2-ac001dfe	H.B. 217 Personal Delivery Devices	US	2018	in_force	2	partial_or_excerpt	delivery_robots
R2-9e5d133d	Automated Vehicle Testing Act (23 V.S.A. Chapter 41, §§ 4201-4203)	US	2019	in_force	3	substantive_document	autonomous_road_vehicles
R2-5d8a8c40	20 V.S.A. § 4622. Law enforcement use of drones	US	2016	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-95eadaba	§ 46.2-908.1:1. Personal delivery devices	US	2020	in_force	3	substantive_document	delivery_robots
R2-4076f1d4	§ 19.2-60.1. Use of unmanned aircraft systems by public bodies; search warrant required	US	2026	in_force	2	substantive_document	drones_uas; security_policing_robots
R2-115aae90	§ 18.2-121.3. Trespass with an unmanned aircraft system; penalty	US	2025	in_force	2	substantive_document	drones_uas
R2-31fedef2	Delaware Code Title 11, Chapter 5, Subchapter VII: Specific Offenses Against Public Health, Order an	US	2025	in_force	1	substantive_document	drones_uas

Table 33b lists the 195 principal binding instruments in the embodied set: 182 in force, 9 superseded and 4 adopted but not yet in force; 175 substantive captures and 20 partial; 168 rated robot-relevant; issued above all by the United States (56), the European Union (34), Japan (22), the United Kingdom (11) and Canada (9), with 5 from the international layer and 4 each from Malaysia and Australia. On the comparable unit of every binding text (Table S01, 291 rows) the forms are 207 regulations or decrees, 71 statutes, 9 mandatory standards, 3 court decisions and 1 treaty, the last captured only as a depositary status page and therefore outside the principal unit. The five binding instruments of the original collection remain, now alongside road-traffic and automated-driving law, civil-aviation and drone regulations, occupational-safety rules for industrial robots and machinery, and product-safety and medical-device regulation. Status is one field, and it has limits the audit exposed: the Machinery Regulation (EU) 2023/1230 is now coded in force (it applies from January 2027), and the Machinery Directive 2006/42/EC, which governs machinery until that date, is coded in force only after a correction of the audit code that had been applied mechanically and marked it superseded; “in force” and “applicable” are not separated and no applicability date is recorded.

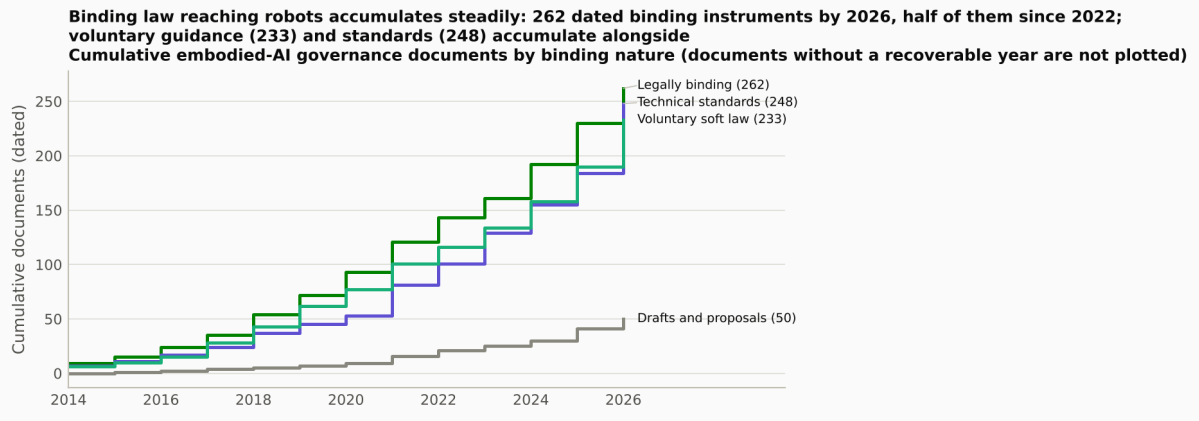


Figure 06. Cumulative embodied-AI governance documents by binding nature.

Figure 6 accumulates dated embodied legal texts by binding nature (every text of instrument class, not the principal unit): 262 binding texts carry a date up to 2026, 143 of them up to 2022; voluntary guidance reaches 233, technical standards 248 and drafts 50. Two cautions apply. Consolidated statutes harvested from legal databases carry the enactment year or the version year as the annotation

found them (Japan's road-traffic and vehicle laws of the 1950s appear at enactment; several of their consolidated texts, Australia's Civil Aviation Safety Regulations and British Columbia's Motor Vehicle Act appear at the 2025 or 2026 compilation date), so the curve mixes the two and the earliest years are not reported as dates of first robot regulation. And the texts that enter the curve include the ancillary and catalogue captures excluded from the headline unit.

5.2 Development over time

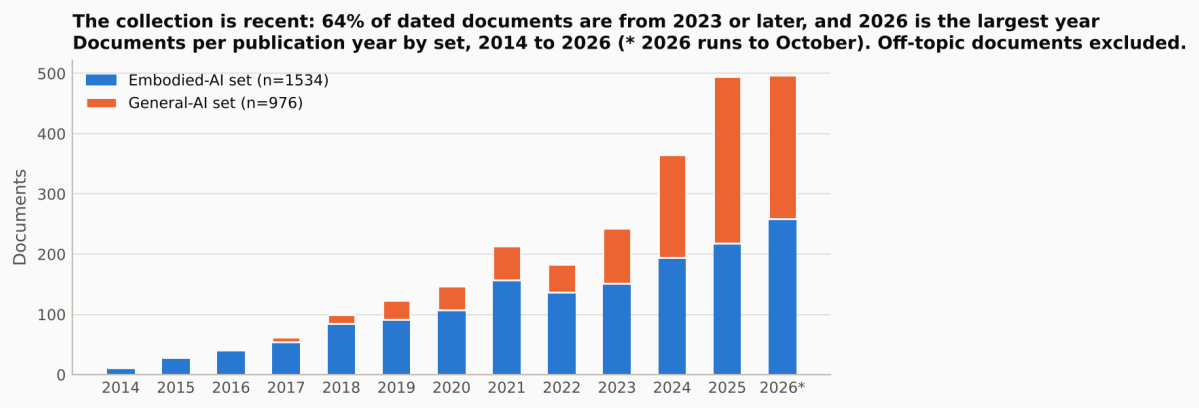


Figure 02. Documents per publication year by set, 2010 to 2026 (2026 is partial).

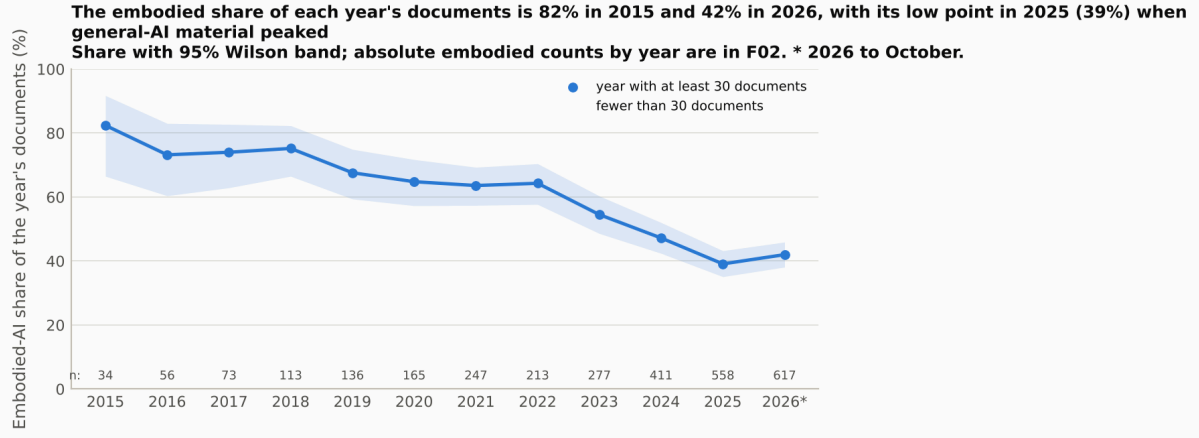


Figure 05. Share of each year's documents that fall in the embodied-AI set, with 95% Wilson intervals.

The collection is recent: a tenth of usable documents date from 2017 or earlier, the median is 2024, 38% date from 2025 or later and 2026 (to October) is the largest year with 617 usable documents, many of them consolidated texts dated at consolidation. The embodied share of each year's documents falls from 82% in 2015 (28 of 34) to 39% in 2025 (218 of 558) and 42% in 2026 (259 of 617; logistic odds ratio 0.84 per year, interval 0.81 to 0.86), because general-AI strategies and AI Act material arrive in volume from 2023; absolute embodied counts rise every year (137 in 2022, 259 in the first ten months of 2026). Time analyses are restricted to 2014 to 2026 and treat 2026 as partial.

Table 03. Embodied categories: document counts and first-appearance years (by document type)

category	level1	n_docs	first_year_any	first_year_governance	n_governance	first_year_research	n_research	median_year
autonomous_road_vehicles	Vehicles & field	427	1951	1951	290	2013	63	2022
drones_uas	Vehicles & field	421	1952	1952	332	2019	32	2023
social_companion_robots	Service & social	244	2006	2013	17	2006	219	2022
industrial_manipulators	Industrial & logistics	218	1972	1972	115	2016	51	2022
care_assistive_robots	Service & social	217	1953	1953	58	2006	140	2022
professional_service_robots	Service & social	208	2007	2013	82	2013	89	2023
military_autonomous_weapons	Security & military	192	2013	2013	74	2016	53	2023
humanoid_general_purpose	Humanoid & general-purpose	168	2013	2013	32	2015	115	2024
surgical_medical_robots	Medical & body-integrated	146	1993	1993	48	2015	74	2023
collaborative_robots	Industrial & logistics	114	1972	1972	41	2017	32	2023
mobile_industrial_logistics	Industrial & logistics	114	1972	1972	71	2017	16	2024
agricultural_robots	Vehicles & field	82	1965	1965	49	2016	12	2023
domestic_household_robots	Service & social	81	2006	2013	31	2006	39	2022
rehabilitation_exoskeleton_prosthetic	Medical & body-integrated	66	1990	1990	39	2018	15	2021

category	level1	n_docs	first_year_any	first_year_governance	n_governance	first_year_research	n_research	median_year
delivery_robots	Industrial & logistics	60	1960	1960	41	2016	11	2023
security_policing_robots	Security & military	58	2013	2013	28	2016	16	2023
educational_robots	Service & social	51	2015	2016	11	2015	35	2023
autonomous_maritime_or_rail	Vehicles & field	49	2001	2001	36	2018	5	2023
space_or_extreme_environment	Vehicles & field	44	2012	2012	23	2015	16	2021
other_embodied	Other embodied	43	1993	1993	15	2016	24	2022
sex_robots	Service & social	28	2006	2017	4	2006	23	2022
biohybrid_or_cyborg	Medical & body-integrated	13	2016	2026	1	2016	12	2022
toys_entertainment_robots	Service & social	11	2006	2016	5	2006	6	2018

Table 3 gives each robot type's document counts and the earliest year of any document and of any governance document. The governance "first years" of the machine classes (road vehicles 1951, unmanned aircraft 1952, care robots 1953, agricultural machinery 1965, industrial manipulators 1972) are the enactment years of consolidated statutes whose robot provisions were added decades later, and are not reported as findings; version dating of consolidated texts was not done. The discourse categories are informative: care, companion, domestic and humanoid robots enter governance documents in 2013, educational robots in 2016, intimate robots in 2017; biohybrid systems have research attention and one draft. Median years show where attention is now: humanoids and logistics robots 2024, most other types 2022 to 2023.

5.3 Geography and language

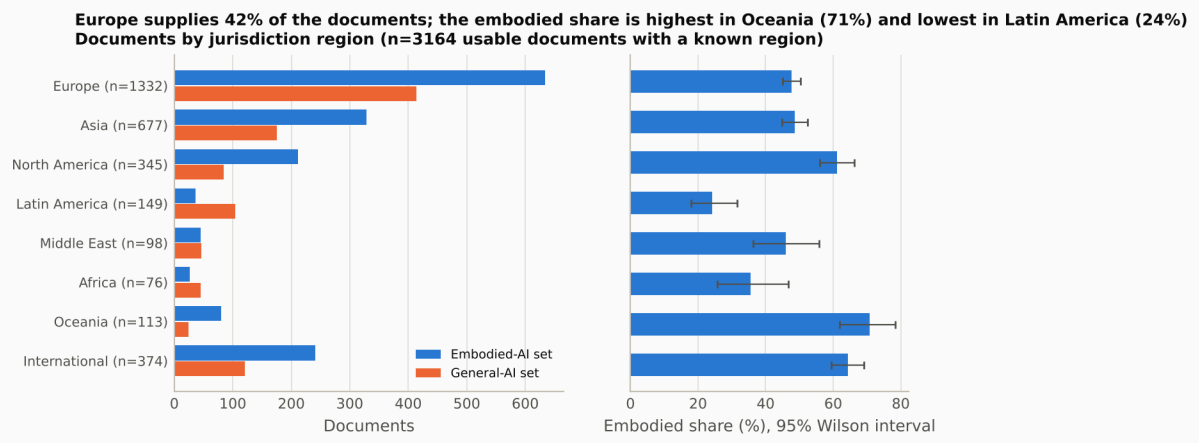


Figure 07. Documents by jurisdiction region and set (left) and the embodied share within each region (right).

Europe supplies 1,332 usable documents (42%), of which 634 are embodied (48%); Asia 677 (329); the international layer 374 (241); North America 345 (211); Latin America and the Caribbean 149 (36); Oceania 113 (80); the Middle East 98 (45); Africa 76 (27). The embodied share varies by region (Cramér's V 0.19), highest in Oceania (71%), the international layer (64%) and North America (61%) and lowest in Latin America (24%) and Africa (36%): Europe and North America contribute embodied documents through research and vehicle and drone law; the international layer through standards; Latin America and Africa still mainly through general-AI instruments.

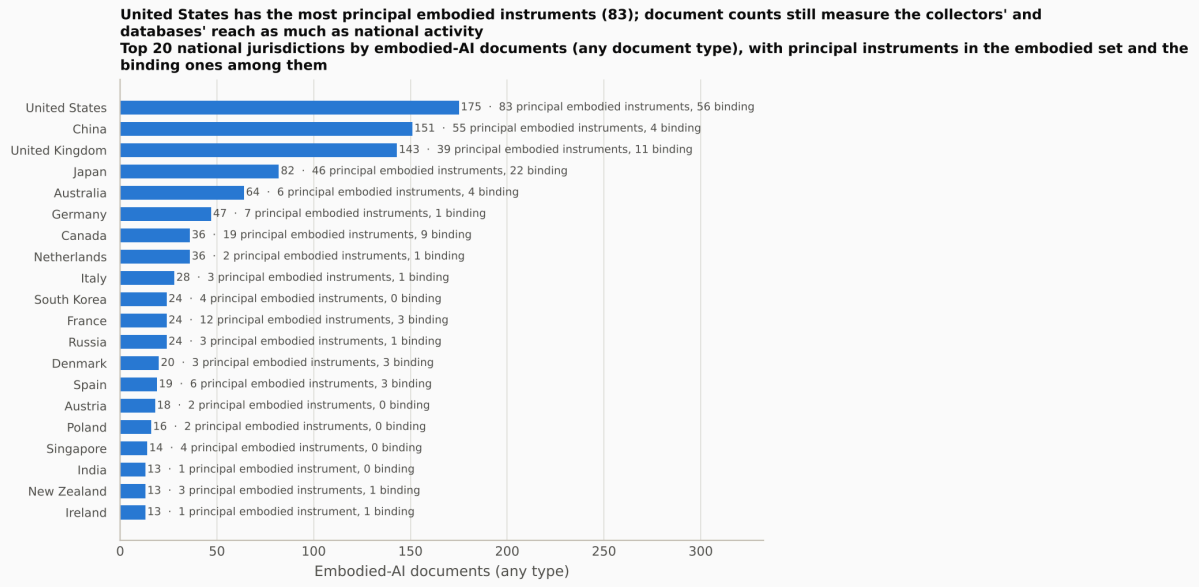


Figure 08. Top 20 national jurisdictions by embodied-AI documents.

Table 04. Documents by jurisdiction country (national jurisdictions only; EU and international bodies excluded)

country	country_name	region	docs	embodied	gov_instruments	embodied_instruments_principal	binding_embodied	embodied_instruments	binding	research	median
US	United States	North America	286	175	154	83	56	94	110	68	
CN	China	Asia	305	151	201	55	4	127	46	20	
GB	United Kingdom	Europe	233	143	78	39	11	49	35	77	
JP	Japan	Asia	148	82	104	46	22	49	64	26	
AU	Australia	Oceania	87	64	15	6	4	47	6	8	
DE	Germany	Europe	51	47	8	7	1	11	1	38	
CA	Canada	North America	59	36	29	19	9	21	10	8	
NL	Netherlands	Europe	38	36	2	2	1	3	1	32	
IT	Italy	Europe	36	28	8	3	1	4	2	24	
KR	South Korea	Asia	50	24	4	4	0	4	0	10	
FR	France	Europe	37	24	14	12	3	13	3	7	
RU	Russia	Europe	28	24	3	3	1	2	1	24	
DK	Denmark	Europe	20	20	3	3	3	3	3	17	
ES	Spain	Europe	39	19	12	6	3	6	3	11	
AT	Austria	Europe	20	18	2	2	0	3	0	12	
PL	Poland	Europe	29	16	4	2	0	4	1	12	
SG	Singapore	Asia	39	14	18	4	0	5	1	2	
IN	India	Asia	25	13	5	1	0	2	0	12	
NZ	New Zealand	Oceania	21	13	9	3	1	4	1	6	
IE	Ireland	Europe	15	13	2	1	1	2	2	10	
SE	Sweden	Europe	14	12	2	1	1	1	1	7	
BR	Brazil	Latin America & Caribbean	49	11	28	6	2	6	2	2	
MY	Malaysia	Asia	23	9	8	4	4	4	4	4	
TW	Taiwan	Asia	15	9	2	1	0	4	1	3	
FI	Finland	Europe	10	8	0	0	0	0	0	9	
ID	Indonesia	Asia	29	7	1	0	0	0	0	17	
AE	United Arab Emirates	Middle East	14	7	9	3	3	3	3	1	
BE	Belgium	Europe	7	7	3	3	0	3	0	3	
HU	Hungary	Europe	7	7	2	2	1	2	1	3	
TR	Türkiye	Middle East	13	6	5	1	1	1	1	4	
NG	Nigeria	Africa	12	6	5	2	1	2	1	4	
IQ	Iraq	Middle East	8	6	1	0	0	0	0	6	
CH	Switzerland	Europe	7	6	0	0	0	0	0	6	
IR	Iran	Middle East	7	6	0	0	0	0	0	6	
AR	Argentina	Latin America & Caribbean	22	5	11	3	2	5	5	1	
PE	Peru	Latin America & Caribbean	13	5	6	3	2	3	3	1	
KE	Kenya	Africa	14	4	3	0	0	0	0	3	
IL	Israel	Middle East	12	4	8	2	0	2	0	2	
SA	Saudi Arabia	Middle East	12	4	7	0	0	0	1	3	
ZA	South Africa	Africa	11	4	4	1	0	2	0	4	
CZ	Czechia	Europe	7	4	2	2	0	2	0	3	
JO	Jordan	Middle East	7	4	3	0	0	0	0	4	
LU	Luxembourg	Europe	6	4	2	1	0	1	0	3	
PT	Portugal	Europe	6	4	0	0	0	0	0	5	
NO	Norway	Europe	5	4	1	1	0	1	0	4	
RO	Romania	Europe	4	4	0	0	0	0	0	4	
PH	Philippines	Asia	6	3	4	2	0	2	0	0	
TH	Thailand	Asia	6	3	2	0	0	0	1	0	
GR	Greece	Europe	5	3	0	0	0	1	0	2	
UA	Ukraine	Europe	5	3	0	0	0	0	0	4	
IS	Iceland	Europe	4	3	3	2	2	2	2	1	
HK	Hong Kong	Asia	3	3	0	0	0	0	0	3	
TZ	Tanzania	Africa	3	3	2	2	2	2	2	0	

country	country_name	region	docs	embodied	gov_instruments	embodied_instruments_principal	binding_embodied	embodied_instruments	binding	research	median
CO	Colombia	Latin America & Caribbean	12	2	3	0	0	0	0	2	
VN	Vietnam	Asia	12	2	3	2	2	2	3	1	
EG	Egypt	Africa	5	2	2	0	0	0	0	0	
MX	Mexico	Latin America & Caribbean	5	2	3	2	2	2	2	2	
QA	Qatar	Middle East	5	2	3	1	0	2	0	0	
SV	El Salvador	Latin America & Caribbean	4	2	2	1	1	1	2	0	
EC	Ecuador	Latin America & Caribbean	3	2	2	2	1	2	1	0	
FJ	Fiji	Oceania	3	2	1	1	1	1	1	0	
LB	Lebanon	Middle East	3	2	1	0	0	0	0	2	
BG	Bulgaria	Europe	2	2	0	0	0	0	0	2	
DZ	Algeria	Africa	2	2	0	0	0	0	0	2	
HR	Croatia	Europe	2	2	0	0	0	0	0	2	
MM	Myanmar	Asia	2	2	0	0	0	0	0	1	
GH	Ghana	Africa	10	1	3	1	1	1	1	1	
UY	Uruguay	Latin America & Caribbean	10	1	3	1	1	1	1	0	
BH	Bahrain	Middle East	6	1	4	0	0	0	0	1	
OM	Oman	Middle East	5	1	1	1	1	1	1	0	
PA	Panama	Latin America & Caribbean	5	1	1	1	0	1	0	0	
RW	Rwanda	Africa	5	1	3	1	1	1	2	0	
CR	Costa Rica	Latin America & Caribbean	4	1	2	0	0	0	0	0	
JM	Jamaica	Latin America & Caribbean	3	1	2	0	0	0	0	0	
LT	Lithuania	Europe	3	1	0	0	0	0	0	2	
PK	Pakistan	Asia	3	1	0	0	0	0	0	1	
TT	Trinidad and Tobago	Latin America & Caribbean	3	1	1	1	1	1	1	0	
UG	Uganda	Africa	3	1	1	1	1	1	1	2	
UZ	Uzbekistan	Asia	3	1	0	0	0	0	0	2	
BD	Bangladesh	Asia	2	1	1	0	0	0	0	0	
EE	Estonia	Europe	2	1	1	0	0	0	0	0	
GE	Georgia	Middle East	2	1	0	0	0	0	0	1	
HN	Honduras	Latin America & Caribbean	2	1	0	0	0	0	0	0	
KH	Cambodia	Asia	2	1	0	0	0	0	0	0	
MU	Mauritius	Africa	2	1	1	1	0	1	0	0	
PG	Papua New Guinea	Oceania	2	1	2	1	1	1	2	0	
SK	Slovakia	Europe	2	1	1	1	0	1	0	1	
BN	Brunei Darussalam	Asia	1	1	0	0	0	0	0	0	
BW	Botswana	Africa	1	1	0	0	0	1	0	0	
BY	Belarus	Europe	1	1	0	0	0	0	0	1	
KY	Cayman Islands	Latin America & Caribbean	1	1	1	1	0	1	0	0	
LK	Sri Lanka	Asia	1	1	0	0	0	0	0	1	
NP	Nepal	Asia	1	1	0	0	0	0	0	0	
PS	Palestine	Middle East	1	1	0	0	0	0	0	1	
RS	Serbia	Europe	1	1	0	0	0	0	0	1	

country	country_name	region	docs	embodied	gov_instruments	embodied_instruments_principal	binding_embodied	embodied_instruments	binding	research	median
CL	Chile	Latin America & Caribbean	6	0	3	0	0	0	0	0	
ZM	Zambia	Africa	3	0	0	0	0	0	0	0	
CU	Cuba	Latin America & Caribbean	2	0	1	0	0	0	0	0	
KW	Kuwait	Middle East	2	0	0	0	0	0	0	0	
AD	Andorra	Europe	1	0	1	0	0	0	0	0	
BB	Barbados	Latin America & Caribbean	1	0	0	0	0	0	0	0	
CI	Côte d'Ivoire	Africa	1	0	0	0	0	0	0	0	
CY	Cyprus	Middle East	1	0	1	0	0	0	0	0	
DO	Dominican Republic	Latin America & Caribbean	1	0	0	0	0	0	0	0	
GY	Guyana	Latin America & Caribbean	1	0	1	0	0	0	0	0	
HT	Haiti	Latin America & Caribbean	1	0	0	0	0	0	0	0	
LS	Lesotho	Africa	1	0	0	0	0	0	0	0	
LV	Latvia	Europe	1	0	0	0	0	0	0	1	
MA	Morocco	Africa	1	0	0	0	0	0	0	0	
ME	Montenegro	Europe	1	0	0	0	0	0	0	0	
MK	North Macedonia	Europe	1	0	0	0	0	0	0	0	
MN	Mongolia	Asia	1	0	0	0	0	0	0	1	
MT	Malta	Europe	1	0	1	0	0	0	0	0	
SI	Slovenia	Europe	1	0	0	0	0	0	0	1	
TC	Turks and Caicos Islands	Latin America & Caribbean	1	0	0	0	0	0	0	1	
TN	Tunisia	Africa	1	0	0	0	0	0	0	0	

Across 116 countries with any document, 76 contribute a principal instrument, 56 a principal instrument that reaches robots and 38 a principal binding instrument that reaches robots (Table S08). Among national jurisdictions the United States has 83 principal embodied instruments (56 binding), China 55 (4), Japan 46 (22), the United Kingdom 39 (11), Canada 19 (9) and France 12 (3); Australia has 6 (4) on the principal unit against 47 embodied legal texts on the comparable unit. China (305 documents), the United States (286) and the United Kingdom (233) lead on documents; counts measure the collectors' and databases' reach as much as national activity, and China's position rests on one full-text harvest of the State Council policy library (271 of its 305 documents are in Chinese, master table).

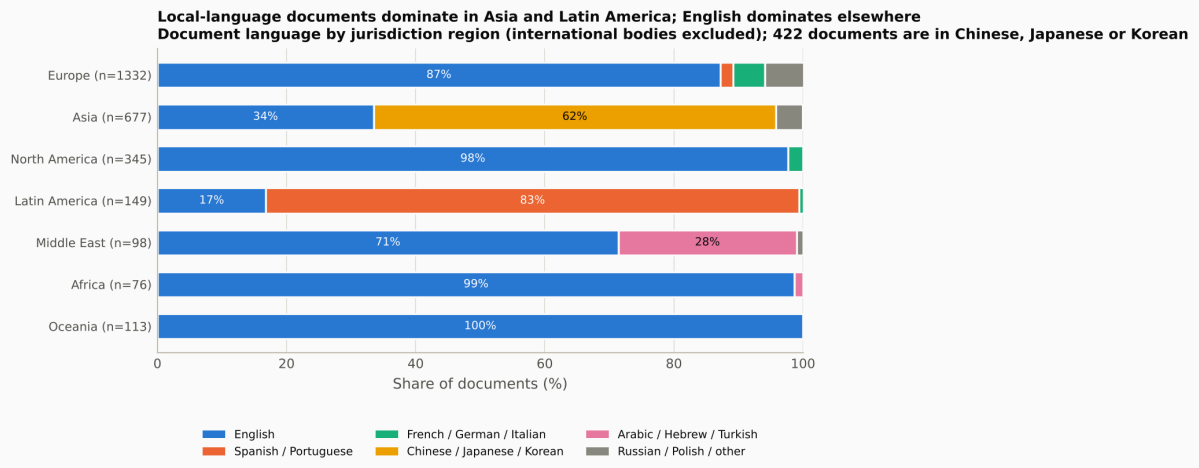


Figure 10. Document language by jurisdiction region.

Table 05. Most frequent venues among research outputs (annotated venue field)

venue	n	embodied	median_citations
arXiv	18	18	18
International Journal of Social Robotics	15	15	93.5
arXiv (cs.RO)	13	13	253
Frontiers in Robotics and AI	13	13	16.5
Ethics and Information Technology	10	10	4

venue	n	embodied	median_citations
AI & Society	7	6	13
Frontiers in Artificial Intelligence and Applications, Vol. 397 (Robophilosophy 2024), IOS Press	6	6	1
日本ロボット学会誌 (Journal of the Robotics Society of Japan)	5	5	0
Frontiers in Artificial Intelligence and Applications (IOS Press)	4	4	1
Science and Engineering Ethics	4	4	nan
ICRES 2022: 7th International Conference on Robot Ethics and Standards	4	4	0
Philosophy & Technology	4	3	nan
Frontiers in Artificial Intelligence and Applications, Vol. 366 (Robophilosophy 2022), IOS Press	4	4	1
AI & SOCIETY	4	3	nan
International Robotics & Automation Journal	3	3	3
ICRES 2019: International Conference on Robot Ethics and Standards, London	3	3	1
Frontiers in Artificial Intelligence and Applications, Vol. 335: Culturally Sustainable Social Robotics (IOS Press)	3	3	3
University of Virginia Libra (Online Archive of University of Virginia Scholarship)	3	3	0
Journal of the Robotics Society of Japan	3	3	71
arXiv (cs.CY)	3	2	nan
AI and Ethics	3	2	nan
IEEE Technology and Society Magazine	2	2	6.5
HRI '24 Companion (ACM/IEEE International Conference on Human-Robot Interaction)	2	2	6.5
International Journal of Advanced Computer Science and Applications (IJACSA)	2	2	44.5
European Journal of Risk Regulation	2	2	14.5

English is 75% of usable documents (2,373); Chinese (278), Japanese (107) and Korean are 62% of Asia's documents (422), Spanish and Portuguese 83% of Latin America's (123). Non-English documents are more often legal texts or reports (68% governance documents against 49%) but, among governance documents, less often binding (33% against 45%; Table 19): the multilingual legal record and the English research literature remain different populations. Research venues are robot-ethics venues; median OpenAlex citations are 1.

6. Part B. What is governed

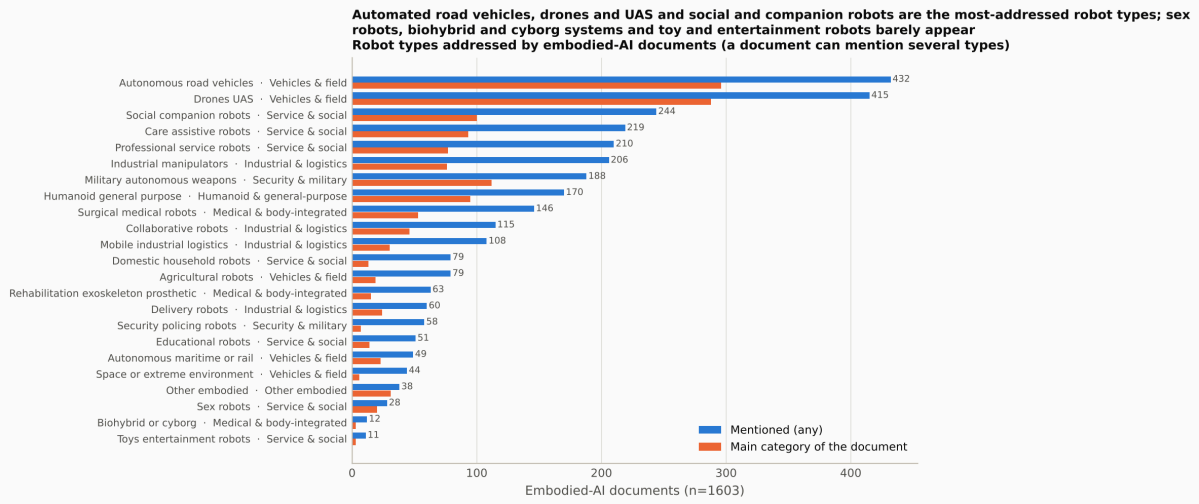


Figure 11. Embodied-AI categories addressed: any mention and primary category (embodied-AI set).

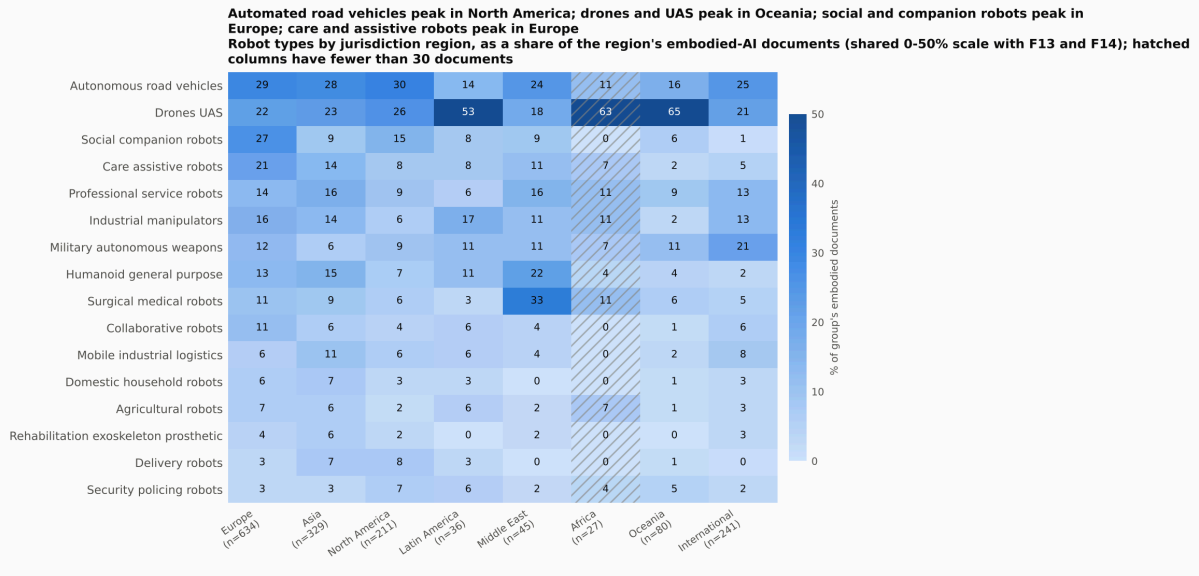


Figure 12. Embodied categories by jurisdiction region, as a share of the region's embodied-AI documents.

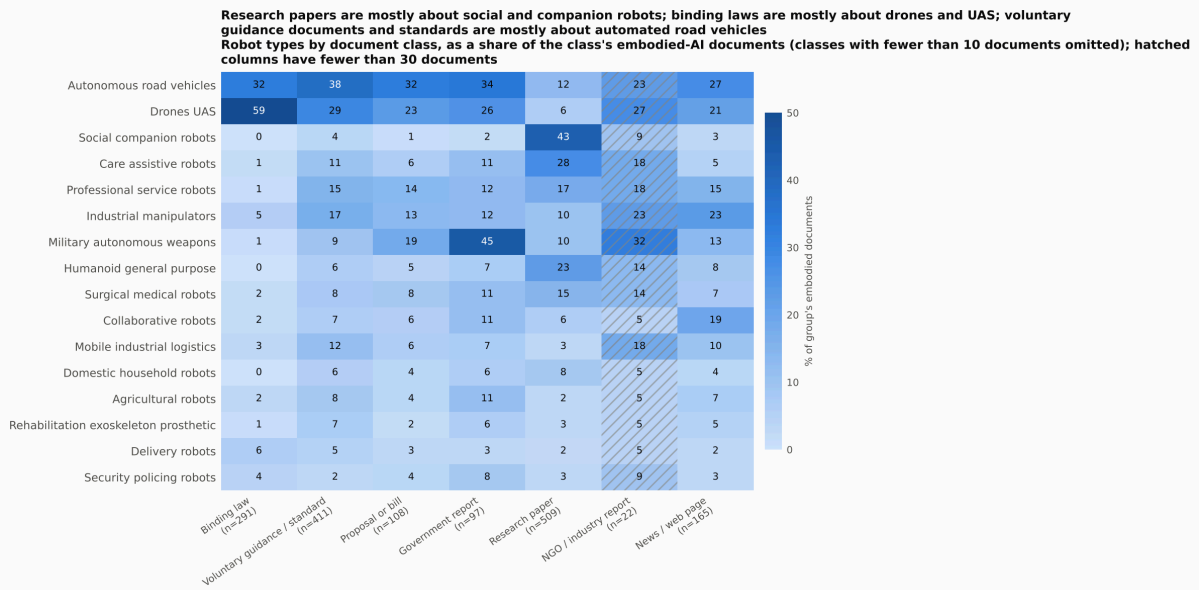


Figure 13. Embodied categories by document class (share of each class's embodied-AI documents).

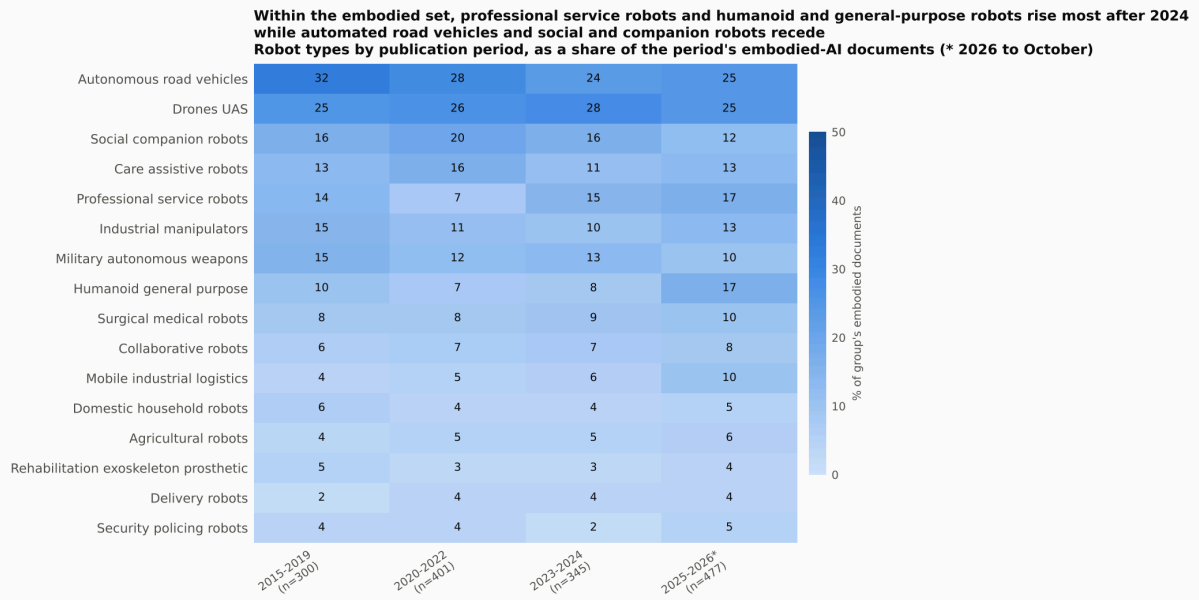


Figure 14. Embodied categories by publication period (share of the period's embodied-AI documents).

Autonomous road vehicles are the most addressed type (432 documents, 27% of the embodied set), then drones (415), companion and social robots (244), care robots (219), professional service robots (210), industrial manipulators (206), military weapons (188), humanoids (170), surgical robots (146) and collaborative robots (115). Regions attend to different machines (Figure 12): social robots are a European (27%) and North American (15%) topic; humanoids lead in the Middle East (22%) and Asia (15%); surgical robots in the Middle East (33%); drones in Oceania (65%), Africa (63%) and Latin America (53%). Genres attend to different machines (Figure 13): research is social-robot heavy (43%); binding texts address drones (59%) and vehicles (32%); soft-law texts vehicles (38%) and industrial arms (17%); drafts vehicles (32%) and military systems (19%). Over time within the embodied set (Figures 14 and 42): social robots fall from 20% of documents in 2020 to 2022 to 12% in 2025 to 2026; humanoids rise from 8% to 17% and mobile logistics robots from 5% to 11%; vehicles hold at 24 to 28%. The humanoid and logistics rise is partly a harvest effect: the gov.cn harvest searched for “humanoid robots” and “embodied intelligence”, and no within-round version of Figure 42 exists, so the period shares are reported as composition-free with respect to genre only.

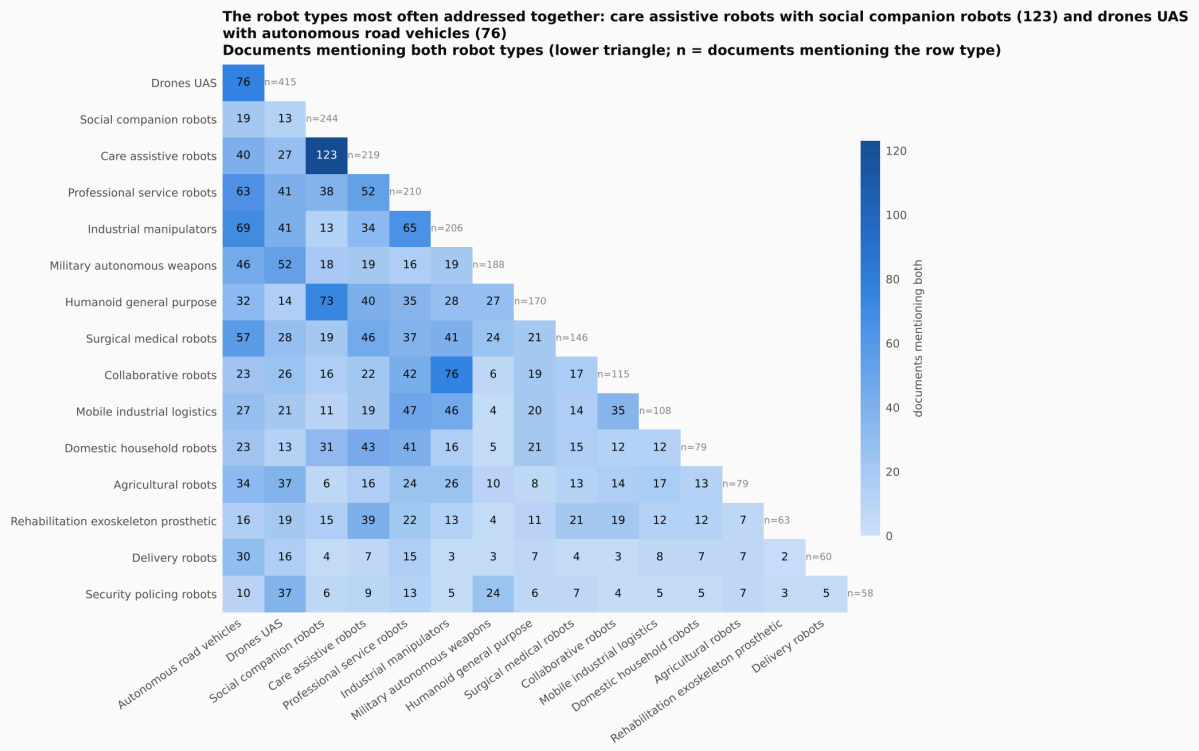


Figure 16. Co-occurrence of embodied categories within documents (counts of documents carrying both labels).

Table 06. Most frequent embodied-category pairs (lift = observed/expected under independence)

category_a	category_b	together	lift	jaccard
social_companion_robots	care_assistive_robots	123	3.69	0.36
autonomous_road_vehicles	drones_uas	76	0.68	0.1
industrial_manipulators	collaborative_robots	76	5.14	0.31
social_companion_robots	humanoid_general_purpose	73	2.82	0.21
autonomous_road_vehicles	industrial_manipulators	69	1.24	0.12
professional_service_robots	industrial_manipulators	65	2.41	0.19
autonomous_road_vehicles	professional_service_robots	63	1.11	0.11
autonomous_road_vehicles	surgical_medical_robots	57	1.45	0.11
drones_uas	military_autonomous_weapons	52	1.07	0.09
care_assistive_robots	professional_service_robots	52	1.81	0.14
professional_service_robots	mobile_industrial_logistics	47	3.32	0.17
industrial_manipulators	mobile_industrial_logistics	46	3.31	0.17
autonomous_road_vehicles	military_autonomous_weapons	46	0.91	0.08
care_assistive_robots	surgical_medical_robots	46	2.31	0.14
care_assistive_robots	domestic_household_robots	43	3.98	0.17
professional_service_robots	collaborative_robots	42	2.79	0.15
industrial_manipulators	surgical_medical_robots	41	2.19	0.13
professional_service_robots	domestic_household_robots	41	3.96	0.17
drones_uas	industrial_manipulators	41	0.77	0.07
drones_uas	professional_service_robots	41	0.75	0.07
care_assistive_robots	humanoid_general_purpose	40	1.72	0.11
autonomous_road_vehicles	care_assistive_robots	40	0.68	0.07
care_assistive_robots	rehabilitation_exoskeleton_prosthetic	39	4.53	0.16
social_companion_robots	professional_service_robots	38	1.19	0.09
drones_uas	security_policing_robots	37	2.46	0.08

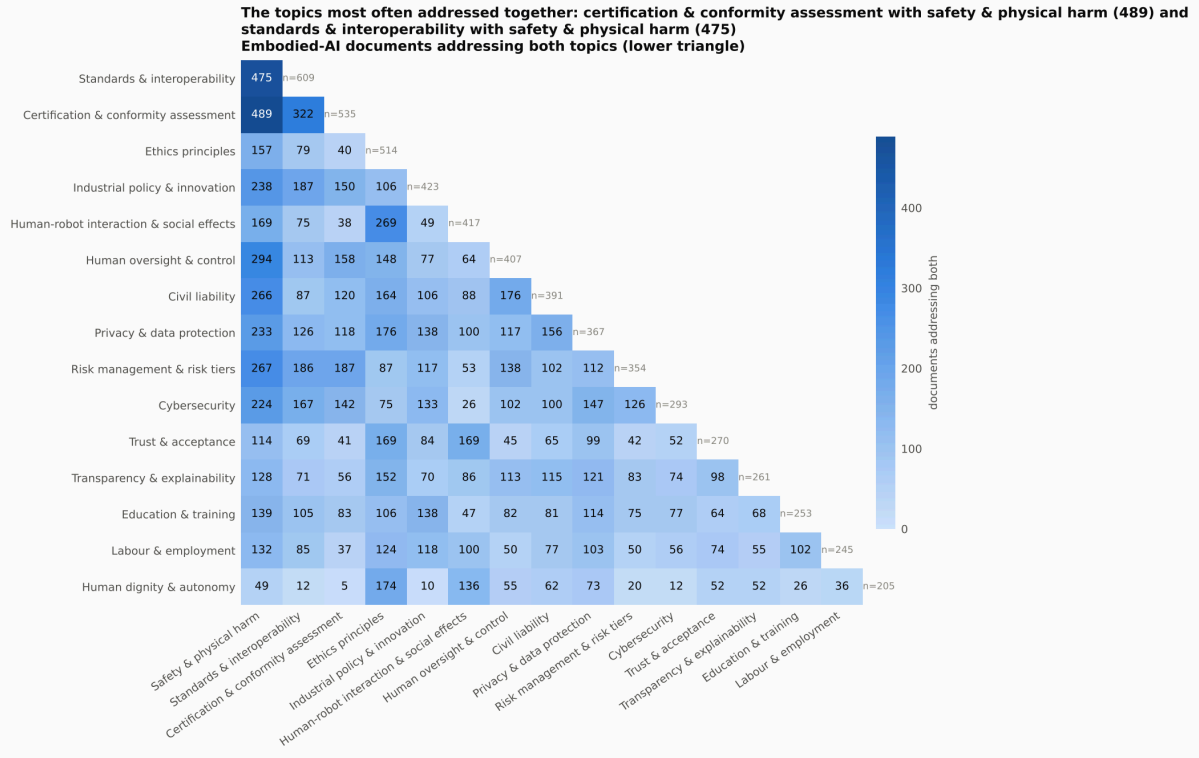


Figure 17. Co-occurrence of governance topics in embodied-AI documents.

Companion and care robots co-occur in 123 documents (Jaccard 0.36) and industrial manipulators and collaborative robots in 76 (lift 5.1); vehicles and drones co-occur in 76 documents but less often than independence predicts (lift 0.68), because the sector laws round 2 added address one machine at a time. Topics form two poles, ethics with human-robot interaction and physical safety with liability, standards and oversight (Figure 17).

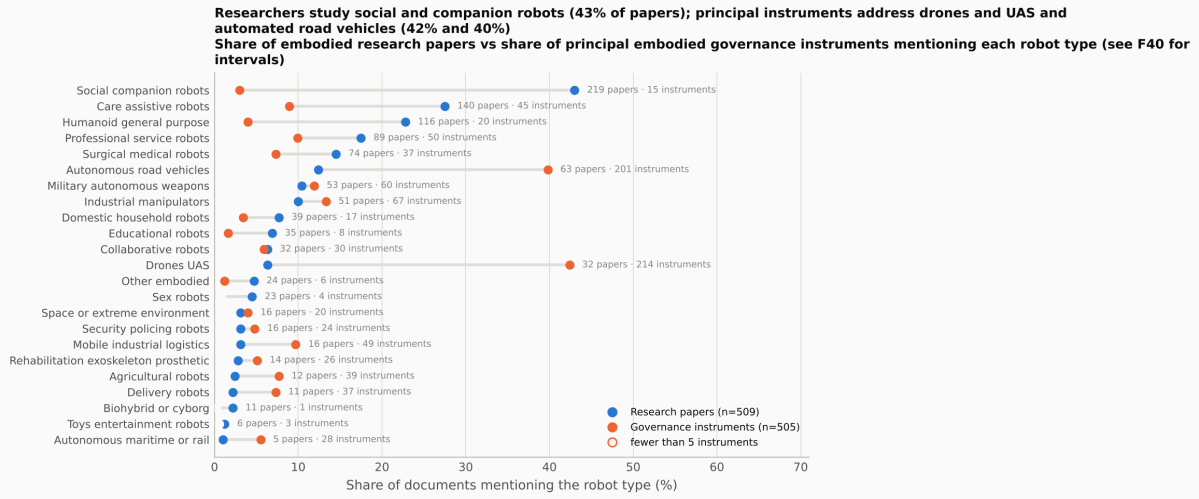


Figure 18. Research attention versus governance attention by embodied category (share of research outputs vs share of principal governance instruments).

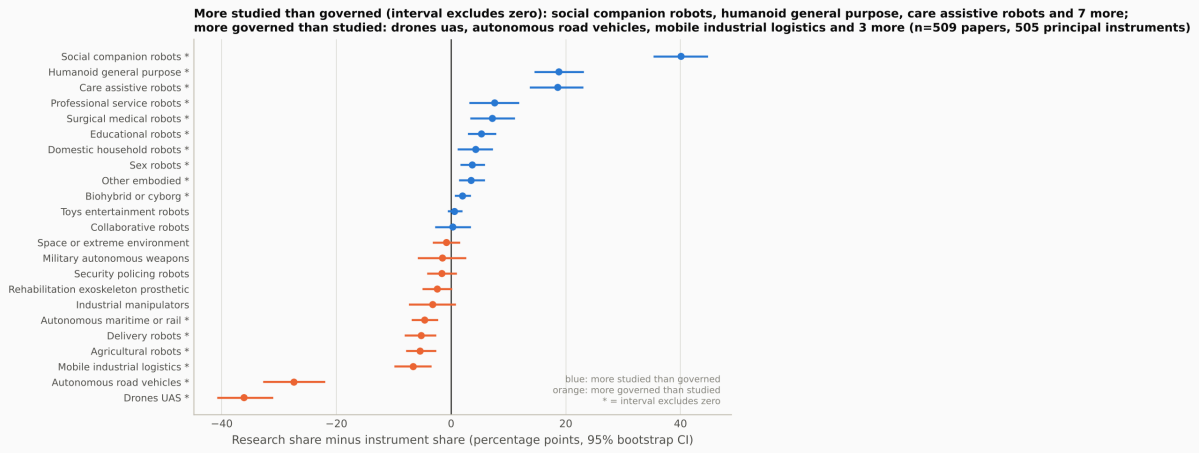


Figure 40. Governance-gap index with 95% bootstrap intervals (research share minus principal-instrument share by robot type); the title names the categories whose interval excludes zero.

Table 07. Governance-gap index per category: share among embodied research outputs (n=509) minus share among embodied governance instruments (n=505)

category	level1	research_pct	n_research	instrument_pct	n_instruments	binding_instruments	gap_index
social_companion_robots	Service & social	43	219	3	15	0	40.1
humanoid_general_purpose	Humanoid & general-purpose	22.8	116	4	20	0	18.8
care_assistive_robots	Service & social	27.5	140	8.9	45	4	18.6
professional_service_robots	Service & social	17.5	89	9.9	50	2	7.6
surgical_medical_robots	Medical & body-integrated	14.5	74	7.3	37	6	7.2
educational_robots	Service & social	6.9	35	1.6	8	1	5.3
domestic_household_robots	Service & social	7.7	39	3.4	17	0	4.3
sex_robots	Service & social	4.5	23	0.8	4	0	3.7
other_embodied	Other embodied	4.7	24	1.2	6	3	3.5
biohybrid_or_cyborg	Medical & body-integrated	2.2	11	0.2	1	0	2
toys_entertainment_robots	Service & social	1.2	6	0.6	3	0	0.6
collaborative_robots	Industrial & logistics	6.3	32	5.9	30	6	0.3
space_or_extreme_environment	Vehicles & field	3.1	16	4	20	2	-0.8
military_autonomous_weapons	Security & military	10.4	53	11.9	60	3	-1.5
security_policing_robots	Security & military	3.1	16	4.8	24	11	-1.6
rehabilitation_exoskeleton_prosthetic	Medical & body-integrated	2.8	14	5.1	26	2	-2.4
industrial_manipulators	Industrial & logistics	10	51	13.3	67	13	-3.2
autonomous_maritime_or_rail	Vehicles & field	1	5	5.5	28	1	-4.6
delivery_robots	Industrial & logistics	2.2	11	7.3	37	15	-5.2
agricultural_robots	Vehicles & field	2.4	12	7.7	39	7	-5.4
mobile_industrial_logistics	Industrial & logistics	3.1	16	9.7	49	9	-6.6
autonomous_road_vehicles	Vehicles & field	12.4	63	39.8	201	67	-27.4
drones_uas	Vehicles & field	6.3	32	42.4	214	107	-36.1

Table 22. Governance-gap index with 95% bootstrap confidence intervals (2,000 resamples of research outputs n=509 and governance instruments n=505 separately)

category	level1	research_pct	n_research	instrument_pct	n_instruments	gap_index	ci_low	ci_high	ci_excludes_zero
social_companion_robots	Service & social	43	219	3	15	40.1	35.5	44.6	True
humanoid_general_purpose	Humanoid & general-purpose	22.8	116	4	20	18.8	14.7	23	True
care_assistive_robots	Service & social	27.5	140	8.9	45	18.6	13.9	22.9	True
professional_service_robots	Service & social	17.5	89	9.9	50	7.6	3.4	11.7	True
surgical_medical_robots	Medical & body-integrated	14.5	74	7.3	37	7.2	3.5	11	True
educational_robots	Service & social	6.9	35	1.6	8	5.3	3.1	7.7	True
domestic_household_robots	Service & social	7.7	39	3.4	17	4.3	1.3	7.1	True
sex_robots	Service & social	4.5	23	0.8	4	3.7	1.8	5.7	True
other_embodied	Other embodied	4.7	24	1.2	6	3.5	1.6	5.7	True
biohybrid_or_cyborg	Medical & body-integrated	2.2	11	0.2	1	2	0.8	3.3	True
toys_entertainment_robots	Service & social	1.2	6	0.6	3	0.6	-0.4	1.8	False
collaborative_robots	Industrial & logistics	6.3	32	5.9	30	0.3	-2.6	3.3	False
space_or_extreme_environment	Vehicles & field	3.1	16	4	20	-0.8	-3	1.4	False
military_autonomous_weapons	Security & military	10.4	53	11.9	60	-1.5	-5.6	2.5	False
security_policing_robots	Security & military	3.1	16	4.8	24	-1.6	-4	0.8	False
rehabilitation_exoskeleton_prosthetic	Medical & body-integrated	2.8	14	5.1	26	-2.4	-4.8	-0	False
industrial_manipulators	Industrial & logistics	10	51	13.3	67	-3.2	-7.2	0.7	False
autonomous_maritime_or_rail	Vehicles & field	1	5	5.5	28	-4.6	-6.7	-2.4	True
delivery_robots	Industrial & logistics	2.2	11	7.3	37	-5.2	-7.9	-2.8	True
agricultural_robots	Vehicles & field	2.4	12	7.7	39	-5.4	-7.7	-2.8	True
mobile_industrial_logistics	Industrial & logistics	3.1	16	9.7	49	-6.6	-9.7	-3.6	True
autonomous_road_vehicles	Vehicles & field	12.4	63	39.8	201	-27.4	-32.6	-22.1	True
drones_uas	Vehicles & field	6.3	32	42.4	214	-36.1	-40.6	-31.2	True

The governance-gap index (research share minus principal-instrument share; Tables 7 and 22) separates three groups, and the bootstrap intervals of 16 of the 23 categories exclude zero. Studied far more than governed: social and companion robots (+40 points, interval 36 to 45; 15 instruments, none binding), care robots (+19, 14 to 23; 45 instruments, 4 binding), humanoids (+19, 15 to 23; 20 instruments, none binding), professional service and surgical robots (+8 and +7) and educational, domestic and intimate robots (+4 to +5). Near parity: collaborative robots, military weapons, security robots, exoskeletons and industrial manipulators (−3, interval −7 to +1). Governed far more than studied: drones (−36, interval −41 to −31; 214 instruments, 107 binding), autonomous road vehicles (−27, interval −33 to −22; 201 instruments, 67 binding) and, by small margins, logistics, agricultural, delivery (37 instruments, 15 binding) and

maritime robots. The ordering is the same as on the v3 unit (supplement S10), so it does not depend on the ancillary captures. The pattern has an explanation: vehicles, drones and field machines inherited transport and aviation regimes, whereas social robots raise questions (deception, attachment, dignity, vulnerable users) without a regulatory home of their own. General regimes do apply to them (general product-safety law, toy-safety rules, data protection, the AI Act's transparency and vulnerability provisions, the consumer-robot standard UL 3300); the gap is in robot-specific instruments, not in the absence of any law, and the Toy Safety Regulation (EU) 2025/2509 and California SB 243 are among the instruments located by collectors and not read (Part F).

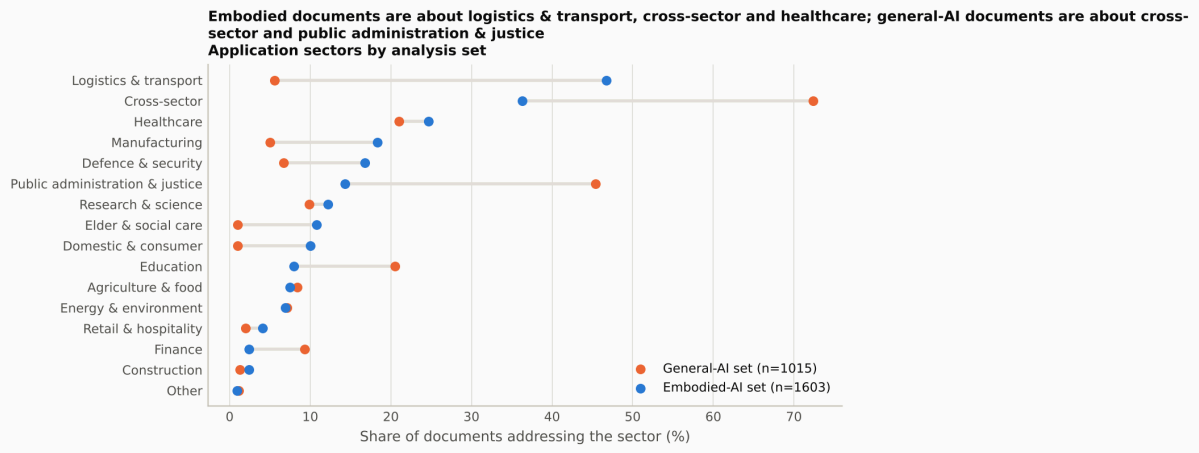


Figure 19. Application sectors by set.

Embodied documents are sectoral (logistics and transport 47%, health 25%, manufacturing 18%, defence 17%, elder and social care 11%); general-AI documents are cross-cutting (72%) and about public administration (45%).

7. Part C. How robots are governed: principles, tools, framings

7.1 Principles: pooled, within genre, within round, adjusted

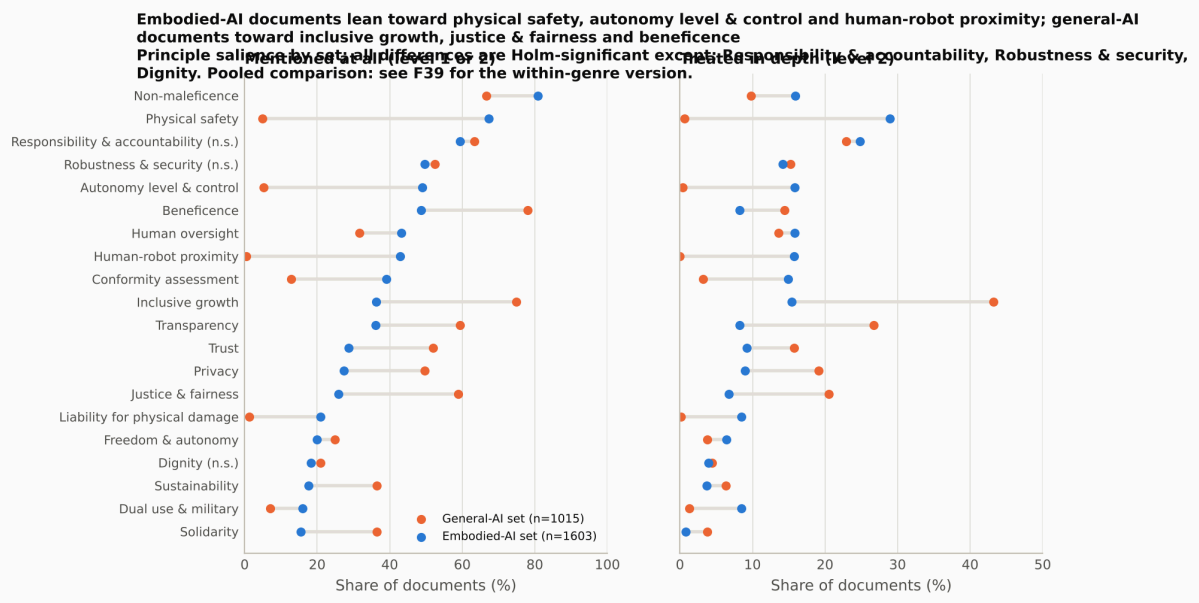


Figure 20. Principle salience by set; asterisks mark differences that remain significant after Holm correction.

Table 08. Principle presence (level >= 1) and substantive treatment (level 2): embodied-AI set (n=1603) vs general-AI set (n=1015); chi-square/Fisher tests with Holm correction

principle	label	group	embodied_any_pct	general_any_pct	embodied_substantive_pct	general_substantive_pct	diff_any_pp	test
non_maleficence	Non-maleficence	core	80.8	66.6	15.9	9.8	14.2	chi-square
physical_safety	Physical safety	embodied-specific	67.3	4.9	28.9	0.6	62.4	chi-square
responsibility_accountability	Responsibility & accountability	core	59.3	63.3	24.8	22.9	-4	chi-square
robustness_security	Robustness & security	core	49.7	52.4	14.2	15.2	-2.7	chi-square
autonomy_level_control	Autonomy level & control	embodied-specific	49	5.2	15.8	0.4	43.7	chi-square

principle	label	group	embodied_any_pct	general_any_pct	embodied_substantive_pct	general_substantive_pct	diff_any_pp	test
beneficence	Beneficence	core	48.7	78	8.2	14.4	-29.4	chi-square
human_oversight	Human oversight	core	43.2	31.7	15.8	13.6	11.5	chi-square
human_robot_proximity	Human-robot proximity	embodied-specific	42.9	0.4	15.7	0	42.5	chi-square
conformity_assessment_certification	Conformity assessment	embodied-specific	39.1	12.8	14.9	3.2	26.3	chi-square
inclusive_growth	Inclusive growth	core	36.2	74.9	15.4	43.2	-38.6	chi-square
transparency	Transparency	core	36.1	59.4	8.2	26.7	-23.4	chi-square
trust	Trust	core	28.7	51.9	9.2	15.7	-23.2	chi-square
privacy	Privacy	core	27.3	49.7	9	19.1	-22.4	chi-square
justice_fairness	Justice & fairness	core	25.9	58.9	6.7	20.5	-33	chi-square
liability_for_physical_damage	Liability for physical damage	embodied-specific	21	1.3	8.5	0.1	19.7	chi-square
freedom_autonomy	Freedom & autonomy	core	19.9	24.8	6.4	3.8	-4.9	chi-square
dignity	Dignity	core	18.2	20.9	3.9	4.4	-2.7	chi-square
sustainability	Sustainability	core	17.7	36.5	3.7	6.3	-18.7	chi-square
dual_use_military	Dual use & military	embodied-specific	15.9	7	8.5	1.3	8.9	chi-square
solidarity	Solidarity	core	15.4	36.4	0.8	3.8	-20.9	chi-square

Pooled over all documents, 17 of 20 principle differences between the sets survive Holm correction (accountability, robustness and dignity do not). Pooled comparisons are, however, mostly comparisons of genres and, in this collection, of collection rounds. Four further analyses settle what survives.

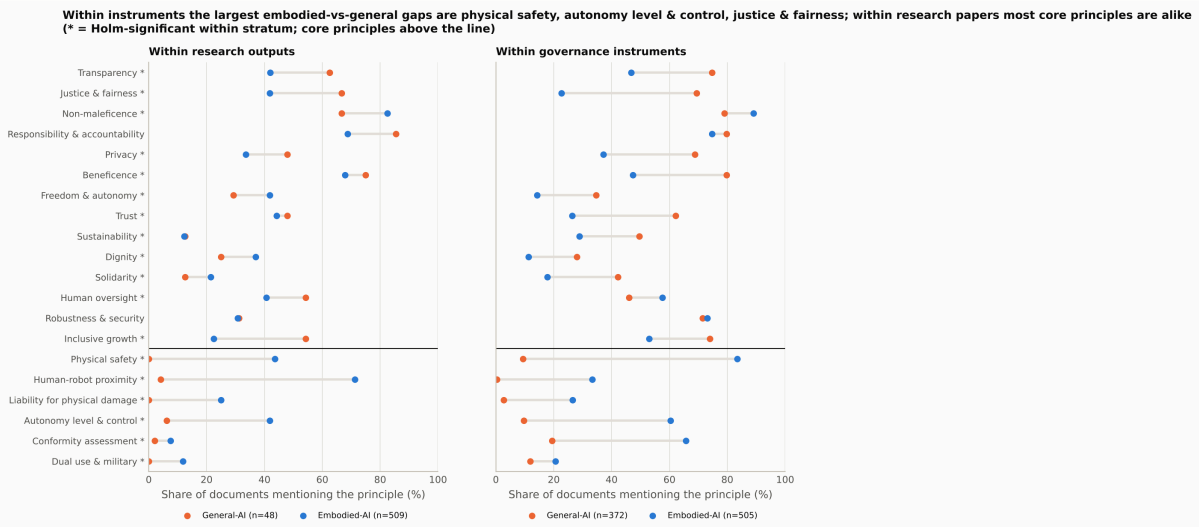


Figure 39. Embodied vs general principle presence within research outputs and within governance instruments (genre-stratified).

Table 21. Embodied vs general comparison within each document genre (Holm correction within stratum)

stratum	item	kind	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm
Research outputs	Transparency	principle	509	48	42	62.5	-20.5	chi-square	0.0099	0.3175
Research outputs	Justice & fairness	principle	509	48	41.8	66.7	-24.8	chi-square	0.0016	0.0521
Research outputs	Non-maleficence	principle	509	48	82.5	66.7	15.8	chi-square	0.0128	0.3953
Research outputs	Responsibility & accountability	principle	509	48	68.8	85.4	-16.7	chi-square	0.0247	0.7155
Research outputs	Privacy	principle	509	48	33.6	47.9	-14.3	chi-square	0.0669	1
Research outputs	Beneficence	principle	509	48	67.8	75	-7.2	chi-square	0.3864	1
Research outputs	Freedom & autonomy	principle	509	48	41.8	29.2	12.7	chi-square	0.1198	1

stratum	item	kind	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm
Research outputs	Trust	principle	509	48	44.2	47.9	-3.7	chi-square	0.7317	1
Research outputs	Sustainability	principle	509	48	12.2	12.5	-0.3	chi-square	1	1
Research outputs	Dignity	principle	509	48	36.9	25	11.9	chi-square	0.1361	1
Research outputs	Solidarity	principle	509	48	21.4	12.5	8.9	chi-square	0.2033	1
Research outputs	Human oversight	principle	509	48	40.7	54.2	-13.5	chi-square	0.097	1
Research outputs	Robustness & security	principle	509	48	30.8	31.2	-0.4	chi-square	1	1
Research outputs	Inclusive growth	principle	509	48	22.4	54.2	-31.8	chi-square	0	0.0001
Research outputs	Physical safety	principle	509	48	43.6	0	43.6	chi-square	0	0
Research outputs	Human-robot proximity	principle	509	48	71.3	4.2	67.1	chi-square	0	0
Research outputs	Liability for physical damage	principle	509	48	25	0	25	chi-square	0.0002	0.0058
Research outputs	Autonomy level & control	principle	509	48	41.8	6.2	35.6	chi-square	0	0.0001
Research outputs	Conformity assessment	principle	509	48	7.5	2.1	5.4	Fisher	0.2372	1
Research outputs	Dual use & military	principle	509	48	11.8	0	11.8	chi-square	0.0229	0.6878
Research outputs	Audits impact assessment	mechanism	509	48	11.6	12.5	-0.9	chi-square	1	1
Research outputs	Conformity assessment certification	mechanism	509	48	3.3	2.1	1.3	Fisher	1	1
Research outputs	Data governance requirements	mechanism	509	48	3.7	4.2	-0.4	Fisher	0.7005	1
Research outputs	Education training programs	mechanism	509	48	9.2	14.6	-5.3	Fisher	0.2102	1
Research outputs	Funding or incentives	mechanism	509	48	2.4	8.3	-6	Fisher	0.0408	1
Research outputs	Human oversight requirement	mechanism	509	48	5.1	4.2	0.9	Fisher	1	1
Research outputs	Incident reporting	mechanism	509	48	2.4	0	2.4	Fisher	0.6122	1
Research outputs	Insurance requirement	mechanism	509	48	2	2.1	-0.1	Fisher	1	1
Research outputs	International cooperation	mechanism	509	48	5.3	10.4	-5.1	Fisher	0.1821	1
Research outputs	Liability regime	mechanism	509	48	12	6.2	5.7	chi-square	0.34	1
Research outputs	Licensing registration	mechanism	509	48	1.6	4.2	-2.6	Fisher	0.2102	1
Research outputs	Oversight body creation	mechanism	509	48	2.8	8.3	-5.6	Fisher	0.0604	1
Research outputs	Prohibitions bans	mechanism	509	48	2.2	0	2.2	Fisher	0.611	1
Research outputs	Risk based tiering	mechanism	509	48	2	8.3	-6.4	Fisher	0.0255	0.7155
Research outputs	Self regulation codes	mechanism	509	48	6.5	6.2	0.2	Fisher	1	1
Research outputs	Technical standards reference	mechanism	509	48	9.8	4.2	5.7	Fisher	0.2973	1
Research outputs	Testing sandboxes	mechanism	509	48	3.7	4.2	-0.4	Fisher	0.7005	1
Research outputs	Transparency disclosure labelling	mechanism	509	48	5.7	10.4	-4.7	Fisher	0.2011	1
Governance instruments	Transparency	principle	505	372	46.7	74.7	-28	chi-square	0	0
Governance instruments	Justice & fairness	principle	505	372	22.6	69.4	-46.8	chi-square	0	0
Governance instruments	Non-maleficence	principle	505	372	89.1	79	10.1	chi-square	0.0001	0.0006
Governance instruments	Responsibility & accountability	principle	505	372	74.7	79.8	-5.2	chi-square	0.0857	0.2952
Governance instruments	Privacy	principle	505	372	37.2	68.8	-31.6	chi-square	0	0
Governance instruments	Beneficence	principle	505	372	47.3	79.8	-32.5	chi-square	0	0
Governance instruments	Freedom & autonomy	principle	505	372	14.3	34.7	-20.4	chi-square	0	0
Governance instruments	Trust	principle	505	372	26.3	62.1	-35.8	chi-square	0	0
Governance instruments	Sustainability	principle	505	372	28.9	49.5	-20.6	chi-square	0	0
Governance instruments	Dignity	principle	505	372	11.3	28	-16.7	chi-square	0	0
Governance instruments	Solidarity	principle	505	372	17.8	42.2	-24.4	chi-square	0	0
Governance instruments	Human oversight	principle	505	372	57.6	46	11.7	chi-square	0.0008	0.0073
Governance instruments	Robustness & security	principle	505	372	73.1	71.5	1.6	chi-square	0.6631	0.6631

stratum	item	kind	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm
Governance instruments	Inclusive growth	principle	505	372	52.9	73.9	-21.1	chi-square	0	0
Governance instruments	Physical safety	principle	505	372	83.4	9.4	74	chi-square	0	0
Governance instruments	Human-robot proximity	principle	505	372	33.3	0.3	33	chi-square	0	0
Governance instruments	Liability for physical damage	principle	505	372	26.5	2.7	23.8	chi-square	0	0
Governance instruments	Autonomy level & control	principle	505	372	60.4	9.7	50.7	chi-square	0	0
Governance instruments	Conformity assessment	principle	505	372	65.7	19.4	46.4	chi-square	0	0
Governance instruments	Dual use & military	principle	505	372	20.6	11.8	8.8	chi-square	0.0009	0.0073
Governance instruments	Audits impact assessment	mechanism	505	372	42.6	57.5	-15	chi-square	0	0.0002
Governance instruments	Conformity assessment certification	mechanism	505	372	42	11.8	30.2	chi-square	0	0
Governance instruments	Data governance requirements	mechanism	505	372	20.4	36.3	-15.9	chi-square	0	0
Governance instruments	Education training programs	mechanism	505	372	35	52.7	-17.6	chi-square	0	0
Governance instruments	Funding or incentives	mechanism	505	372	30.1	46.2	-16.1	chi-square	0	0
Governance instruments	Human oversight requirement	mechanism	505	372	26.7	21	5.8	chi-square	0.059	0.2952
Governance instruments	Incident reporting	mechanism	505	372	24.4	15.9	8.5	chi-square	0.0029	0.02
Governance instruments	Insurance requirement	mechanism	505	372	15.4	1.9	13.6	chi-square	0	0
Governance instruments	International cooperation	mechanism	505	372	34.1	55.9	-21.9	chi-square	0	0
Governance instruments	Liability regime	mechanism	505	372	11.3	5.4	5.9	chi-square	0.0033	0.02
Governance instruments	Licensing registration	mechanism	505	372	43.8	10.2	33.5	chi-square	0	0
Governance instruments	Oversight body creation	mechanism	505	372	37.6	56.2	-18.6	chi-square	0	0
Governance instruments	Prohibitions bans	mechanism	505	372	21	11.8	9.2	chi-square	0.0005	0.0052
Governance instruments	Risk based tiering	mechanism	505	372	18.2	23.1	-4.9	chi-square	0.0895	0.2952
Governance instruments	Self regulation codes	mechanism	505	372	7.3	23.7	-16.3	chi-square	0	0
Governance instruments	Technical standards reference	mechanism	505	372	65	34.9	30	chi-square	0	0
Governance instruments	Testing sandboxes	mechanism	505	372	36.4	30.4	6.1	chi-square	0.0716	0.2952
Governance instruments	Transparency disclosure labelling	mechanism	505	372	20.4	42.2	-21.8	chi-square	0	0
Government reports	Transparency	principle	97	148	57.7	69.6	-11.9	chi-square	0.0774	1
Government reports	Justice & fairness	principle	97	148	55.7	75	-19.3	chi-square	0.0026	0.0563
Government reports	Non-maleficence	principle	97	148	90.7	64.9	25.9	chi-square	0	0.0003
Government reports	Responsibility & accountability	principle	97	148	80.4	69.6	10.8	chi-square	0.0825	1
Government reports	Privacy	principle	97	148	41.2	61.5	-20.2	chi-square	0.0029	0.0612
Government reports	Beneficence	principle	97	148	69.1	83.1	-14	chi-square	0.0156	0.2805
Government reports	Freedom & autonomy	principle	97	148	19.6	34.5	-14.9	chi-square	0.0175	0.298
Government reports	Trust	principle	97	148	53.6	68.9	-15.3	chi-square	0.022	0.352
Government reports	Sustainability	principle	97	148	34	43.9	-9.9	chi-square	0.1576	1
Government reports	Dignity	principle	97	148	27.8	23.6	4.2	chi-square	0.5573	1
Government reports	Solidarity	principle	97	148	28.9	45.9	-17.1	chi-square	0.0109	0.2189
Government reports	Human oversight	principle	97	148	71.1	39.2	31.9	chi-square	0	0.0001
Government reports	Robustness & security	principle	97	148	63.9	63.5	0.4	chi-square	1	1

stratum	item	kind	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm
Government reports	Inclusive growth	principle	97	148	67	85.8	-18.8	chi-square	0.0008	0.022
Government reports	Physical safety	principle	97	148	76.3	2.7	73.6	chi-square	0	0
Government reports	Human-robot proximity	principle	97	148	32	0	32	chi-square	0	0
Government reports	Liability for physical damage	principle	97	148	37.1	1.4	35.8	chi-square	0	0
Government reports	Autonomy level & control	principle	97	148	69.1	4.1	65	chi-square	0	0
Government reports	Conformity assessment	principle	97	148	52.6	14.9	37.7	chi-square	0	0
Government reports	Dual use & military	principle	97	148	48.5	8.1	40.3	chi-square	0	0
Government reports	Audits impact assessment	mechanism	97	148	45.4	43.2	2.1	chi-square	0.8455	1
Government reports	Conformity assessment certification	mechanism	97	148	18.6	10.8	7.7	chi-square	0.127	1
Government reports	Data governance requirements	mechanism	97	148	14.4	18.9	-4.5	chi-square	0.4606	1
Government reports	Education training programs	mechanism	97	148	39.2	50.7	-11.5	chi-square	0.1021	1
Government reports	Funding or incentives	mechanism	97	148	27.8	54.7	-26.9	chi-square	0.0001	0.0017
Government reports	Human oversight requirement	mechanism	97	148	36.1	11.5	24.6	chi-square	0	0.0003
Government reports	Incident reporting	mechanism	97	148	12.4	2	10.3	chi-square	0.0024	0.0562
Government reports	Insurance requirement	mechanism	97	148	11.3	1.4	10	chi-square	0.0018	0.0434
Government reports	International cooperation	mechanism	97	148	59.8	50	9.8	chi-square	0.1698	1
Government reports	Liability regime	mechanism	97	148	17.5	4.1	13.5	chi-square	0.0009	0.0232
Government reports	Licensing registration	mechanism	97	148	14.4	2	12.4	chi-square	0.0005	0.0135
Government reports	Oversight body creation	mechanism	97	148	32	47.3	-15.3	chi-square	0.0243	0.3643
Government reports	Prohibitions bans	mechanism	97	148	27.8	6.8	21.1	chi-square	0	0.0004
Government reports	Risk based tiering	mechanism	97	148	16.5	16.2	0.3	chi-square	1	1
Government reports	Self regulation codes	mechanism	97	148	12.4	12.2	0.2	chi-square	1	1
Government reports	Technical standards reference	mechanism	97	148	36.1	20.9	15.1	chi-square	0.0137	0.2607
Government reports	Testing sandboxes	mechanism	97	148	34	25	9	chi-square	0.1664	1
Government reports	Transparency disclosure labelling	mechanism	97	148	20.6	22.3	-1.7	chi-square	0.878	1
News and communications	Transparency	principle	165	380	17.6	39.7	-22.2	chi-square	0	0
News and communications	Justice & fairness	principle	165	380	12.7	42.9	-30.2	chi-square	0	0
News and communications	Non-maleficence	principle	165	380	74.5	55.3	19.3	chi-square	0	0.0007
News and communications	Responsibility & accountability	principle	165	380	41.8	41.8	-0	chi-square	1	1
News and communications	Privacy	principle	165	380	10.9	28.2	-17.2	chi-square	0	0.0004
News and communications	Beneficence	principle	165	380	58.8	77.4	-18.6	chi-square	0	0.0004
News and communications	Freedom & autonomy	principle	165	380	6.1	11.6	-5.5	chi-square	0.068	0.6799
News and communications	Trust	principle	165	380	21.2	36.6	-15.4	chi-square	0.0006	0.0106
News and communications	Sustainability	principle	165	380	20	25.8	-5.8	chi-square	0.1789	1
News and communications	Dignity	principle	165	380	5.5	13.2	-7.7	chi-square	0.0121	0.1694
News and communications	Solidarity	principle	165	380	7.3	32.4	-25.1	chi-square	0	0
News and communications	Human oversight	principle	165	380	31.5	13.2	18.4	chi-square	0	0
News and communications	Robustness & security	principle	165	380	33.3	32.1	1.2	chi-square	0.8558	1

stratum	item	kind	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm
News and communications	Inclusive growth	principle	165	380	53.3	76.8	-23.5	chi-square	0	0
News and communications	Physical safety	principle	165	380	67.9	1.1	66.8	chi-square	0	0
News and communications	Human-robot proximity	principle	165	380	35.8	0	35.8	chi-square	0	0
News and communications	Liability for physical damage	principle	165	380	12.1	0.3	11.9	chi-square	0	0
News and communications	Autonomy level & control	principle	165	380	38.8	1.3	37.5	chi-square	0	0
News and communications	Conformity assessment	principle	165	380	31.5	6.8	24.7	chi-square	0	0
News and communications	Dual use & military	principle	165	380	17	3.2	13.8	chi-square	0	0
News and communications	Audits impact assessment	mechanism	165	380	13.3	15	-1.7	chi-square	0.7074	1
News and communications	Conformity assessment certification	mechanism	165	380	16.4	3.4	12.9	chi-square	0	0
News and communications	Data governance requirements	mechanism	165	380	4.2	5.5	-1.3	chi-square	0.6799	1
News and communications	Education training programs	mechanism	165	380	17	33.7	-16.7	chi-square	0.0001	0.0021
News and communications	Funding or incentives	mechanism	165	380	24.8	30.5	-5.7	chi-square	0.2143	1
News and communications	Human oversight requirement	mechanism	165	380	11.5	5.3	6.3	chi-square	0.0155	0.2013
News and communications	Incident reporting	mechanism	165	380	6.7	2.6	4	chi-square	0.0448	0.493
News and communications	Insurance requirement	mechanism	165	380	6.7	0	6.7	Fisher	0	0
News and communications	International cooperation	mechanism	165	380	24.8	34.7	-9.9	chi-square	0.0294	0.3525
News and communications	Liability regime	mechanism	165	380	4.8	1.1	3.8	Fisher	0.0093	0.1402
News and communications	Licensing registration	mechanism	165	380	21.8	1.6	20.2	chi-square	0	0
News and communications	Oversight body creation	mechanism	165	380	15.8	34.7	-19	chi-square	0	0.0003
News and communications	Prohibitions bans	mechanism	165	380	10.9	9.7	1.2	chi-square	0.7928	1
News and communications	Risk based tiering	mechanism	165	380	5.5	14.7	-9.3	chi-square	0.0034	0.0546
News and communications	Self regulation codes	mechanism	165	380	5.5	10.5	-5.1	chi-square	0.0821	0.7387
News and communications	Technical standards reference	mechanism	165	380	23	8.9	14.1	chi-square	0	0.0004
News and communications	Testing sandboxes	mechanism	165	380	18.2	13.2	5	chi-square	0.1642	1
News and communications	Transparency disclosure labelling	mechanism	165	380	4.2	13.2	-8.9	chi-square	0.003	0.0502

Within principal instruments (505 vs 372; Table 21), the embodied-specific considerations differ sharply: physical safety 83% vs 9%, autonomy-level control 60% vs 10%, conformity assessment 66% vs 19%, proximity 33% vs 0.3%, liability for physical damage 27% vs 3%, dual use 21% vs 12% (all Holm-significant). Ten of fourteen core principles are also raised less often in robot instruments (transparency 47% vs 75%, justice and fairness 23% vs 69%, privacy 37% vs 69%, beneficence 47% vs 80%, trust 26% vs 62%, inclusive growth 53% vs 74%, sustainability, solidarity, freedom and dignity); non-maleficence (89% vs 79%) and human oversight (58% vs 46%) are raised more often; accountability (75% vs 80%) and robustness (73% vs 72%) do not differ. Within research papers (509 vs 48), four of the six embodied-specific items differ (conformity assessment and dual use do not), inclusive growth is lower in robot papers (22% vs 54%) and no other core principle differs.

Table 20b. Principle presence (level 1 or 2), embodied-AI set vs general-AI set, within each collection round and within governance instruments of each round (Holm within stratum): shows which contrasts depend on the round-2 additions

stratum	principle	group	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm	significant_holm_05
Round 1, all usable documents	Transparency	core	675	803	43.7	59.4	-15.7	chi-square	0	0	True
Round 1, all usable documents	Justice & fairness	core	675	803	43.1	59.2	-16	chi-square	0	0	True
Round 1, all usable documents	Non-maleficence	core	675	803	81.9	68.9	13.1	chi-square	0	0	True
Round 1, all usable documents	Responsibility & accountability	core	675	803	66.1	63	3.1	chi-square	0.2422	0.2422	False
Round 1, all usable documents	Privacy	core	675	803	37.3	50.4	-13.1	chi-square	0	0	True
Round 1, all usable documents	Beneficence	core	675	803	70.4	79.6	-9.2	chi-square	0.0001	0.0003	True

stratum	principle	group	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm	significant_holm_05
Round 1, all usable documents	Freedom & autonomy	core	675	803	38.2	25.2	13.1	chi-square	0	0	True
Round 1, all usable documents	Trust	core	675	803	46.5	51.3	-4.8	chi-square	0.0747	0.1494	False
Round 1, all usable documents	Sustainability	core	675	803	19.6	33.9	-14.3	chi-square	0	0	True
Round 1, all usable documents	Dignity	core	675	803	33.6	21.7	12	chi-square	0	0	True
Round 1, all usable documents	Solidarity	core	675	803	23.7	35.4	-11.7	chi-square	0	0	True
Round 1, all usable documents	Human oversight	core	675	803	40.9	32.8	8.1	chi-square	0.0015	0.0058	True
Round 1, all usable documents	Robustness & security	core	675	803	39.7	49.9	-10.2	chi-square	0.0001	0.0005	True
Round 1, all usable documents	Inclusive growth	core	675	803	34.8	75.5	-40.7	chi-square	0	0	True
Round 1, all usable documents	Physical safety	embodied-specific	675	803	48	2.9	45.1	chi-square	0	0	True
Round 1, all usable documents	Human-robot proximity	embodied-specific	675	803	63.3	0.4	62.9	chi-square	0	0	True
Round 1, all usable documents	Liability for physical damage	embodied-specific	675	803	23.9	0.4	23.5	chi-square	0	0	True
Round 1, all usable documents	Autonomy level & control	embodied-specific	675	803	41.5	5.2	36.3	chi-square	0	0	True
Round 1, all usable documents	Conformity assessment	embodied-specific	675	803	14.4	10.7	3.7	chi-square	0.0405	0.1214	False
Round 1, all usable documents	Dual use & military	embodied-specific	675	803	15.3	5.5	9.8	chi-square	0	0	True
Round 2, all usable documents	Transparency	core	928	212	30.5	59.4	-28.9	chi-square	0	0	True
Round 2, all usable documents	Justice & fairness	core	928	212	13.4	58	-44.7	chi-square	0	0	True
Round 2, all usable documents	Non-maleficence	core	928	212	80.1	58	22	chi-square	0	0	True
Round 2, all usable documents	Responsibility & accountability	core	928	212	54.3	64.2	-9.8	chi-square	0.0115	0.0344	True
Round 2, all usable documents	Privacy	core	928	212	19.9	46.7	-26.8	chi-square	0	0	True
Round 2, all usable documents	Beneficence	core	928	212	32.9	72.2	-39.3	chi-square	0	0	True
Round 2, all usable documents	Freedom & autonomy	core	928	212	6.6	23.6	-17	chi-square	0	0	True
Round 2, all usable documents	Trust	core	928	212	15.7	54.2	-38.5	chi-square	0	0	True
Round 2, all usable documents	Sustainability	core	928	212	16.4	46.2	-29.8	chi-square	0	0	True
Round 2, all usable documents	Dignity	core	928	212	7	17.9	-10.9	chi-square	0	0	True
Round 2, all usable documents	Solidarity	core	928	212	9.4	40.1	-30.7	chi-square	0	0	True
Round 2, all usable documents	Human oversight	core	928	212	44.9	27.8	17.1	chi-square	0	0	True
Round 2, all usable documents	Robustness & security	core	928	212	57	61.8	-4.8	chi-square	0.2313	0.4518	False
Round 2, all usable documents	Inclusive growth	core	928	212	37.3	72.6	-35.4	chi-square	0	0	True
Round 2, all usable documents	Physical safety	embodied-specific	928	212	81.4	12.7	68.6	chi-square	0	0	True
Round 2, all usable documents	Human-robot proximity	embodied-specific	928	212	28	0.5	27.5	chi-square	0	0	True
Round 2, all usable documents	Liability for physical damage	embodied-specific	928	212	18.9	4.7	14.1	chi-square	0	0	True
Round 2, all usable documents	Autonomy level & control	embodied-specific	928	212	54.4	5.2	49.2	chi-square	0	0	True
Round 2, all usable documents	Conformity assessment	embodied-specific	928	212	57.1	20.8	36.4	chi-square	0	0	True
Round 2, all usable documents	Dual use & military	embodied-specific	928	212	16.4	12.7	3.6	chi-square	0.2259	0.4518	False
Round 1, governance instruments	Transparency	core	61	245	83.6	83.3	0.3	chi-square	1	1	False
Round 1, governance instruments	Justice & fairness	core	61	245	68.9	76.3	-7.5	chi-square	0.2989	1	False
Round 1, governance instruments	Non-maleficence	core	61	245	91.8	89	2.8	chi-square	0.681	1	False

stratum	principle	group	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm	significant_holm_05
Round 1, governance instruments	Responsibility & accountability	core	61	245	75.4	86.9	-11.5	chi-square	0.0417	0.5008	False
Round 1, governance instruments	Privacy	core	61	245	77	80.8	-3.8	chi-square	0.6313	1	False
Round 1, governance instruments	Beneficence	core	61	245	91.8	84.9	6.9	chi-square	0.2323	1	False
Round 1, governance instruments	Freedom & autonomy	core	61	245	47.5	42.9	4.7	chi-square	0.6062	1	False
Round 1, governance instruments	Trust	core	61	245	80.3	69	11.3	chi-square	0.1109	1	False
Round 1, governance instruments	Sustainability	core	61	245	62.3	52.7	9.6	chi-square	0.2264	1	False
Round 1, governance instruments	Dignity	core	61	245	34.4	33.9	0.5	chi-square	1	1	False
Round 1, governance instruments	Solidarity	core	61	245	52.5	46.1	6.3	chi-square	0.4571	1	False
Round 1, governance instruments	Human oversight	core	61	245	65.6	56.7	8.8	chi-square	0.2677	1	False
Round 1, governance instruments	Robustness & security	core	61	245	93.4	78.4	15.1	chi-square	0.0117	0.1632	False
Round 1, governance instruments	Inclusive growth	core	61	245	90.2	76.3	13.8	chi-square	0.0277	0.3598	False
Round 1, governance instruments	Physical safety	embodied-specific	61	245	57.4	6.5	50.8	chi-square	0	0	True
Round 1, governance instruments	Human-robot proximity	embodied-specific	61	245	39.3	0.4	38.9	Fisher	0	0	True
Round 1, governance instruments	Liability for physical damage	embodied-specific	61	245	24.6	0.8	23.8	Fisher	0	0	True
Round 1, governance instruments	Autonomy level & control	embodied-specific	61	245	49.2	11.8	37.3	chi-square	0	0	True
Round 1, governance instruments	Conformity assessment	embodied-specific	61	245	59	18.4	40.6	chi-square	0	0	True
Round 1, governance instruments	Dual use & military	embodied-specific	61	245	37.7	10.2	27.5	chi-square	0	0	True
Round 2, governance instruments	Transparency	core	444	127	41.7	58.3	-16.6	chi-square	0.0013	0.0092	True
Round 2, governance instruments	Justice & fairness	core	444	127	16.2	55.9	-39.7	chi-square	0	0	True
Round 2, governance instruments	Non-maleficence	core	444	127	88.7	59.8	28.9	chi-square	0	0	True
Round 2, governance instruments	Responsibility & accountability	core	444	127	74.5	66.1	8.4	chi-square	0.078	0.1561	False
Round 2, governance instruments	Privacy	core	444	127	31.8	45.7	-13.9	chi-square	0.0052	0.0311	True
Round 2, governance instruments	Beneficence	core	444	127	41.2	70.1	-28.9	chi-square	0	0	True
Round 2, governance instruments	Freedom & autonomy	core	444	127	9.7	18.9	-9.2	chi-square	0.0072	0.0359	True
Round 2, governance instruments	Trust	core	444	127	18.9	48.8	-29.9	chi-square	0	0	True
Round 2, governance instruments	Sustainability	core	444	127	24.3	43.3	-19	chi-square	0	0.0004	True
Round 2, governance instruments	Dignity	core	444	127	8.1	16.5	-8.4	chi-square	0.0086	0.0359	True

stratum	principle	group	n_embodied	n_general	embodied_pct	general_pct	diff_pp	test	p	p_holm	significant_holm_05
Round 2, governance instruments	Solidarity	core	444	127	13.1	34.6	-21.6	chi-square	0	0	True
Round 2, governance instruments	Human oversight	core	444	127	56.5	25.2	31.3	chi-square	0	0	True
Round 2, governance instruments	Robustness & security	core	444	127	70.3	58.3	12	chi-square	0.0146	0.0439	True
Round 2, governance instruments	Inclusive growth	core	444	127	47.7	69.3	-21.5	chi-square	0	0.0003	True
Round 2, governance instruments	Physical safety	embodied-specific	444	127	86.9	15	72	chi-square	0	0	True
Round 2, governance instruments	Human-robot proximity	embodied-specific	444	127	32.4	0	32.4	chi-square	0	0	True
Round 2, governance instruments	Liability for physical damage	embodied-specific	444	127	26.8	6.3	20.5	chi-square	0	0	True
Round 2, governance instruments	Autonomy level & control	embodied-specific	444	127	61.9	5.5	56.4	chi-square	0	0	True
Round 2, governance instruments	Conformity assessment	embodied-specific	444	127	66.7	21.3	45.4	chi-square	0	0	True
Round 2, governance instruments	Dual use & military	embodied-specific	444	127	18.2	15	3.3	chi-square	0.4679	0.4679	False

Table 20b repeats the contrast within each collection round. Within the round-1 instruments alone (61 embodied vs 245 general), none of the fourteen core principles differs after Holm correction (transparency 84% vs 83%, privacy 77% vs 81%, justice 69% vs 76%, accountability 75% vs 87% with Holm p = 0.50), while all six embodied-specific items do (physical safety 57% vs 7%, conformity 59% vs 18%, autonomy-level control 49% vs 12%, proximity 39% vs 0.4%, dual use 38% vs 10%, liability 25% vs 1%). Within the round-2 instruments (444 vs 127), thirteen core principles differ, ten of them lower in robot instruments (justice 16% vs 56%, beneficence 41% vs 70%, trust 19% vs 49%, solidarity, inclusive growth, sustainability, transparency 42% vs 58%, privacy, freedom, dignity) and three higher (non-maleficence, human oversight, robustness). The "ethical core deficit" of robot rules therefore exists in the database-harvested sector law of round 2 and not in the collector-found instruments of round 1; it is a composition effect, and the within-round-1 result reproduces the v2 reading that the core does not differ once genre is held constant. Across all documents the picture is mixed in round 1 (twelve core principles differ, four of them higher in embodied documents) and uniform in round 2 (thirteen differ, all but two lower), so the pooled all-document contrasts of Table 8 should be read with Table 20b beside them.

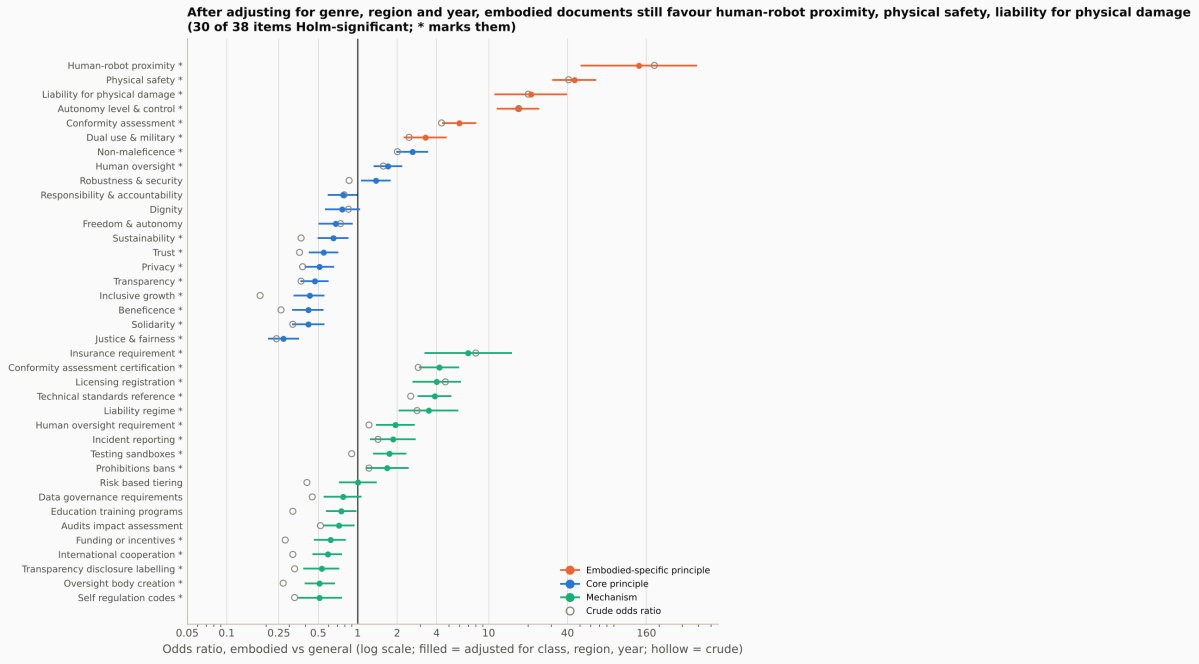


Figure 38. Crude and covariate-adjusted odds ratios (embodied vs general) for principles and mechanisms, adjusting for document class, region and year.

Table 20. Embodied-AI set vs general-AI set: crude and covariate-adjusted odds ratios (logistic regression adjusting for document class, jurisdiction region, collection round and publication year; usable documents with a known year)

item	group	n	events	events_embodied	events_general	sparse_cell	crude_or	adjusted_or	ci_low	ci_high	p	p_holm	significant_holm_05
principle: Transparency	core	2526	1154	564	590	0	0.37	0.47	0.37	0.59	0	0	True

item	group	n	events	events_embodied	events_general	sparse_cell	crude_or	adjusted_or	ci_low	ci_high	p	p_holm	significant_holm_05
principle: Justice & fairness	core	2526	991	409	582	0	0.24	0.27	0.21	0.35	0	0	True
principle: Non-maleficence	core	2526	1916	1254	662	0	2.01	2.61	2.01	3.41	0	0	True
principle: Responsibility & accountability	core	2526	1544	915	629	0	0.79	0.77	0.6	0.98	0.0336	0.1345	False
principle: Privacy	core	2526	929	433	496	0	0.38	0.51	0.4	0.65	0	0	True
principle: Beneficence	core	2526	1527	760	767	0	0.26	0.42	0.32	0.54	0	0	True
principle: Freedom & autonomy	core	2526	568	316	252	0	0.74	0.68	0.51	0.9	0.008	0.0644	False
principle: Trust	core	2526	975	455	520	0	0.36	0.55	0.43	0.7	0	0	True
principle: Sustainability	core	2526	637	276	361	0	0.37	0.65	0.5	0.84	0.0009	0.01	True
principle: Dignity	core	2526	500	291	209	0	0.85	0.76	0.57	1.03	0.0765	0.2295	False
principle: Solidarity	core	2526	598	241	357	0	0.32	0.42	0.32	0.55	0	0	True
principle: Human oversight	core	2526	989	670	319	0	1.57	1.7	1.34	2.15	0	0.0002	True
principle: Robustness & security	core	2526	1298	774	524	0	0.86	1.38	1.08	1.76	0.0096	0.067	False
principle: Inclusive growth	core	2526	1308	567	741	0	0.18	0.43	0.33	0.55	0	0	True
principle: Physical safety	embodied-specific	2526	1082	1036	46	0	40.75	45.18	31.2	65.42	0	0	True
principle: Human-robot proximity	embodied-specific	2526	672	668	4	1	184.04	140.4	51.13	385.58	0	0	True
principle: Liability for physical damage	embodied-specific	2526	343	330	13	0	20.04	20.96	11.23	39.12	0	0	True
principle: Autonomy level & control	embodied-specific	2526	809	757	52	0	16.96	16.69	11.67	23.87	0	0	True
principle: Conformity assessment	embodied-specific	2526	722	599	123	0	4.37	5.94	4.47	7.89	0	0	True
principle: Dual use & military	embodied-specific	2526	323	252	71	0	2.47	3.28	2.28	4.72	0	0	True
mechanism: Audits impact assessment	mechanism	2526	718	nan	nan	nan	0.52	0.72	0.56	0.93	0.0106	0.067	False
mechanism: Conformity assessment certification	mechanism	2526	381	nan	nan	nan	2.9	4.19	2.98	5.88	0	0	True
mechanism: Data governance requirements	mechanism	2526	352	nan	nan	nan	0.45	0.77	0.56	1.05	0.1013	0.2295	False
mechanism: Education training programs	mechanism	2526	710	nan	nan	nan	0.32	0.75	0.58	0.96	0.0236	0.1181	False
mechanism: Funding or incentives	mechanism	2526	629	nan	nan	nan	0.28	0.62	0.47	0.8	0.0003	0.004	True
mechanism: Human oversight requirement	mechanism	2526	344	nan	nan	nan	1.22	1.94	1.4	2.68	0.0001	0.0009	True
mechanism: Incident reporting	mechanism	2526	237	nan	nan	nan	1.43	1.86	1.26	2.73	0.0017	0.0173	True

item	group	n	events	events_embodied	events_general	sparse_cell	crude_or	adjusted_or	ci_low	ci_high	p	p_holm	significant_holm_05
mechanism: Insurance requirement	mechanism	2526	128	nan	nan	nan	7.96	6.98	3.28	14.86	0	0	True
mechanism: International cooperation	mechanism	2526	749	nan	nan	nan	0.32	0.59	0.46	0.75	0	0.0004	True
mechanism: Liability regime	mechanism	2526	178	nan	nan	nan	2.84	3.47	2.08	5.78	0	0	True
mechanism: Licensing registration	mechanism	2526	387	nan	nan	nan	4.67	4.01	2.66	6.06	0	0	True
mechanism: Oversight body creation	mechanism	2526	687	nan	nan	nan	0.27	0.51	0.4	0.66	0	0	True
mechanism: Prohibitions bans	mechanism	2526	266	nan	nan	nan	1.22	1.68	1.17	2.42	0.0054	0.0484	True
mechanism: Risk based tiering	mechanism	2526	310	nan	nan	nan	0.41	1	0.73	1.38	0.984	0.984	False
mechanism: Self regulation codes	mechanism	2526	245	nan	nan	nan	0.33	0.51	0.35	0.75	0.0005	0.0057	True
mechanism: Technical standards reference	mechanism	2526	876	nan	nan	nan	2.54	3.86	2.92	5.09	0	0	True
mechanism: Testing sandboxes	mechanism	2526	518	nan	nan	nan	0.9	1.75	1.33	2.31	0.0001	0.0009	True
mechanism: Transparency disclosure labelling	mechanism	2526	424	nan	nan	nan	0.33	0.53	0.39	0.71	0	0.0003	True

Table 34. Core principles: Mantel-Haenszel odds ratios (embodied vs general) stratified by document class, with Holm correction

principle	strata	mh_odds_ratio	ci_low	ci_high	p	p_holm
Transparency	7	0.28	0.23	0.34	0	0
Justice & fairness	7	0.16	0.13	0.2	0	0
Non-maleficence	7	1.78	1.45	2.18	0	0
Responsibility & accountability	7	0.55	0.46	0.67	0	0
Privacy	7	0.26	0.22	0.32	0	0
Beneficence	7	0.26	0.21	0.31	0	0
Freedom & autonomy	7	0.4	0.31	0.5	0	0
Trust	7	0.29	0.24	0.36	0	0
Sustainability	7	0.43	0.35	0.52	0	0
Dignity	7	0.53	0.42	0.68	0	0
Solidarity	7	0.3	0.24	0.38	0	0
Human oversight	7	1.38	1.15	1.66	0.0003	0.0006
Robustness & security	7	0.94	0.78	1.14	0.5592	0.5592
Inclusive growth	7	0.29	0.24	0.35	0	0

Adjusting for document class, region, collection round and year (Table 20; 2,526 usable documents with a known year), the embodied-specific considerations keep large odds ratios: physical safety 45 (interval 31 to 65), liability for physical damage 21 (11 to 39), autonomy-level control 17 (12 to 24), conformity assessment 6.0 (4.5 to 7.9) and dual use 3.3 (2.3 to 4.7); human-robot proximity rests on 4 general documents and is flagged as a sparse cell, so its odds ratio is not estimable and is not quoted. Among the core principles, ten of fourteen differ after Holm correction: non-maleficence (2.6) and human oversight (1.7) are higher in embodied documents; transparency (0.47), justice and fairness (0.27), privacy (0.51), beneficence (0.42), trust (0.55), sustainability (0.65), solidarity (0.42) and inclusive growth (0.43) lower; accountability (0.77), freedom (0.68), dignity (0.76) and robustness (1.38) do not differ. Without round as a covariate (v3) thirteen differed. The Mantel-Haenszel analysis stratified by class (Table 34) agrees with the pooled within-genre tables but does not control for round, and is reported for completeness.

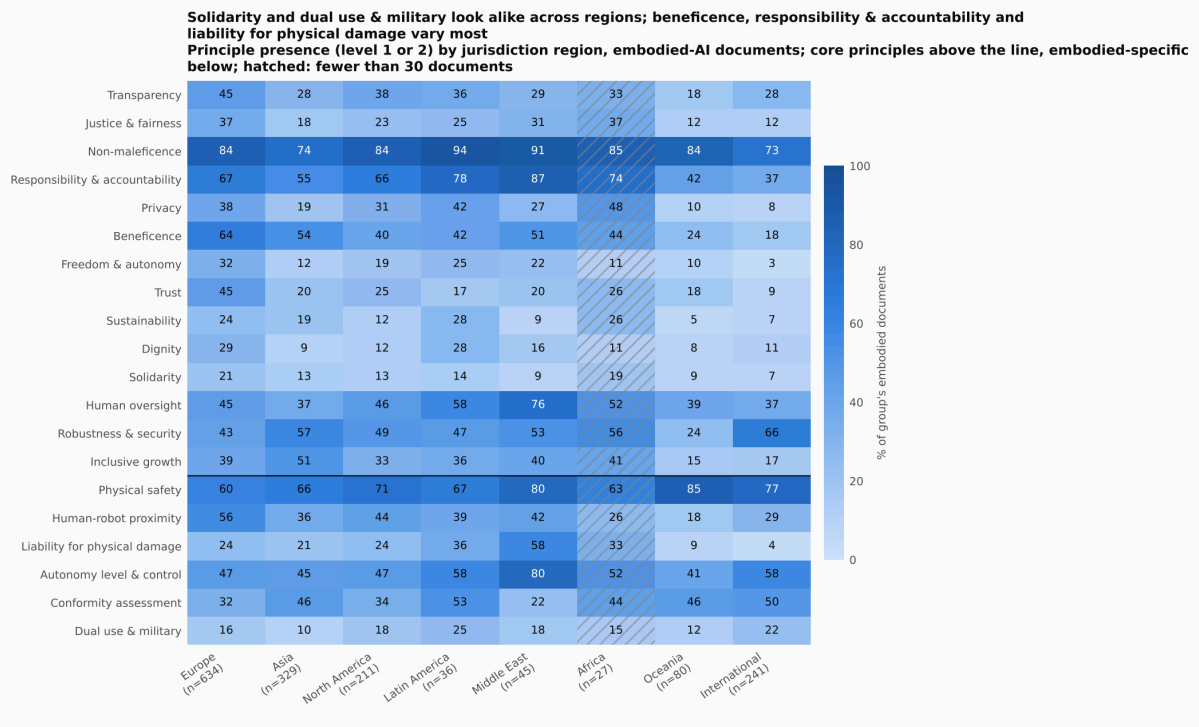


Figure 21. Principle presence by jurisdiction region among embodied-AI documents (core principles above the line, embodied-specific below).

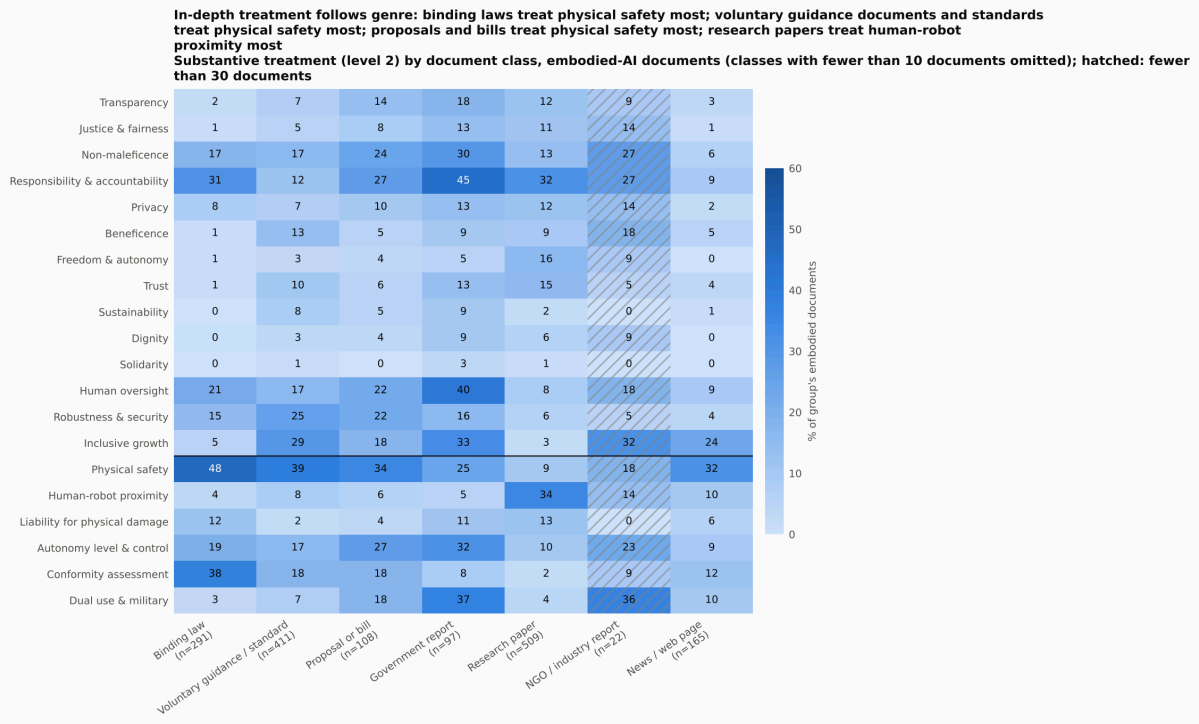


Figure 22. Substantive treatment (level 2) of principles by document class among embodied-AI documents.

By region (Figure 21), liability for physical damage runs from 4% (international layer) to 58% (Middle East), human oversight from 37% to 76% and autonomy-level control from 41% (Oceania) to 80% (Middle East); beneficence runs from 18% to 64% and trust from 9% to 45%, but the region profiles mix genres (Section 8). By class (Figure 22), in-depth treatment follows function: binding texts operationalise physical safety (48%) and conformity assessment (38%); drafts accountability and autonomy-level control (27% each); research proximity (34%) and accountability (32%).

7.2 Operationalisation of the embodied-specific considerations

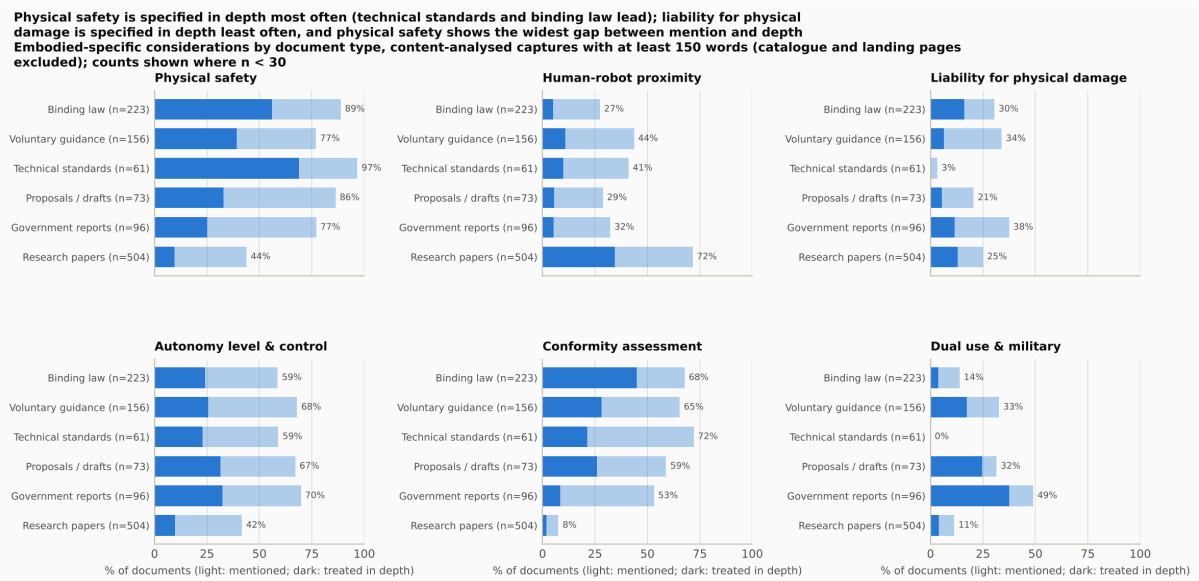


Figure 23. Embodied-specific considerations by document type, content-analysed captures of at least 150 words only (catalogue and landing pages excluded): share mentioning (light) and treating in depth (dark); counts shown where n is below 30.

Table 09. Embodied-specific principles by instrument type (embodied-AI documents, content-analysed captures only; catalogue and landing pages excluded)

group	n	principle	label	any_pct	substantive_pct
Binding instruments	195	physical_safety	Physical safety	89.2	61.5
Binding instruments	195	human_robot_proximity	Human-robot proximity	29.7	5.6
Binding instruments	195	liability_for_physical_damage	Liability for physical damage	33.3	17.9
Binding instruments	195	autonomy_level_control	Autonomy level & control	61	24.6
Binding instruments	195	conformity_assessment_certification	Conformity assessment	70.3	47.2
Binding instruments	195	dual_use_military	Dual use & military	15.9	4.1
Soft-law instruments	207	physical_safety	Physical safety	73.4	32.9
Soft-law instruments	207	human_robot_proximity	Human-robot proximity	37.7	9.7
Soft-law instruments	207	liability_for_physical_damage	Liability for physical damage	27.5	4.8
Soft-law instruments	207	autonomy_level_control	Autonomy level & control	57	20.3
Soft-law instruments	207	conformity_assessment_certification	Conformity assessment	60.9	24.6
Soft-law instruments	207	dual_use_military	Dual use & military	25.1	13.5
Technical standards	62	physical_safety	Physical safety	95.2	67.7
Technical standards	62	human_robot_proximity	Human-robot proximity	40.3	9.7
Technical standards	62	liability_for_physical_damage	Liability for physical damage	3.2	0
Technical standards	62	autonomy_level_control	Autonomy level & control	59.7	22.6
Technical standards	62	conformity_assessment_certification	Conformity assessment	71	21
Technical standards	62	dual_use_military	Dual use & military	0	0
Proposals / drafts	77	physical_safety	Physical safety	84.4	32.5
Proposals / drafts	77	human_robot_proximity	Human-robot proximity	27.3	5.2
Proposals / drafts	77	liability_for_physical_damage	Liability for physical damage	19.5	5.2
Proposals / drafts	77	autonomy_level_control	Autonomy level & control	66.2	31.2
Proposals / drafts	77	conformity_assessment_certification	Conformity assessment	57.1	24.7
Proposals / drafts	77	dual_use_military	Dual use & military	32.5	24.7
Research outputs	509	physical_safety	Physical safety	43.6	9.4
Research outputs	509	human_robot_proximity	Human-robot proximity	71.3	34.4
Research outputs	509	liability_for_physical_damage	Liability for physical damage	25	12.8
Research outputs	509	autonomy_level_control	Autonomy level & control	41.8	9.6
Research outputs	509	conformity_assessment_certification	Conformity assessment	7.5	1.8
Research outputs	509	dual_use_military	Dual use & military	11.8	4.3

Table 35. Governance documents (n=351): label prevalence under the v1 excerpt annotation and the v2 extended, metadata-blind annotation, with per-document agreement

label	v1_excerpt_pct	v2_extended_blind_pct	agreement
legally binding (binding_nature)	12	14	0.952
in force (instrument_status)	8.8	10.5	0.966
is governance document	99.7	98	0.977
any embodied category	14.5	17.9	0.926
relevance >= 2	9.7	9.7	0.983

label	v1_excerpt_pct	v2_extended_blind_pct	agreement
physical safety mentioned	14.5	16.2	0.937
physical safety in depth	4.3	4.3	1
autonomy-level control mentioned	12	17.9	0.906
autonomy-level control in depth	0.9	2	0.989
conformity assessment mechanism	11.4	15.1	0.923
risk-based tiering mechanism	19.7	25.1	0.895
technical standards referenced	35.6	43.3	0.872
liability regime mechanism	3.7	7.1	0.96
mechanisms per document (mean)	4.87	5.64	nan
topics per document (mean)	6.36	6.83	nan
robot types per document (mean)	0.43	0.68	nan
named instruments per document (mean)	1.25	1.78	nan

Table 9 and Figure 23 show which document types move the six considerations from mention to specification, computed on content-analysed captures only (Table 9 on the instrument classes of the embodied set with catalogue and landing pages excluded; Figure 23 on every capture of at least 150 words including government reports, so the two n differ). Principal binding instruments (n = 195) mention physical safety in 89% of cases and specify it in 62%, conformity assessment in 70% and 47%, autonomy-level control in 61% and 25%, liability for physical damage in 33% and 18%, proximity in 30% and 6%. Soft-law instruments (n = 207) mention physical safety in 73% and specify it in 33%, autonomy-level control in 57% and 21%. Drafts (n = 77) mention physical safety in 84% and specify it in 33%, autonomy-level control in 66% and 31%, dual use in 33% and 25% (the autonomous-weapons drafts). The 62 technical standards with a content-analysed scope page name physical safety in 95% and treat it in depth in 68%, autonomy-level control in 60% and 23%, liability in 3% (never in depth) and dual use never; these shares describe scope statements of 62 standards, not normative text, and the remaining 192 of the 254 standards in the embodied governance documents are catalogue pages that are excluded from every depth statistic. Autonomy-level control is therefore the consideration with the widest gap between mention and specification in binding law (61 points mentioned, 25 specified), and the v2 observation that nothing yet specified it no longer holds. Table 35 (unchanged since v2) records what the extended, blind re-annotation of governance documents changed: mechanisms per document rose from 4.9 to 5.6 and robot types from 0.4 to 0.7, while binding and in-force labels changed for fewer than 5% of documents.

7.3 Tools: governance mechanisms

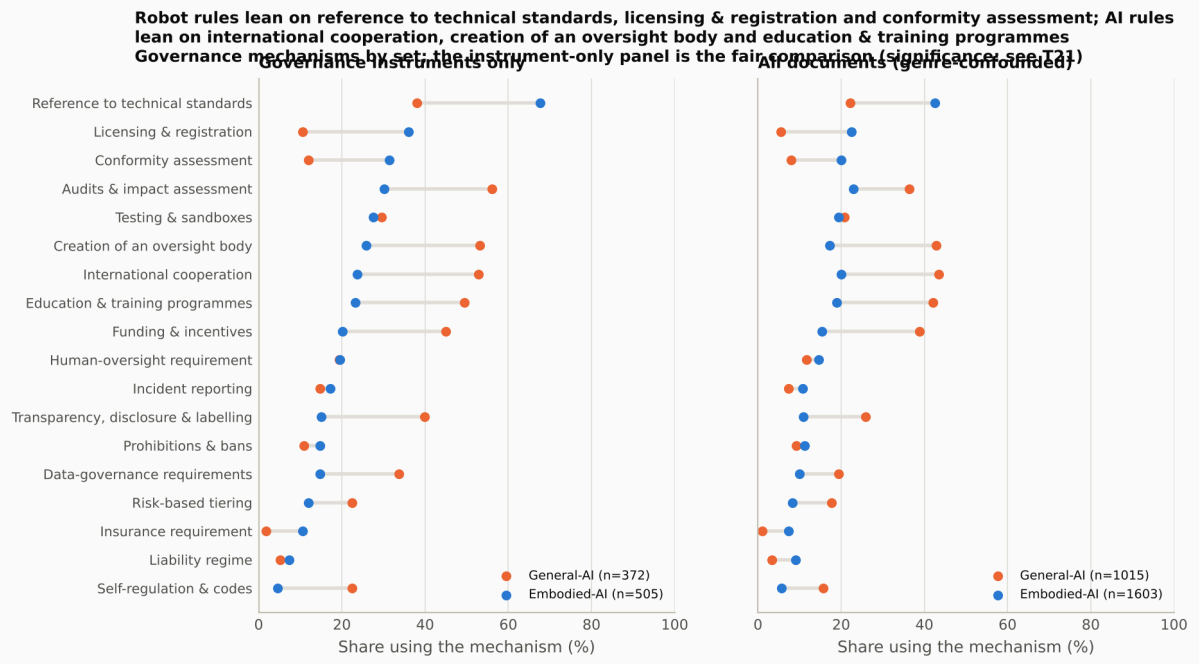


Figure 24. Governance mechanisms by set: principal governance instruments only (left, the fair comparison) and all documents (right, genre-confounded).

Table 10. Governance mechanisms by set (all usable documents; instrument-only shares for governance documents, embodied n=822, general n=409)

mechanism	embodied_pct	general_pct	embodied_instruments_pct	general_instruments_pct	test	p	p_holm
technical_standards_reference	42.6	22.2	67.6	38.1	chi-square	0	0
audits_impact_assessment	23	36.4	30.2	56	chi-square	0	0
licensing_registration	22.4	5.5	36	10.5	chi-square	0	0
conformity_assessment_certification	20	8	31.4	12	chi-square	0	0
international_cooperation	20	43.5	23.7	52.8	chi-square	0	0

mechanism	embodied_pct	general_pct	embodied_instruments_pct	general_instruments_pct	test	p	p_holm
testing_sandboxes	19.4	20.8	27.5	29.6	chi-square	0.4149	0.4149
education_training_programs	19	42	23.2	49.4	chi-square	0	0
oversight_body_creation	17.3	42.8	25.8	53.1	chi-square	0	0
funding_or_incentives	15.4	38.8	20.2	45	chi-square	0	0
human_oversight_requirement	14.6	11.7	19.5	19.3	chi-square	0.0415	0.1245
prohibitions_bans	11.2	9.2	14.8	10.8	chi-square	0.1052	0.2105
transparency_disclosure_labelling	10.9	25.8	15	39.9	chi-square	0	0
incident_reporting	10.7	7.3	17.2	14.7	chi-square	0.0041	0.0164
data_governance_requirements	9.9	19.4	14.7	33.7	chi-square	0	0
liability_regime	9.1	3.3	7.4	5.1	chi-square	0	0
risk_based_tiering	8.3	17.7	11.9	22.5	chi-square	0	0
insurance_requirement	7.4	1	10.6	1.7	chi-square	0	0
self_regulation_codes	5.7	15.7	4.6	22.5	chi-square	0	0

Among principal instruments (505 vs 372; Table 21), robot instruments refer to technical standards (65% vs 35%) and use licensing and registration (44% vs 10%), conformity assessment (42% vs 12%), insurance requirements (15% vs 2%), incident reporting (24% vs 16%), prohibitions (21% vs 12%) and liability regimes (11% vs 5%) significantly more often; general-AI instruments rely on audits (58% vs 43%), international cooperation (56% vs 34%), oversight bodies (56% vs 38%), education (53% vs 35%), funding (46% vs 30%), labelling (42% vs 20%), data governance (36% vs 20%) and self-regulation (24% vs 7%); human-oversight requirements, risk tiering (18% vs 23%) and testing and sandboxes (36% vs 30%) do not differ. The adjusted models (Table 20) confirm insurance (odds ratio 7.0), conformity assessment (4.2), licensing and registration (4.1), references to standards (3.9), liability regimes (3.5), human-oversight requirements (1.9), incident reporting (1.9), testing and sandboxes (1.7) and prohibitions (1.7) as the mechanisms associated with embodied documents after genre, region, round and year, and oversight bodies (0.51), self-regulation (0.51), labelling (0.53), international cooperation (0.59) and funding (0.62) as those of general-AI documents; audits (0.72), risk tiering (1.00), data governance and education do not differ after correction. No within-round table exists for mechanisms, so these contrasts carry the same caution as the pooled principle contrasts: within the instrument genre the embodied side is 88% round 2.

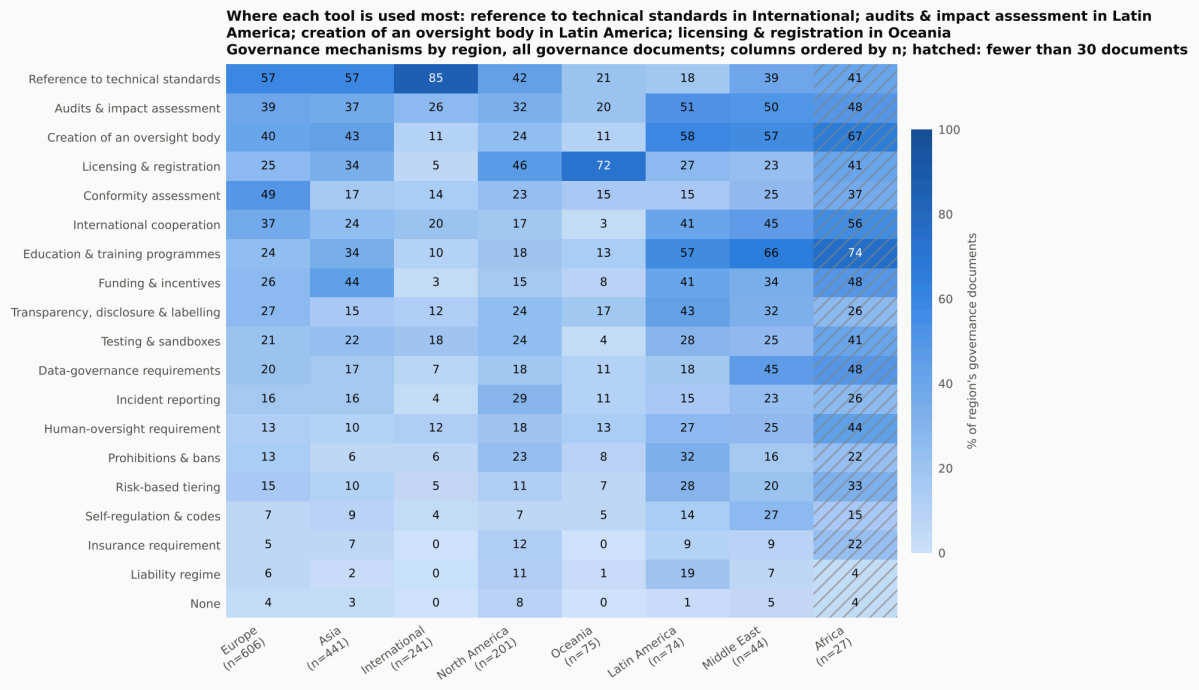


Figure 25. Governance mechanisms by jurisdiction region (all governance documents).

Table 29. Regional divergence of principles and mechanisms on the same population (embodied documents in six regions): bias-corrected Cramer's V with bootstrap 95% CIs, Monte-Carlo chi-square where cells are sparse, Holm correction

item	population	n	v_bias_corrected	ci_low	ci_high	test	p	p_holm
mechanism: Reference to technical standards	embodied documents, 6 regions	1576	0.353	0.31	0.398	chi-square	0	0

item	population	n	v_bias_corrected	ci_low	ci_high	test	p	p_holm
mechanism: Licensing & registration	embodied documents, 6 regions	1576	0.34	0.298	0.399	chi-square	0	0
principle: Beneficence	embodied documents, 6 regions	1576	0.333	0.297	0.387	chi-square	0	0
principle: Trust	embodied documents, 6 regions	1576	0.298	0.255	0.353	chi-square	0	0
mechanism: Liability regime	embodied documents, 6 regions	1576	0.277	0.213	0.354	chi-square	0	0
principle: Freedom & autonomy	embodied documents, 6 regions	1576	0.268	0.227	0.315	chi-square	0	0
principle: Privacy	embodied documents, 6 regions	1576	0.262	0.219	0.306	chi-square	0	0
principle: Responsibility & accountability	embodied documents, 6 regions	1576	0.245	0.202	0.3	chi-square	0	0
principle: Liability for physical damage	embodied documents, 6 regions	1576	0.241	0.21	0.285	chi-square	0	0
principle: Inclusive growth	embodied documents, 6 regions	1576	0.232	0.196	0.288	chi-square	0	0
principle: Human-robot proximity	embodied documents, 6 regions	1576	0.229	0.188	0.285	chi-square	0	0
principle: Dignity	embodied documents, 6 regions	1576	0.224	0.186	0.28	chi-square	0	0
principle: Justice & fairness	embodied documents, 6 regions	1576	0.223	0.185	0.277	chi-square	0	0
mechanism: Funding & incentives	embodied documents, 6 regions	1576	0.216	0.177	0.262	chi-square	0	0
principle: Robustness & security	embodied documents, 6 regions	1576	0.199	0.156	0.257	chi-square	0	0
mechanism: Prohibitions & bans	embodied documents, 6 regions	1576	0.184	0.129	0.26	chi-square	0	0
mechanism: Insurance requirement	embodied documents, 6 regions	1576	0.176	0.143	0.237	chi-square	0	0
principle: Sustainability	embodied documents, 6 regions	1576	0.176	0.145	0.229	chi-square	0	0
principle: Transparency	embodied documents, 6 regions	1576	0.173	0.132	0.236	chi-square	0	0
mechanism: Incident reporting	embodied documents, 6 regions	1576	0.171	0.126	0.237	chi-square	0	0
mechanism: Risk-based tiering	embodied documents, 6 regions	1576	0.164	0.102	0.241	chi-square	0	0
mechanism: Creation of an oversight body	embodied documents, 6 regions	1576	0.159	0.118	0.221	chi-square	0	0
principle: Conformity assessment	embodied documents, 6 regions	1576	0.151	0.111	0.207	chi-square	0	0
principle: Physical safety	embodied documents, 6 regions	1576	0.149	0.11	0.202	chi-square	0	0
principle: Solidarity	embodied documents, 6 regions	1576	0.137	0.091	0.188	chi-square	0	0
principle: Non-maleficence	embodied documents, 6 regions	1576	0.134	0.089	0.196	chi-square	0	0.0001
principle: Human oversight	embodied documents, 6 regions	1576	0.131	0.09	0.189	chi-square	0	0.0001
principle: Autonomy level & control	embodied documents, 6 regions	1576	0.128	0.091	0.187	chi-square	0	0.0002
mechanism: Education & training programmes	embodied documents, 6 regions	1576	0.124	0.083	0.183	chi-square	0	0.0004
mechanism: International cooperation	embodied documents, 6 regions	1576	0.107	0.071	0.167	chi-square	0.0005	0.0044
mechanism: Human-oversight requirement	embodied documents, 6 regions	1576	0.103	0.045	0.178	chi-square	0.001	0.0077
mechanism: Data-governance requirements	embodied documents, 6 regions	1576	0.097	0.06	0.146	chi-square	0.0021	0.0144
principle: Dual use & military	embodied documents, 6 regions	1576	0.092	0.046	0.155	chi-square	0.0038	0.0226
mechanism: Testing & sandboxes	embodied documents, 6 regions	1576	0.087	0.051	0.154	chi-square	0.0062	0.0311
mechanism: Conformity assessment	embodied documents, 6 regions	1576	0.07	0.019	0.138	chi-square	0.034	0.1359
mechanism: Audits & impact assessment	embodied documents, 6 regions	1576	0.059	0	0.137	chi-square	0.0746	0.2237
mechanism: Transparency, disclosure & labelling	embodied documents, 6 regions	1576	0.054	0	0.147	chi-square	0.1019	0.2237
mechanism: Self-regulation & codes	embodied documents, 6 regions	1576	0.028	0	0.108	chi-square (Monte Carlo)	0.8803	0.8803

Table 15. Regional divergence: association between region and principle presence (embodied-AI documents) or mechanism presence (governance documents); Cramer's V with Holm-corrected chi-square p

item	label	cramers_v	p	min_region_pct	max_region_pct	p_holm
principle:beneficence	Beneficence	0.339	0	17.8	63.9	0
mechanism:technical_standards_reference	Technical standards reference	0.333	0	17.6	85.5	0
mechanism:conformity_assessment_certification	Conformity assessment certification	0.333	0	14.1	48.5	0
mechanism:funding_or_incentives	Funding or incentives	0.323	0	2.9	48.1	0
mechanism:licensing_registration	Licensing registration	0.315	0	5.4	72	0
mechanism:education_training_programs	Education training programs	0.307	0	10	74.1	0
principle:trust	Trust	0.304	0	9.1	44.6	0
mechanism:oversight_body_creation	Oversight body creation	0.296	0	10.7	66.7	0
principle:freedom_autonomy	Freedom & autonomy	0.275	0	3.3	32	0
principle:privacy	Privacy	0.269	0	7.9	41.7	0
principle:responsibility_accountability	Responsibility & accountability	0.253	0	36.5	86.7	0
principle:liability_for_physical_damage	Liability for physical damage	0.248	0	3.7	57.8	0
principle:inclusive_growth	Inclusive growth	0.24	0	15	51.1	0
principle:human_robot_proximity	Human-robot proximity	0.237	0	17.5	55.5	0
mechanism:international_cooperation	International cooperation	0.233	0	2.7	55.6	0
principle:dignity	Dignity	0.232	0	7.5	28.7	0
principle:justice_fairness	Justice & fairness	0.231	0	12.4	37.1	0
mechanism:prohibitions_bans	Prohibitions bans	0.214	0	5.9	32.4	0
principle:robustness_security	Robustness & security	0.208	0	23.8	66.4	0
mechanism:data_governance_requirements	Data governance requirements	0.195	0	7.1	48.1	0
mechanism:liability_regime	Liability regime	0.193	0.0002	0.4	18.9	0.001
mechanism:transparency_disclosure_labelling	Transparency disclosure labelling	0.19	0	11.6	43.2	0
principle:sustainability	Sustainability	0.186	0	5	27.8	0
principle:transparency	Transparency	0.184	0	17.5	45.4	0
mechanism:incident_reporting	Incident reporting	0.181	0	4.1	28.9	0
mechanism:risk_based_tiering	Risk based tiering	0.177	0	4.6	33.3	0
mechanism:insurance_requirement	Insurance requirement	0.17	0.0002	0	22.2	0.001
mechanism:human_oversight_requirement	Human oversight requirement	0.166	0	10	44.4	0
principle:conformity_assessment_certification	Conformity assessment	0.163	0	22.2	52.8	0
principle:physical_safety	Physical safety	0.161	0	60.3	85	0
principle:solidarity	Solidarity	0.15	0	7.5	21.5	0
principle:non_maleficence	Non-maleficence	0.147	0	72.6	94.4	0.0001
principle:human_oversight	Human oversight	0.145	0	36.9	75.6	0.0001
mechanism:audits_impact_assessment	Audits impact assessment	0.143	0	20	51.4	0.0001
mechanism:self_regulation_codes	Self regulation codes	0.142	0	4.1	27.3	0.0001
principle:autonomy_level_control	Autonomy level & control	0.142	0	41.2	80	0.0001
mechanism:testing_sandboxes	Testing sandboxes	0.12	0.001	4	40.7	0.0019
principle:dual_use_military	Dual use & military	0.111	0.0038	9.7	25	0.0038

On a common population (1,576 embodied documents in six regions; Table 29), 34 of 38 items diverge by region after Holm correction (bias-corrected V 0.09 to 0.35): references to standards (0.35), licensing and registration (0.34), beneficence (0.33), trust (0.30), liability regimes (0.28), freedom, privacy, accountability and liability for physical damage (0.24 to 0.27) lead; the conformity-assessment mechanism (0.07), audits (0.06), labelling (0.05) and self-regulation (0.03) do not differ detectably. The common population holds genre constant across regions in name only: the international layer is standards and has no research outputs, the Middle East is 69% research, Oceania is drone regulation. Table 29b repeats the statistics within one genre at a time, keeping the regions with at least 20 documents of that genre. Within research outputs (484 papers from Europe, Asia, North America and the Middle East), 2 of 14 core principles differ by region after Holm correction (human oversight, V 0.23, from 28% in North America to 77% in the Middle East; inclusive growth, 0.17), 4 of 6 embodied-specific considerations do (liability for physical damage 0.28, human-robot proximity 0.25, autonomy-level control 0.20, physical safety 0.17) and 1 of 11 mechanisms (liability regimes, 0.38); beneficence and transparency have V 0.00, justice 0.03, privacy 0.09 and trust 0.11. Within principal instruments (476 from Europe, Asia, North America, Latin America and the international layer), 12 of 14 core principles differ (beneficence 0.43, sustainability 0.34, inclusive growth 0.34, trust 0.28, privacy 0.24) and 14 of 18 mechanisms (funding 0.39, standards references 0.26, prohibitions 0.25), while physical safety, human-robot proximity, autonomy-level control, conformity assessment and dual use do not. The pattern reverses between genres: research papers share an ethical vocabulary and differ on the body, instruments share the body and differ on values and tools. The within-instrument contrast is itself a mix of instrument types, with the standards-heavy international layer at the low end of every value item (beneficence 23%, trust 9%) and Asia's strategies and guidelines at the high end (beneficence 75%, inclusive growth 74%), so the v3 statement that regions differ on values is retained only in this qualified form: among principal instruments, and not separable from the kinds of instrument each region's databases supply. The research-only distance matrix (Table 31) agrees: Europe and North America are 0.01 apart on research outputs.

Table 29b. Regional divergence within one genre (research outputs, n=484; regions with at least 20 documents of that genre: Europe, Asia, North America, Middle East): bias-corrected Cramer's V, Holm correction

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
research outputs	principle:transparency	Transparency	484	0	0.82	37.5	44	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:justice_fairness	Justice & fairness	484	0.026	0.3428	32.1	45.3	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:non_maleficence	Non-maleficence	484	0.081	0.1017	73.4	93.5	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:responsibility_accountability	Responsibility & accountability	484	0.146	0.0041	59.4	93.5	0.0976	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:privacy	Privacy	484	0.092	0.0691	19.4	37.7	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:beneficence	Beneficence	484	0	0.6855	58.1	70.3	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:freedom_autonomy	Freedom & autonomy	484	0.112	0.0287	29	46.6	0.5749	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:trust	Trust	484	0.11	0.0315	22.6	47.9	0.5984	False	Europe (305), Asia (84), North America (64), Middle East (31)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
research outputs	principle:sustainability	Sustainability	484	0.055	0.2175	3.2	17.2	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:dignity	Dignity	484	0.13	0.0111	22.6	42	0.2543	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:solidarity	Solidarity	484	0.055	0.2143	9.7	23.4	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:human_oversight	Human oversight	484	0.232	0	28.1	77.4	0.0001	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:robustness_security	Robustness & security	484	0.126	0.0137	27.2	44	0.3011	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:inclusive_growth	Inclusive growth	484	0.166	0.0009	17.2	36.9	0.0237	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:physical_safety	Physical safety	484	0.17	0.0007	39	74.2	0.0182	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:human_robot_proximity	Human-robot proximity	484	0.254	0	48.4	79	0	True	Europe (305), Asia (84), North America (64), Middle East (31)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
research outputs	principle:liability_for_physical_damage	Liability for physical damage	484	0.277	0	19.3	67.7	0	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:autonomy_level_control	Autonomy level & control	484	0.202	0	29.7	77.4	0.0013	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:conformity_assessment_certification	Conformity assessment	484	0.103	0.3094	0	13.1	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	principle:dual_use_military	Dual use & military	484	0	0.9762	9.7	12.5	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:audits_impact_assessment	Audits impact assessment	484	0.092	0.0695	8.9	19	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:conformity_assessment_certification	Conformity assessment certification	484	0.067	0.6301	0	6	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:data_governance_requirements	Data governance requirements	484	0	0.9385	0	4.8	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:education_training_programs	Education training programs	484	0.06	0.1921	7.2	14.1	1	False	Europe (305), Asia (84), North America (64), Middle East (31)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
research outputs	mechanism:human_oversight_requirement	Human oversight requirement	484	0.165	0.027	2.6	13.1	0.5669	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:international_cooperation	International cooperation	484	0.06	0.6778	3.1	9.7	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:liability_regime	Liability regime	484	0.381	0	4.7	58.1	0	True	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:self_regulation_codes	Self regulation codes	484	0	0.9555	3.2	9.5	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:technical_standards_reference	Technical standards reference	484	0.082	0.0985	0	14.3	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:testing_sandboxes	Testing sandboxes	484	0.113	0.2384	0	9.5	1	False	Europe (305), Asia (84), North America (64), Middle East (31)
research outputs	mechanism:transparency_disclosure_labelling	Transparency disclosure labelling	484	0.108	0.2674	3.1	10.7	1	False	Europe (305), Asia (84), North America (64), Middle East (31)

Table 29b. Regional divergence within one genre (principal instruments, n=476; regions with at least 20 documents of that genre: Europe, Asia, North America, Latin America & Caribbean, International): bias-corrected Cramer's V, Holm correction

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	principle:transparency	Transparency	476	0.176	0.0009	37.8	60.6	0.0154	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), International (64)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	principle:justice_fairness	Justice & fairness	476	0.2	0.0001	9.8	34.1	0.0025	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:non_maleficence	Non-maleficence	476	0.118	0.0315	81.5	93.5	0.2833	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:responsibility_accountability	Responsibility & accountability	476	0.215	0	51.6	95.2	0.0008	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:privacy	Privacy	476	0.241	0	15.6	57.1	0.0001	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:beneficence	Beneficence	476	0.425	0	19.6	74.8	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:freedom_autonomy	Freedom & autonomy	476	0.208	0.0001	4.7	24.7	0.0015	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:trust	Trust	476	0.284	0	9.4	43.5	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:sustainability	Sustainability	476	0.335	0	5.9	42	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	principle:dignity	Dignity	476	0.184	0.0005	3.9	19.4	0.0093	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:solidarity	Solidarity	476	0.169	0.0015	5.9	24.1	0.0218	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:human_oversight	Human oversight	476	0.169	0.0015	42	76.2	0.0218	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:robustness_security	Robustness & security	476	0.091	0.0943	63.7	82.8	0.3133	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:inclusive_growth	Inclusive growth	476	0.335	0	23.4	73.9	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:physical_safety	Physical safety	476	0.141	0.0091	73.9	93.8	0.0912	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:human_robot_proximity	Human-robot proximity	476	0.024	0.3683	26.5	47.6	0.7043	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:liability_for_physical_damage	Liability for physical damage	476	0.214	0	4.7	47.6	0.0008	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	principle:autonomy_level_control	Autonomy level & control	476	0.114	0.0377	52.1	71.9	0.3017	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:conformity_assessment_certification	Conformity assessment	476	0.143	0.0083	51	72.4	0.0908	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	principle:dual_use_military	Dual use & military	476	0.108	0.0486	11.8	33.3	0.3133	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:audits_impact_assessment	Audits impact assessment	476	0.207	0.0001	27.5	61.9	0.0015	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:conformity_assessment_certification	Conformity assessment certification	476	0.158	0.0032	29.4	51.8	0.0379	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:data_governance_requirements	Data governance requirements	476	0.162	0.0025	9.4	30	0.0324	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:education_training_programs	Education training programs	476	0.228	0	15.6	47.6	0.0002	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:funding_or_incentives	Funding or incentives	476	0.389	0	0	56.3	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	mechanism:human_oversight_requirement	Human oversight requirement	476	0.109	0.0476	21.8	52.4	0.3133	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:incident_reporting	Incident reporting	476	0.17	0.0014	6.2	38.1	0.0216	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:insurance_requirement	Insurance requirement	476	0.18	0.0006	0	33.3	0.0117	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:international_cooperation	International cooperation	476	0.203	0.0001	16.7	45.3	0.002	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:liability_regime	Liability regime	476	0.239	0	1.6	33.3	0.0001	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:licensing_registration	Licensing registration	476	0.234	0	18.8	66.7	0.0001	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:oversight_body_creation	Oversight body creation	476	0.24	0	15.6	52.4	0.0001	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:prohibitions_bans	Prohibitions bans	476	0.246	0	9.2	52.4	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)

population	item	label	n	cramers_v	p	min_region_pct	max_region_pct	p_holm	significant_holm_05	regions
principal instruments	mechanism:risk_based_tiering	Risk based tiering	476	0.234	0	6.2	47.6	0.0001	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:self_regulation_codes	Self regulation codes	476	0.029	0.3521	3.1	10.1	0.7043	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:technical_standards_reference	Technical standards reference	476	0.258	0	33.3	76.6	0	True	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:testing_sandboxes	Testing sandboxes	476	0.102	0.0621	23.8	43.7	0.3133	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)
principal instruments	mechanism:transparency_disclosure_labelling	Transparency disclosure labelling	476	0.11	0.0448	13.4	28.6	0.3133	False	Europe (170), Asia (119), North America (102), Latin America & Caribbean (21), Internationa (64)

Table 24. K-means on mechanism vectors of governance documents (n=1652): silhouette and bootstrap stability by k (exploratory)

k	silhouette	mean_adjusted_rand_vs_bootstrap	smallest_cluster
2	0.201	0.883	669
3	0.204	0.87	369
4	0.157	0.912	293
5	0.164	0.855	193
6	0.167	0.634	176

K-means bundles of mechanisms (Table 24) have silhouettes of 0.20 or less at every k and are not interpreted.

7.4 Framing, stance and the personhood debate

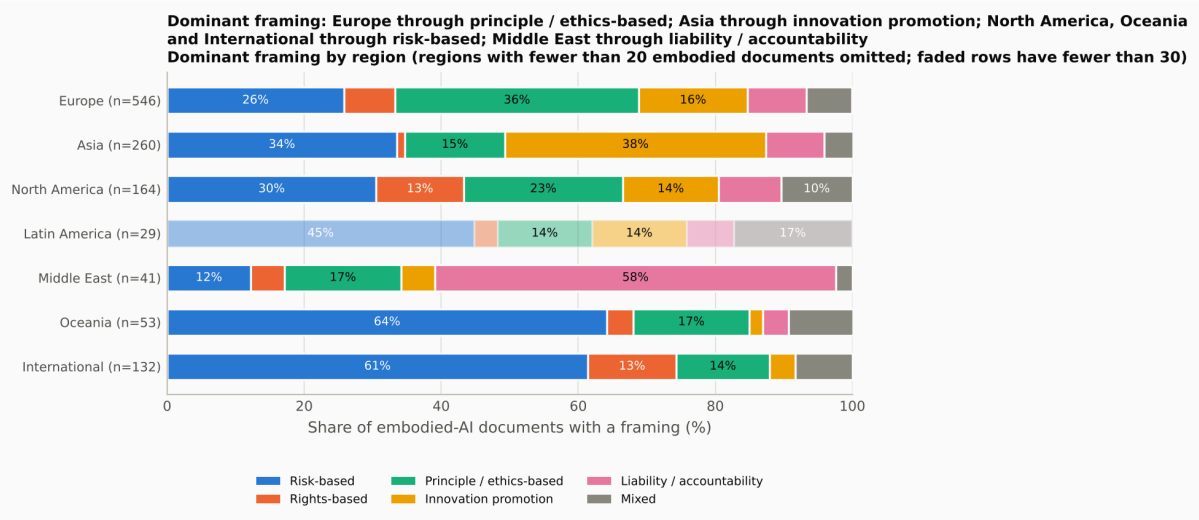


Figure 26. Dominant governance framing by region (embodied-AI documents with a framing).

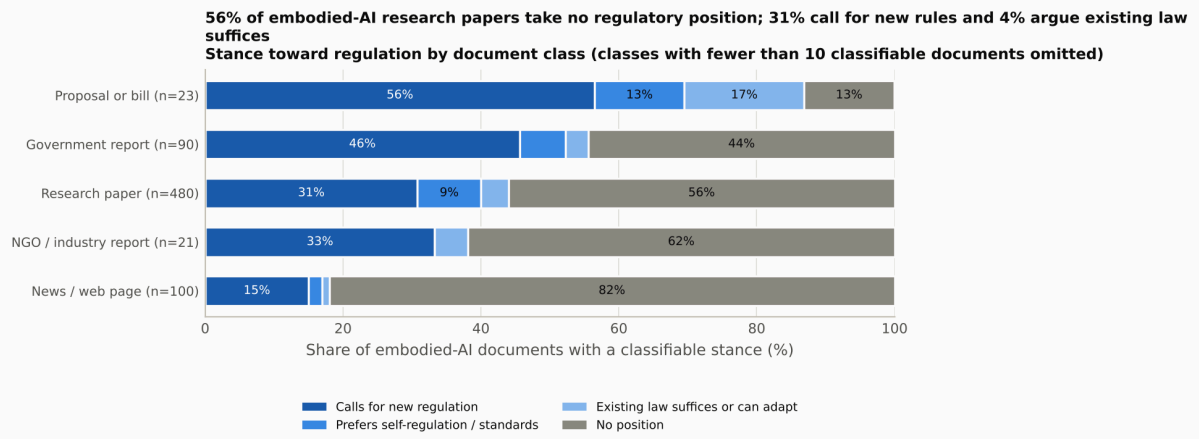


Figure 27. Stance toward regulation by document class (embodied-AI documents).

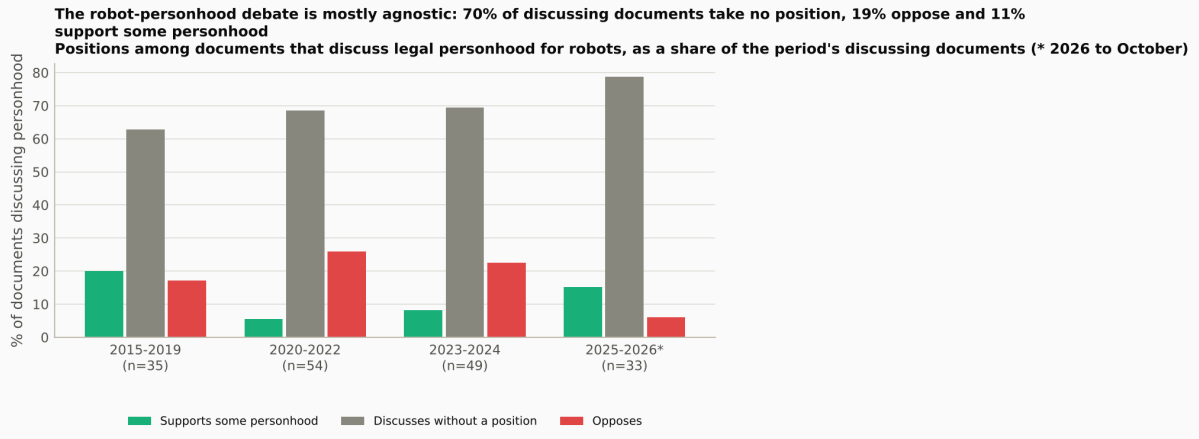


Figure 28. Positions on robot legal personhood over time (documents that discuss the question).

Table 12. Robot legal personhood: positions by jurisdiction region (documents discussing it, n=174)

study_region	discusses_neutral	opposes	supports
Africa	4	1	1
Asia	23	2	3
Europe	57	23	6
International	3	3	0
Latin America & Caribbean	1	1	0
Middle East	13	2	4
North America	18	1	4
Oceania	2	1	1

Framing follows the document mix each region contributes. Principle-or-ethics framing leads in Europe (36% of 546 framed embodied documents, risk-based 26%); North America is risk-based (31%) before ethics (23%; n = 164); Asia is innovation-framed (38%) and risk-based (34%) with ethics at 15% (n = 260); Oceania (64% of 53), the international layer (61% of 132) and Latin America (45% of 29) are risk-based; the Middle East is liability-framed (59% of 41, a legal-doctrinal literature on civil liability for smart robots); the association is moderate (Cramér's V 0.24). Among the 480 embodied research papers that take an applicable stance, 56% take no regulatory position, 31% call for new regulation, 9% prefer self-regulation or standards and 4% argue existing law suffices. Legal personhood is discussed in 174 documents (19 supportive, 34 opposed, 121 without a position); its share of embodied documents falls from 10% in 2020 to 2024 to 5% in 2025 to 2026 (adjusted odds ratio 0.39).

7.5 Which instruments are cited

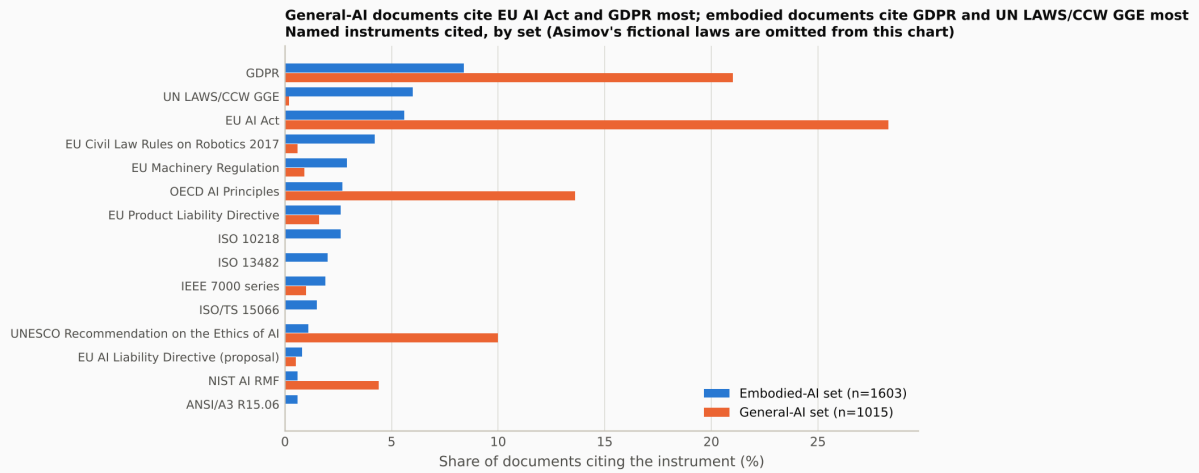


Figure 29. Most cited named governance instruments, embodied-AI set vs general-AI set.

Table 13. Named instruments cited, by set (canonicalised from the annotated reference lists)

instrument	embodied_docs	embodied_pct	general_docs	general_pct
GDPR	134	8.4	213	21
UN LAWS/CCW GGE	96	6	2	0.2
EU AI Act	89	5.6	287	28.3
EU Civil Law Rules on Robotics 2017	68	4.2	6	0.6
EU Machinery Regulation	46	2.9	9	0.9
OECD AI Principles	43	2.7	138	13.6
EU Product Liability Directive	42	2.6	16	1.6
ISO 10218	42	2.6	0	0
Asimov's Laws of Robotics	40	2.5	2	0.2
ISO 13482	32	2	0	0
IEEE 7000 series	30	1.9	10	1
ISO/TS 15066	24	1.5	0	0
UNESCO Recommendation on the Ethics of AI	17	1.1	101	10
EU AI Liability Directive (proposal)	13	0.8	5	0.5
NIST AI RMF	9	0.6	45	4.4
ANSI/A3 R15.06	9	0.6	0	0
ISO/IEC 42001	6	0.4	22	2.2
ISO/IEC 22989	6	0.4	14	1.4
US Executive Order 14110	5	0.3	25	2.5
Korea Intelligent Robots Act	5	0.3	0	0
Council of Europe Framework Convention on AI	5	0.3	29	2.9
Hiroshima Process	4	0.2	26	2.6
Korea AI Basic Act	3	0.2	9	0.9
ISO/IEC 23894	2	0.1	10	1
Japan AI Promotion Act	2	0.1	5	0.5
China Generative AI Measures	2	0.1	6	0.6
Bletchley Declaration	1	0.1	14	1.4
Colorado AI Act	0	0	0	0
Bill of Rights (US AI Blueprint)	0	0	0	0

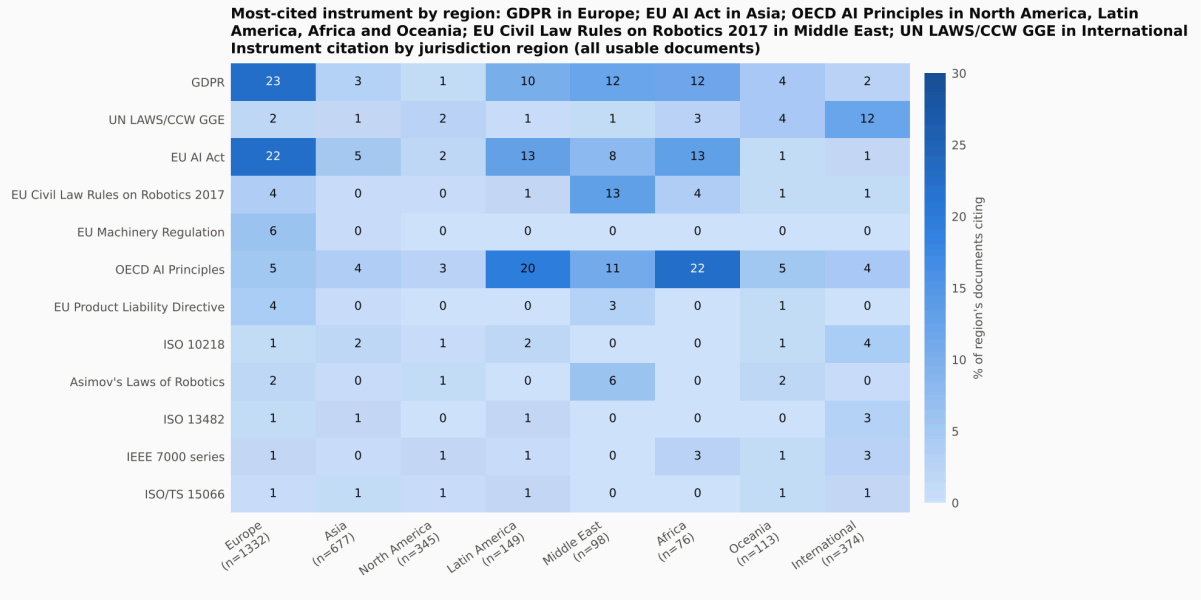


Figure 30. Instrument citation by jurisdiction region (share of all usable documents in the region).

General-AI documents cite the EU AI Act (28%), the GDPR (21%), the OECD principles (14%) and the UNESCO Recommendation (10%). Embodied documents cite the GDPR (8%), the UN CCW process (6%) and the AI Act (6%) and then a robotics canon the general set does not: the Civil Law Rules on Robotics (4% vs 0.6%), the Machinery Regulation (3% vs 0.9%), the Product Liability Directive (2.6%), ISO 10218 (2.6%), ISO 13482 (2.0%), the IEEE 7000 series (1.9%) and ISO/TS 15066 (1.5%). Asimov's fictional laws are cited by 2.5% of embodied documents and are not an instrument. Only 27% of embodied documents cite any named instrument, against 52% of general-AI documents, because standards and sector regulations rarely cite other instruments by name. By region, the AI Act (22%) and the GDPR (23%) are Europe's references, the OECD principles are Latin America's (20%) and Africa's (22%), the Civil Law Rules are cited most in the Middle East (13%) and the CCW process in the international layer (12%). No ISO robot standard is cited by more than 4% of any region's usable documents (Figure 30).

8. Part D. Regional comparison

Table 14. Regional profiles (embodied-AI documents unless stated)

region	documents	embodied_documents	embodied_share_pct	governance_documents	embodied_governance_documents	binding_embodied	research_share_embodie
Europe	1332	634	47.6	606	212	63	
Asia	677	329	48.6	441	199	32	
North America	345	211	61.2	201	115	65	
Latin America & Caribbean	149	36	24.2	74	23	12	
Middle East	98	45	45.9	44	9	5	

region	documents	embodied_documents	embodied_share_pct	governance_documents	embodied_governance_documents	binding_embodied	research_share_embodie
Africa	76	27	35.5	27	12	6	
Oceania	113	80	70.8	75	53	7	
International	374	241	64.4	241	199	5	

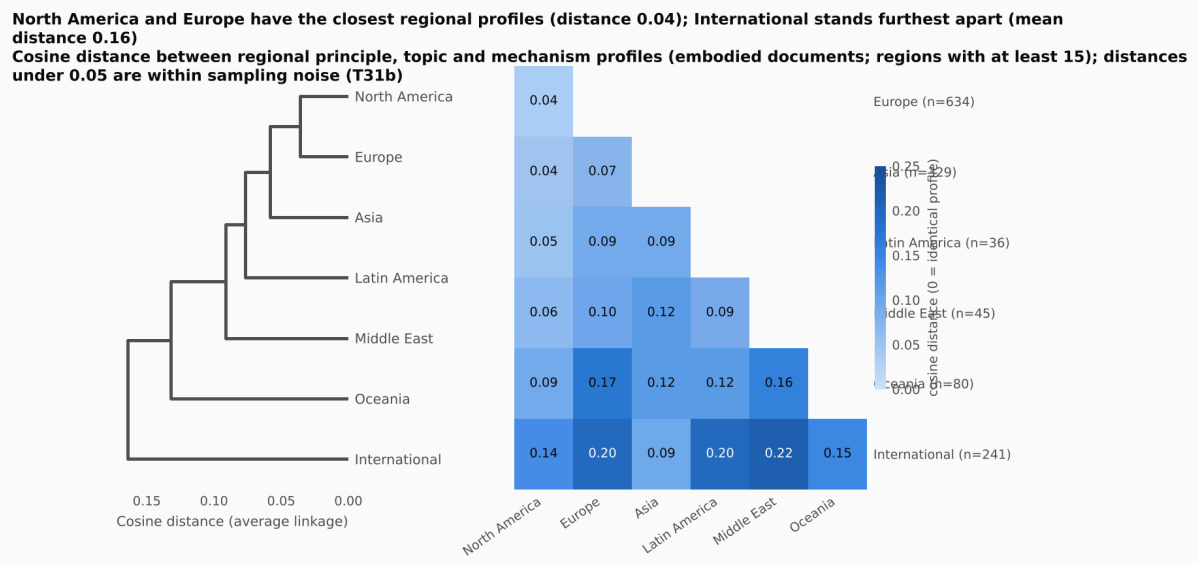


Figure 31. Cosine distance between regional principle, topic and mechanism profiles (embodied documents; regions with at least 30), ordered by average-linkage clustering with n per region; distances below the split-half noise floor (T31b) are within sampling noise.

Table 23. Pairwise cosine distances between regional embodied-AI profiles with 95% bootstrap intervals (300 resamples within region)

region_a	region_b	n_a	n_b	distance	ci_low	ci_high
Europe	North America	634	211	0.038	0.026	0.06
Asia	North America	329	211	0.046	0.034	0.072
North America	Latin America & Caribbean	211	36	0.051	0.048	0.098
North America	Middle East	211	45	0.07	0.054	0.129
Europe	Asia	634	329	0.073	0.059	0.096
Europe	Latin America & Caribbean	634	36	0.09	0.067	0.156
Asia	Latin America & Caribbean	329	36	0.091	0.075	0.149
North America	Oceania	211	80	0.095	0.066	0.159
Asia	International	329	241	0.097	0.08	0.125
Latin America & Caribbean	Middle East	36	45	0.098	0.08	0.171
Europe	Middle East	634	45	0.103	0.072	0.163
Asia	Oceania	329	80	0.121	0.094	0.186
Latin America & Caribbean	Oceania	36	80	0.121	0.086	0.216
Asia	Middle East	329	45	0.125	0.088	0.196
North America	International	211	241	0.139	0.116	0.179
Oceania	International	80	241	0.146	0.12	0.198
Middle East	Oceania	45	80	0.168	0.132	0.263
Europe	Oceania	634	80	0.174	0.13	0.25
Europe	International	634	241	0.198	0.169	0.235
Latin America & Caribbean	International	36	241	0.202	0.167	0.271
Middle East	International	45	241	0.227	0.183	0.299

Table 23b. Nearest-neighbour region in the point estimate and the share of bootstrap resamples in which it is unchanged

region	n	nearest_region	stability
Europe	634	North America	1
Asia	329	North America	1
North America	211	Europe	0.72
Latin America & Caribbean	36	North America	0.95
Middle East	45	North America	0.96
Oceania	80	North America	0.83
International	241	Asia	0.99

Table 31b. Split-half cosine distance within each region (100 random halves): the noise floor below which between-region distances are not distinguishable from sampling variation

region	n	split_half_distance_median	p95
Europe	634	0.005	0.008
Asia	329	0.008	0.013
North America	211	0.015	0.022
Latin America & Caribbean	36	0.067	0.111
Middle East	45	0.046	0.082
Oceania	80	0.033	0.054
International	241	0.009	0.023

Table 31. Cosine distances between regional profiles using research outputs only (regions with at least 15 embodied research outputs: Europe, Asia, North America, Middle East)

region	Europe	Asia	North America	Middle East
Europe	0	0.059	0.01	0.182
Asia	0.059	0	0.067	0.085
North America	0.01	0.067	0	0.189
Middle East	0.182	0.085	0.189	0

Regional profiles are in Table 14. The cosine distances between regional principle, topic and mechanism profiles (Figure 31; regions with at least 30 embodied documents, ordered by clustering, with n per region) place Europe and North America 0.04 apart (bootstrap interval 0.03 to 0.06) and Asia and North America 0.05 apart (0.03 to 0.07), above North America's split-half noise floor of 0.02 (Table 31b); Europe and Asia are 0.07 apart, the Middle East 0.10 from Europe, Oceania 0.17 and the international layer 0.20, and the international layer stands furthest from everyone. Nearest neighbours are stable under bootstrap for Europe, Asia, the international layer, Latin America and the Middle East (95 to 100% of resamples) and less so for Oceania (83%) and North America (72%; Table 23b). Restricting to research papers (Table 31) brings Europe and North America to 0.01 apart, so their separation in the full profile is a difference in instruments (United States and Canadian vehicle and drone law against EU law and standards) and not in the research literature, whereas the Middle East's distinctiveness (0.18 from Europe) is in the research literature itself. The distances are computed on documents, not on principal instruments, and the regions' instrument counts on the principal unit (Table 54) are small outside Europe, Asia and North America.

In words, with the unit stated: Europe and North America share a social- and care-robot research literature with an ethics framing and now a large body of vehicle and drone law (170 and 102 principal embodied instruments; 63 and 65 binding); Asia is instrument-heavy (199 embodied governance documents, 119 principal instruments, 32 binding) and addresses vehicles, drones and industrial robots under innovation and risk framings; the Middle East is a liability-doctrine literature (research is 69% of its 45 embodied documents) with 8 principal instruments; Latin America is a legislative region (21 principal instruments, 12 binding; drones in 53% of its embodied documents) with thin robot research; Africa is drone regulation (63%) and capacity-building strategy (10 principal instruments, 6 binding); Oceania is Australian and New Zealand drone and vehicle regulation (80 embodied documents, 53 governance documents, but only 11 principal instruments and 7 binding, because 46 of its 48 v3 binding rows were register entries for individual CASA instruments); the international layer is the standards layer (199 embodied governance documents, 64 principal instruments, 5 binding).

9. Part E. Trends and the newest documents

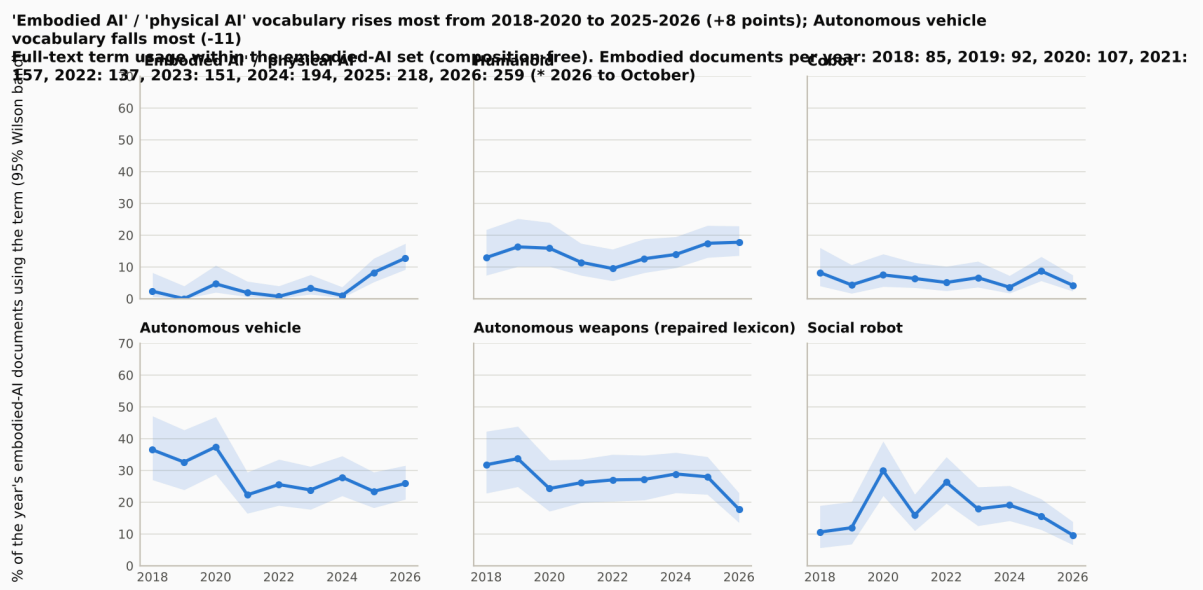


Figure 33. Full-text term usage within the embodied-AI set by year (composition-free), with 95% Wilson bands and documents per year.

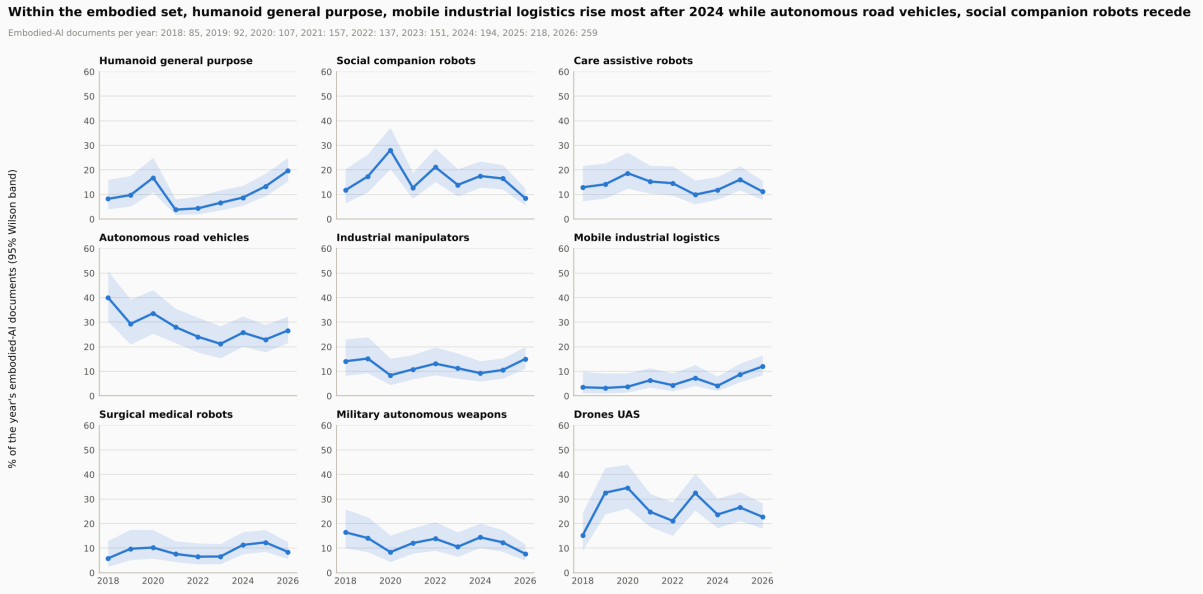


Figure 42. Robot-type shares within the embodied-AI set by year, with 95% Wilson bands (composition-free trends).

Within the embodied set, the terms “embodied AI” or “physical AI” appear in 1% of 2024 documents (2 of 194), 8% of 2025 documents (18 of 218) and 13% of 2026 documents (33 of 259); humanoid vocabulary is present in 9 to 18% of embodied documents in every year since 2018, highest in 2025 and 2026 (17 to 18%); social-robot vocabulary falls from 30% in 2020 to 10% in 2026; autonomous-vehicle vocabulary holds at 22 to 37%. Robot-type shares (Figure 42) show humanoids and logistics robots rising after 2024 and social robots receding; as noted in Part B, part of the 2025 to 2026 humanoid signal comes from the Chinese harvest terms, and a within-round version is not available.

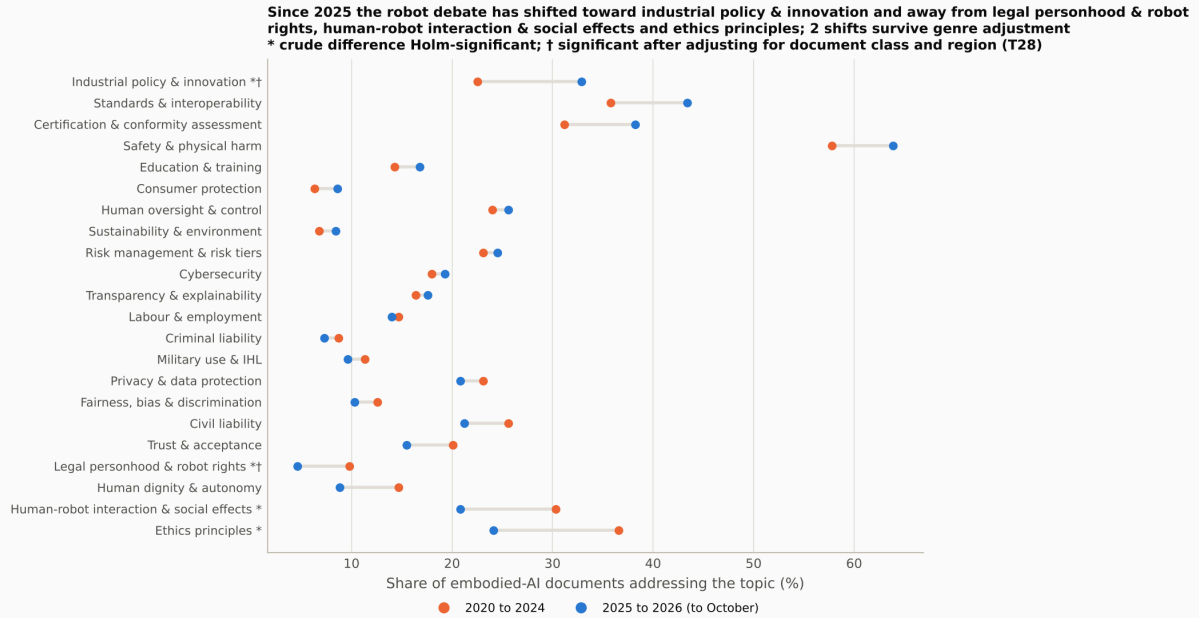


*Figure 34. Governance topics in 2025 to 2026 embodied documents compared with 2020 to 2024: * crude difference Holm-significant; † significant after adjusting for document class and region.*

Table 28. Topic change 2025-26 vs 2020-24 in the embodied set: crude shares, research-only shares, and odds ratios adjusted for document class and region (n=1223)

topic	pct_2020_2024	pct_2025_2026	change_pp	research_only_2020_2024	research_only_2025_2026	adjusted_or	ci_low	ci_high	p	p_holm
Industrial policy & innovation	22.5	32.9	10.4	8.7	13.5	1.69	1.24	2.31	0.001	0.0267
Other	4	5.9	1.8	4	4.3	1.62	0.92	2.84	0.094	1
Children & vulnerable groups	5.9	6.5	0.6	13.4	12.1	1.58	0.95	2.65	0.0792	1
Sustainability & environment	6.8	8.4	1.5	4.7	5	1.34	0.84	2.15	0.2175	1
Consumer protection	6.3	8.6	2.3	5.4	8.5	1.31	0.83	2.07	0.254	1
Transparency & explainability	16.4	17.6	1.3	23.5	33.3	1.29	0.93	1.81	0.1291	1
Certification & conformity assessment	31.2	38.2	6.9	5.1	5.7	1.28	0.96	1.71	0.0922	1
Standards & interoperability	35.8	43.4	7.6	8.7	14.2	1.23	0.9	1.67	0.1948	1
Education & training	14.3	16.8	2.4	9.7	13.5	1.2	0.85	1.7	0.2934	1
Trust & acceptance	20.1	15.5	-4.6	36.5	38.3	1.2	0.83	1.72	0.3271	1
Safety & physical harm	57.8	63.9	6.2	28.2	41.1	1.16	0.88	1.53	0.3044	1
Research ethics	5.8	4.8	-0.9	13.7	12.1	1.16	0.66	2.04	0.6085	1
Risk management & risk tiers	23.1	24.5	1.5	9	17.7	1.11	0.83	1.49	0.4785	1
Labour & employment	14.7	14	-0.7	18.1	14.9	1.11	0.78	1.6	0.5605	1
Cybersecurity	18	19.3	1.3	6.1	10.6	1.1	0.8	1.52	0.5641	1
Intellectual property	1.9	2.5	0.6	1.1	2.8	1.05	0.46	2.4	0.9125	1
Human oversight & control	24	25.6	1.6	15.5	28.4	1	0.75	1.33	0.9943	1
Fairness, bias & discrimination	12.6	10.3	-2.3	18.4	19.1	0.99	0.66	1.47	0.958	1
Privacy & data protection	23.1	20.8	-2.3	25.3	21.3	0.96	0.71	1.3	0.7911	1
Human-robot interaction & social effects	30.3	20.8	-9.5	63.9	53.9	0.86	0.6	1.23	0.404	1

topic	pct_2020_2024	pct_2025_2026	change_pp	research_only_2020_2024	research_only_2025_2026	adjusted_or	ci_low	ci_high	p	p_holm
Taxation & economic policy	1.9	1.5	-0.4	2.2	2.1	0.73	0.27	1.97	0.5396	1
Civil liability	25.6	21.2	-4.4	33.2	36.9	0.72	0.53	0.98	0.0364	0.873
Military use & IHL	11.3	9.6	-1.6	7.6	4.3	0.71	0.46	1.09	0.1183	1
Human dignity & autonomy	14.7	8.8	-5.9	31.8	26.2	0.68	0.44	1.04	0.075	1
Criminal liability	8.7	7.3	-1.4	9.4	7.1	0.67	0.42	1.06	0.0847	1
Ethics principles	36.6	24.1	-12.5	69.3	63.1	0.67	0.49	0.93	0.0179	0.447
Legal personhood & robot rights	9.8	4.6	-5.2	24.2	14.9	0.39	0.22	0.7	0.0015	0.0397

Table 16. Governance topics in embodied-AI documents: 2025-2026 (n=477) vs 2020-2024 (n=746)

topic	pct_2020_2024	pct_2025_2026	change_pp	p	p_holm
industrial_policy_innovation	22.5	32.9	10.4	0.0001	0.0021
standards_interoperability	35.8	43.4	7.6	0.0093	0.2036
certification_conformity_assessment	31.2	38.2	6.9	0.015	0.3155
safety_physical_harm	57.8	63.9	6.2	0.0367	0.7345
education_training	14.3	16.8	2.4	0.2849	1
consumer_protection	6.3	8.6	2.3	0.161	1
other	4	5.9	1.8	0.1784	1
human_oversight_control	24	25.6	1.6	0.5766	1
sustainability_environment	6.8	8.4	1.5	0.3706	1
risk_management_risk_tiers	23.1	24.5	1.5	0.6016	1
cybersecurity	18	19.3	1.3	0.6124	1
transparency_explainability	16.4	17.6	1.3	0.6212	1
children_vulnerable_groups	5.9	6.5	0.6	0.7604	1
intellectual_property	1.9	2.5	0.6	0.5806	1
taxation_economic_policy	1.9	1.5	-0.4	0.7553	1
labor_employment	14.7	14	-0.7	0.7982	1
research_ethics	5.8	4.8	-0.9	0.5608	1
criminal_liability	8.7	7.3	-1.4	0.4536	1
military_use_ihl	11.3	9.6	-1.6	0.424	1
privacy_data_protection	23.1	20.8	-2.3	0.3817	1
fairness_bias_discrimination	12.6	10.3	-2.3	0.2524	1
civil_liability	25.6	21.2	-4.4	0.0885	1
trust_acceptance	20.1	15.5	-4.6	0.0512	0.9724
legal_personhood_robot_rights	9.8	4.6	-5.2	0.0014	0.0345
human_dignity_autonomy	14.7	8.8	-5.9	0.0029	0.0658
human_robot_interaction_social_effects	30.3	20.8	-9.5	0.0003	0.0074
ethics_principles	36.6	24.1	-12.5	0	0.0002

Comparing 477 embodied documents from 2025 to 2026 with 746 from 2020 to 2024 (Table 16), the crude shares of industrial policy (+10 points), standards (+8), conformity assessment (+7) and physical safety (+6) rise and those of ethics principles (−13), human-robot interaction (−10), human dignity (−6) and legal personhood (−5) fall; after Holm correction only industrial policy, ethics principles, human-robot interaction and personhood remain, and cybersecurity, which rose in the original collection, moves by 1 point. Adjusted for document class and region (Table 28; 1,223 documents), industrial policy (odds ratio 1.7, interval 1.2 to 2.3, Holm p = 0.03) and personhood (0.39, 0.22 to 0.70, Holm p = 0.04) survive; ethics principles (0.67, Holm p = 0.45) and human-robot interaction (0.86) do not, so the retreat of ethics discourse is composition. Within research papers alone, industrial policy moves from 9% to 14%. Table 28 does not adjust for collection round, and the 2025 to 2026 layer is where round 2's harvests of 2025 and 2026 policy notices and consolidated texts concentrate, so the industrial-policy rise should be read as a change in which documents were written and collected in 2025 to 2026 as much as a change within genres.

Table 17. Logistic regressions: presence of embodied-specific principles (level >= 1) and two mechanisms on publication year, adjusting for region and document class (embodied-AI documents, 2015-2026)

outcome	n	events	odds_ratio_per_year	ci_low	ci_high	p	p_holm
principle: Physical safety	1496	999	0.999	0.959	1.041	0.9725	0.9725
principle: Human-robot proximity	1496	650	0.937	0.901	0.975	0.0012	0.0099
principle: Liability for physical damage	1496	317	0.938	0.897	0.98	0.0044	0.0306
principle: Autonomy level & control	1496	735	0.979	0.944	1.015	0.246	0.492
principle: Conformity assessment	1496	572	1.049	1.007	1.094	0.0233	0.1164
principle: Dual use & military	1496	244	0.96	0.915	1.008	0.1	0.3001
mechanism: Risk based tiering	1496	124	1.098	1.023	1.178	0.0097	0.0583
mechanism: Conformity assessment certification	1496	291	1.053	1.002	1.106	0.0401	0.1605

Table 30. Trend models (odds ratio per year) under five specifications: collapsed region and class covariates; excluding 2026; stated-date documents only; round-1 documents only; collection round as a covariate

outcome	sensitivity	n	events	odds_ratio_per_year	ci_low	ci_high	p
mechanism: Risk-based tiering	all years, collapsed covariates	1523	129	1.1	1.029	1.176	0.0052
mechanism: Risk-based tiering	excluding partial year 2026	1264	100	1.106	1.018	1.201	0.0166
mechanism: Risk-based tiering	documents with a stated date only	1297	116	1.08	1.007	1.158	0.0316
mechanism: Risk-based tiering	round-1 documents only	663	37	1.306	1.112	1.534	0.0011
mechanism: Risk-based tiering	all years, collection round as covariate	1523	129	1.101	1.03	1.178	0.0048
mechanism: Conformity assessment & certification	all years, collapsed covariates	1523	300	1.049	1.002	1.099	0.042
mechanism: Conformity assessment & certification	excluding partial year 2026	1264	241	1.051	0.994	1.112	0.0826
mechanism: Conformity assessment & certification	documents with a stated date only	1297	261	1.066	1.013	1.122	0.0142
mechanism: Conformity assessment & certification	round-1 documents only	663	48	1.262	1.1	1.449	0.0009
mechanism: Conformity assessment & certification	all years, collection round as covariate	1523	300	1.049	1.002	1.1	0.042
principle: Conformity assessment	all years, collapsed covariates	1523	584	1.044	1.004	1.086	0.0327
principle: Conformity assessment	excluding partial year 2026	1264	466	1.037	0.989	1.088	0.1364
principle: Conformity assessment	documents with a stated date only	1297	515	1.042	0.999	1.088	0.0574
principle: Conformity assessment	round-1 documents only	663	91	1.119	1.018	1.231	0.0195
principle: Conformity assessment	all years, collection round as covariate	1523	584	1.046	1.005	1.088	0.0286
principle: Dual use & military	all years, collapsed covariates	1523	248	0.966	0.923	1.012	0.1447
principle: Dual use & military	excluding partial year 2026	1264	217	0.984	0.933	1.039	0.5687
principle: Dual use & military	documents with a stated date only	1297	217	0.969	0.922	1.019	0.2202
principle: Dual use & military	round-1 documents only	663	102	0.907	0.843	0.976	0.0089
principle: Dual use & military	all years, collection round as covariate	1523	248	0.966	0.923	1.012	0.1415
principle: Physical safety	all years, collapsed covariates	1523	1016	1.003	0.965	1.043	0.871
principle: Physical safety	excluding partial year 2026	1264	837	1.002	0.956	1.05	0.9308
principle: Physical safety	documents with a stated date only	1297	889	1.005	0.963	1.049	0.816
principle: Physical safety	round-1 documents only	663	315	1.024	0.97	1.082	0.3916
principle: Physical safety	all years, collection round as covariate	1523	1016	1.004	0.965	1.045	0.8394
principle: Autonomy level & control	all years, collapsed covariates	1523	749	0.99	0.956	1.025	0.5637
principle: Autonomy level & control	excluding partial year 2026	1264	620	0.985	0.945	1.027	0.4754
principle: Autonomy level & control	documents with a stated date only	1297	660	0.987	0.95	1.024	0.4824
principle: Autonomy level & control	round-1 documents only	663	274	1.002	0.948	1.058	0.9465
principle: Autonomy level & control	all years, collection round as covariate	1523	749	0.99	0.957	1.025	0.5837
principle: Human-robot proximity	all years, collapsed covariates	1523	657	0.935	0.901	0.971	0.0005
principle: Human-robot proximity	excluding partial year 2026	1264	567	0.935	0.894	0.978	0.0032
principle: Human-robot proximity	documents with a stated date only	1297	549	0.921	0.883	0.96	0.0001
principle: Human-robot proximity	round-1 documents only	663	417	1.004	0.946	1.065	0.8997
principle: Human-robot proximity	all years, collection round as covariate	1523	657	0.935	0.9	0.971	0.0005
principle: Liability for physical damage	all years, collapsed covariates	1523	326	0.947	0.909	0.986	0.009
principle: Liability for physical damage	excluding partial year 2026	1264	283	0.958	0.913	1.006	0.0828
principle: Liability for physical damage	documents with a stated date only	1297	289	0.943	0.903	0.986	0.0099
principle: Liability for physical damage	round-1 documents only	663	159	0.998	0.937	1.063	0.9502
principle: Liability for physical damage	all years, collection round as covariate	1523	326	0.947	0.909	0.986	0.009

Trend models on the embodied set (Table 17; 2015 to 2026, 1,496 documents, adjusted for class and region) give, pooled, risk-based tiering rising about 10% a year (odds ratio 1.10, interval 1.02 to 1.18, Holm p = 0.06) and the conformity-assessment mechanism 5% (1.05, Holm p = 0.16), human-robot proximity (0.94, 0.90 to 0.98, Holm p = 0.01) and liability for physical damage (0.94, 0.90 to 0.98, Holm p = 0.03) declining, and physical safety, autonomy-level control and dual use flat. Table 30 shows which of these survive the collection design. Within round-1 documents only (663 documents, the population comparable to v2), risk-based tiering rises 31% a year (1.31, interval 1.11 to 1.53), the conformity-assessment mechanism 26% (1.26, 1.10 to 1.45) and the conformity consideration 12% (1.12, 1.02 to 1.23), reproducing the v2 result; with collection round as a covariate the pooled estimates are 1.10 (1.03 to 1.18, p = 0.005) and 1.05 (1.00 to 1.10, p = 0.04). The v3 statement that the compliance-machinery trend "does not survive" is therefore withdrawn: it is intact in the original population and diluted, not refuted, by the round-2 sector law, whose year profile is flat in these mechanisms. Dual use declines within round 1 (0.91, 0.84 to 0.98) and is flat pooled. Human-robot proximity and liability for physical damage decline pooled (0.94 and 0.95 per year, p = 0.0005 and 0.009) and with round as a covariate (the same values), but are flat within round-1 documents (1.00, interval 0.95 to 1.07, and 1.00, 0.94 to 1.06): the pooled declines coincide with the arrival of round-2 instruments that mention neither, so they are not reported as trends.

The 2025 to 2026 layer of the collection is, read document by document: industrial strategies for humanoids and embodied intelligence (Shanghai, Beijing, Japan's second AI Basic Plan, Israel's draft strategy) and Brazil's humanoid bill; research on oversight and cybersecurity of agentic and multi-robot systems; the conformity layer catching up (ISO/PAS 8800, ISO/IEC TR 5469, ISO 31101, ISO 3691-4, draft ISO 26058, the Commission's draft high-risk guidelines); and, from round 2, the binding sector layer in motion: the UK Automated Vehicles Act 2024 with its marketing-restrictions regulations, Denmark's revised self-driving trial scheme, Japan's consolidated road-traffic and vehicle-safety rules, China's MIIT notice on access, recall and over-the-air upgrades for intelligent connected vehicles, drone regulations in Vietnam, Rwanda, Argentina, Brazil and Peru, and EU regulations on non-road mobile machinery and on amending the AI Act, machinery and aviation rules (Table 59).

10. How different robot types are governed

Table 12. Synthesis by robot group (level-1 taxonomy): documents, instruments and principle emphasis

group	documents	research	governance_instruments	binding	standards	drafts	median_year	physical_safety_any_pct	liability_any_pct	proximity_any_pct	conformity_a
Industrial & logistics	350	88	183	34	59	21	2023	74	23.1	59.1	
Service & social	516	328	127	5	48	20	2023	50	19	75.8	
Medical & body-integrated	197	91	72	7	14	10	2023	79.2	38.1	56.9	
Humanoid & general-purpose	170	116	32	0	10	5	2024	43.5	27.1	72.4	
Vehicles & field	832	96	597	266	140	63	2023	82.2	25.5	25.8	
Security & military	222	64	91	16	1	20	2022	60.8	23.9	33.3	
Other embodied	38	24	10	3	3	3	2022	63.2	21.1	28.9	

Table 13. Synthesis by embodied category: documents, instruments and emphasis

category	docs	instruments	binding	standards	physical_safety_any_pct	liability_any_pct	privacy_any_pct	dignity_any_pct	top_primary_topic
autonomous_road_vehicles	432	292	94	72	82.6	35.4	34.3	11.3	safety_physical_harm
drones_uas	415	321	172	51	82.7	20	35.4	10.1	safety_physical_harm
social_companion_robots	244	17	0	1	35.2	12.7	41	49.6	ethics_principles
care_assistive_robots	219	57	4	12	56.2	22.8	50.7	44.3	ethics_principles
professional_service_robots	210	82	2	38	71	24.3	35.7	16.2	industrial_policy_innova
industrial_manipulators	206	102	16	33	73.3	20.4	34	18.4	safety_physical_harm
military_autonomous_weapons	188	69	3	0	60.6	25.5	28.2	37.8	military_use_ihl
humanoid_general_purpose	170	32	0	10	43.5	27.1	34.7	41.2	legal_personhood_robo
surgical_medical_robots	146	48	6	8	82.9	45.2	46.6	34.2	ethics_principles
collaborative_robots	115	40	6	9	82.6	18.3	35.7	19.1	safety_physical_harm
mobile_industrial_logistics	108	65	9	26	74.1	15.7	20.4	6.5	industrial_policy_innova
agricultural_robots	79	45	7	8	60.8	20.3	49.4	13.9	industrial_policy_innova
domestic_household_robots	79	29	0	13	68.4	26.6	45.6	15.2	industrial_policy_innova
rehabilitation_exoskeleton_prosthetic	63	35	2	8	82.5	27	39.7	28.6	industrial_policy_innova
delivery_robots	60	41	17	1	90	46.7	38.3	6.7	safety_physical_harm

category	docs	instruments	binding	standards	physical_safety_any_pct	liability_any_pct	privacy_any_pct	dignity_any_pct	top_primary_topic
security_policing_robots	58	28	13	1	70.7	20.7	56.9	24.1	privacy_data_protection
educational_robots	51	11	1	2	51	15.7	51	35.3	ethics_principles
autonomous_maritime_or_rail	49	36	1	8	71.4	20.4	30.6	16.3	industrial_policy_innova
space_or_extreme_environment	44	22	2	1	72.7	18.2	36.4	11.4	industrial_policy_innova
other_embodied	38	10	3	3	63.2	21.1	21.1	18.4	ethics_principles
sex_robots	28	4	0	0	28.6	17.9	32.1	85.7	ethics_principles
biohybrid_or_cyborg	12	1	0	0	33.3	25	25	50	ethics_principles
toys_entertainment_robots	11	5	0	2	81.8	36.4	63.6	27.3	privacy_data_protection

Tables S12 and S13 count governance-flagged documents on the comparable unit (any mention of the robot type, every legal text); Table 55 gives the principal-unit counts per group and Table 7 per category, and the counts below use those.

- **Industrial and logistics robots.** 350 documents, 131 principal instruments, 29 binding (Table 55): ISO 10218 and ISO/TS 15066 (captured as well as cited), ISO 3691-4 and draft ISO 26058, Israel's SI 395, Brazil's NR-12, the EU machinery rules, Manitoba's and Alberta's occupational-safety codes and Japan's Ordinance on Industrial Safety and Health; physical safety is present in 74% of documents and proximity in 59%; the dominant tool is the reference to standards, and ISO 10218 and the Machinery Regulation are the instruments most cited for industrial manipulators. Governance here is occupational-safety governance.
- **Service and social robots.** 516 documents and the least governed per document: 82 principal instruments, 5 binding. ISO 13482 and ISO 31101 reach professional service and care robots (50 and 45 principal instruments; 2 and 4 binding); social, educational and domestic robots are reached by 15, 8 and 17 instruments and intimate robots by 4, mainly guidance; none of the 15 social-robot instruments is binding. Proximity (76%) and dignity dominate; the GDPR and ISO 13482 are the instruments most cited, so data-protection law and one voluntary standard govern by default, together with general product, toy and consumer law that the collection does not capture as robot documents.
- **Medical and body-integrated robots.** 197 documents, 56 principal instruments, 7 binding: medical-device law and guidance (EU, FDA, Health Canada, MHRA, MFDS, WHO) and the proposed liability instruments; the highest physical-safety (79%) and liability (38%) presence; the strongest Middle Eastern presence.
- **Humanoid and general-purpose robots.** 170 documents, 20 principal instruments and no binding instrument; the v3 "one binding humanoid instrument" was a 34-word stub of a Beijing development-zone subsidy measure and is now classed as soft industrial policy. Legal personhood is the most common primary topic. The gap between the 2025 to 2026 surge in discourse and the thinness of instruments is widest here; Hangzhou's 2026 embodied-robot regulation and the GB/T 48317 humanoid standard series were located and not read.
- **Vehicles and field robots.** The most instrument-dense group (832 documents, 389 principal instruments, 174 binding): road-traffic and automated-driving acts, civil-aviation and drone rules, type-approval and machinery regulation, plus the AI strategies and the AI Act that reference them. It holds 174 of the 195 principal binding instruments, and its documents carry physical safety (82%) and conformity assessment (54%) more than any other group. Driver-assistance rules (automatic emergency braking type approval) are labelled as automated-vehicle instruments because the taxonomy has no driver-assistance category; this is a known limit.
- **Security and military robots.** 222 documents, 79 principal instruments, 14 binding: the UN CCW process (whose working papers and chair's texts are now classed as position documents, not drafts), national security and policing law, drone and counter-drone rules and humanitarian-law commentary; the highest share of documents calling for new regulation (36%).

11. Part F. Reliability, validation and coverage

11.1 Reliability of the labels

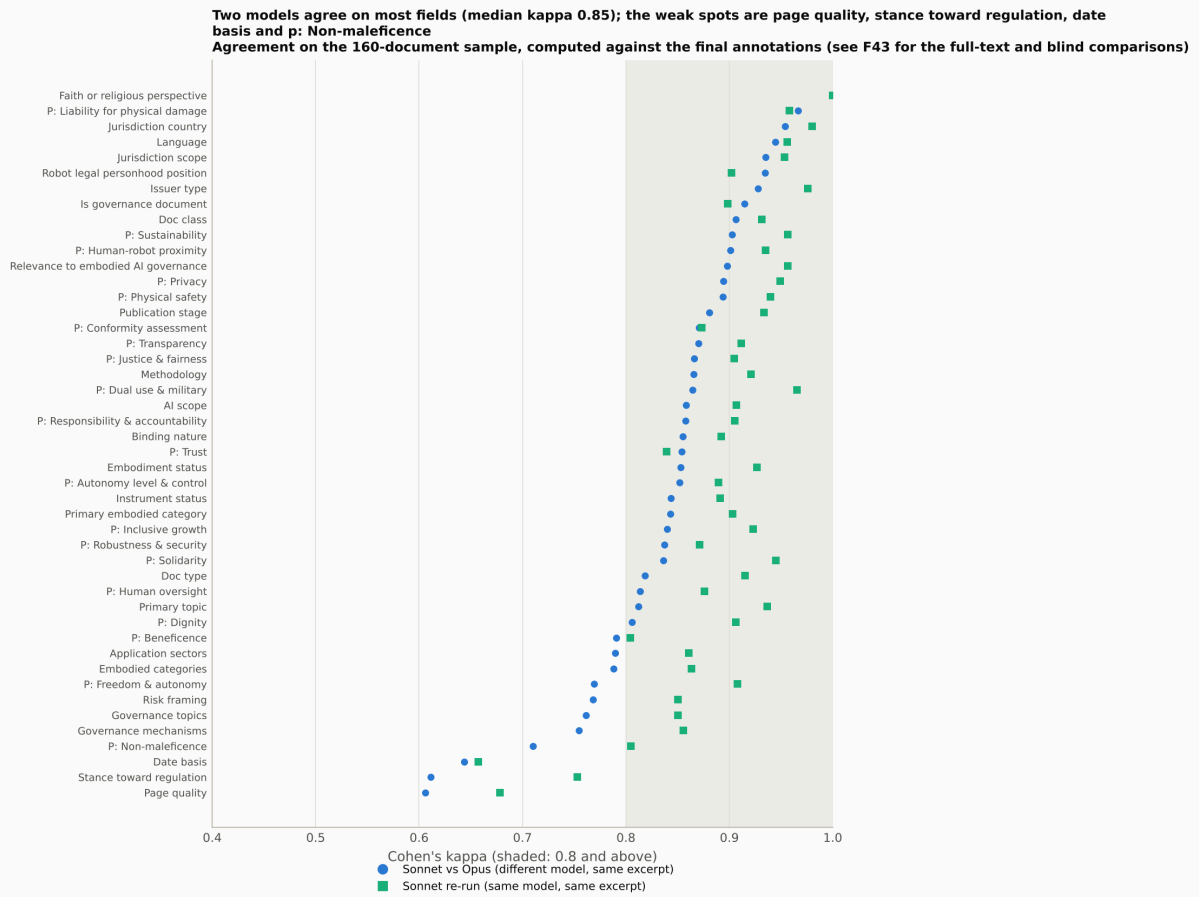


Figure 35. Annotation reliability on the 160-document sample, computed against the final annotations: inter-model kappa and same-model test-retest kappa per field (shaded band marks kappa at or above 0.8).

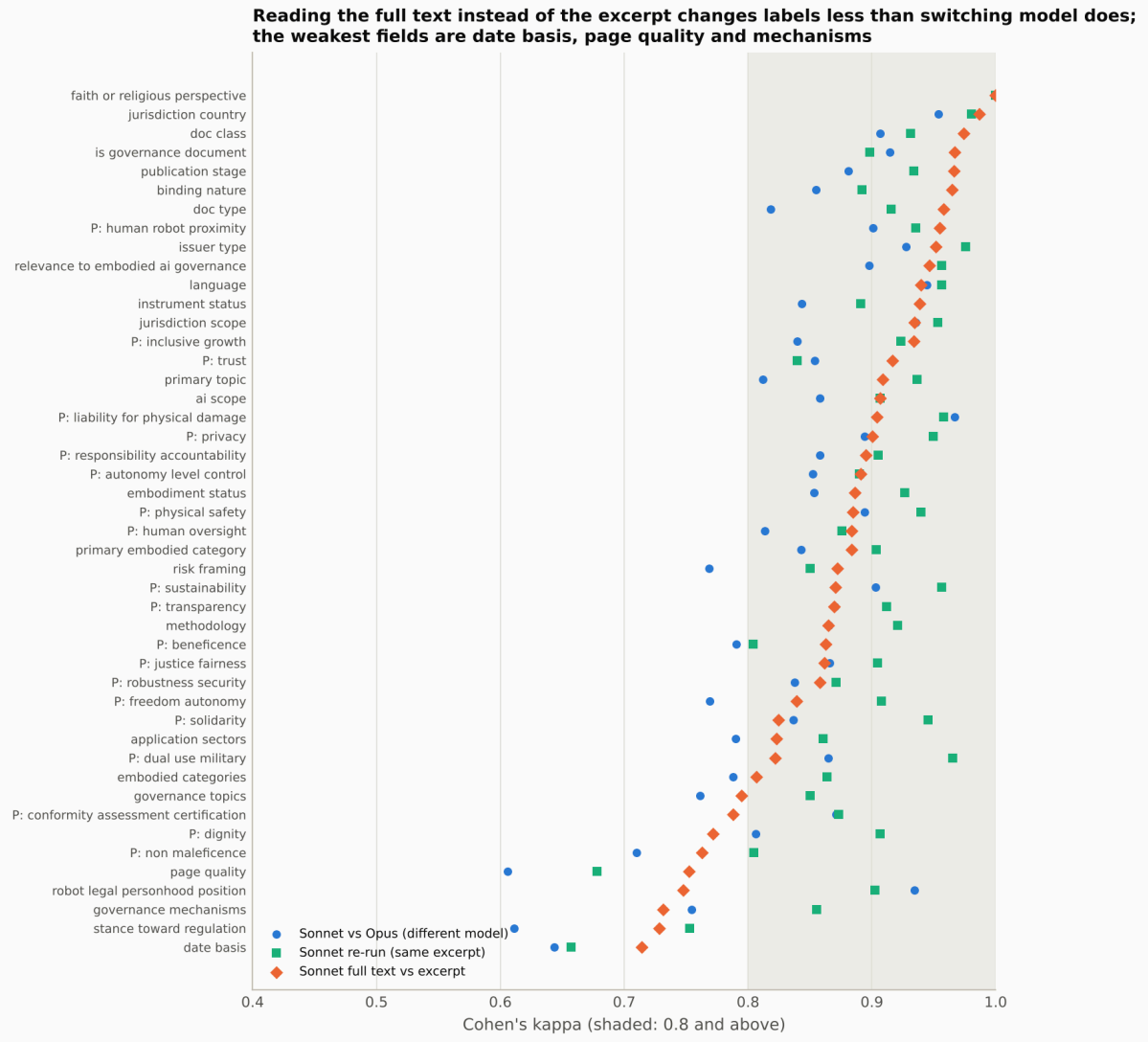


Figure 43. Per-field kappa for three comparisons: different model on the same excerpt, same model re-run, and same model reading the full text instead of the excerpt.

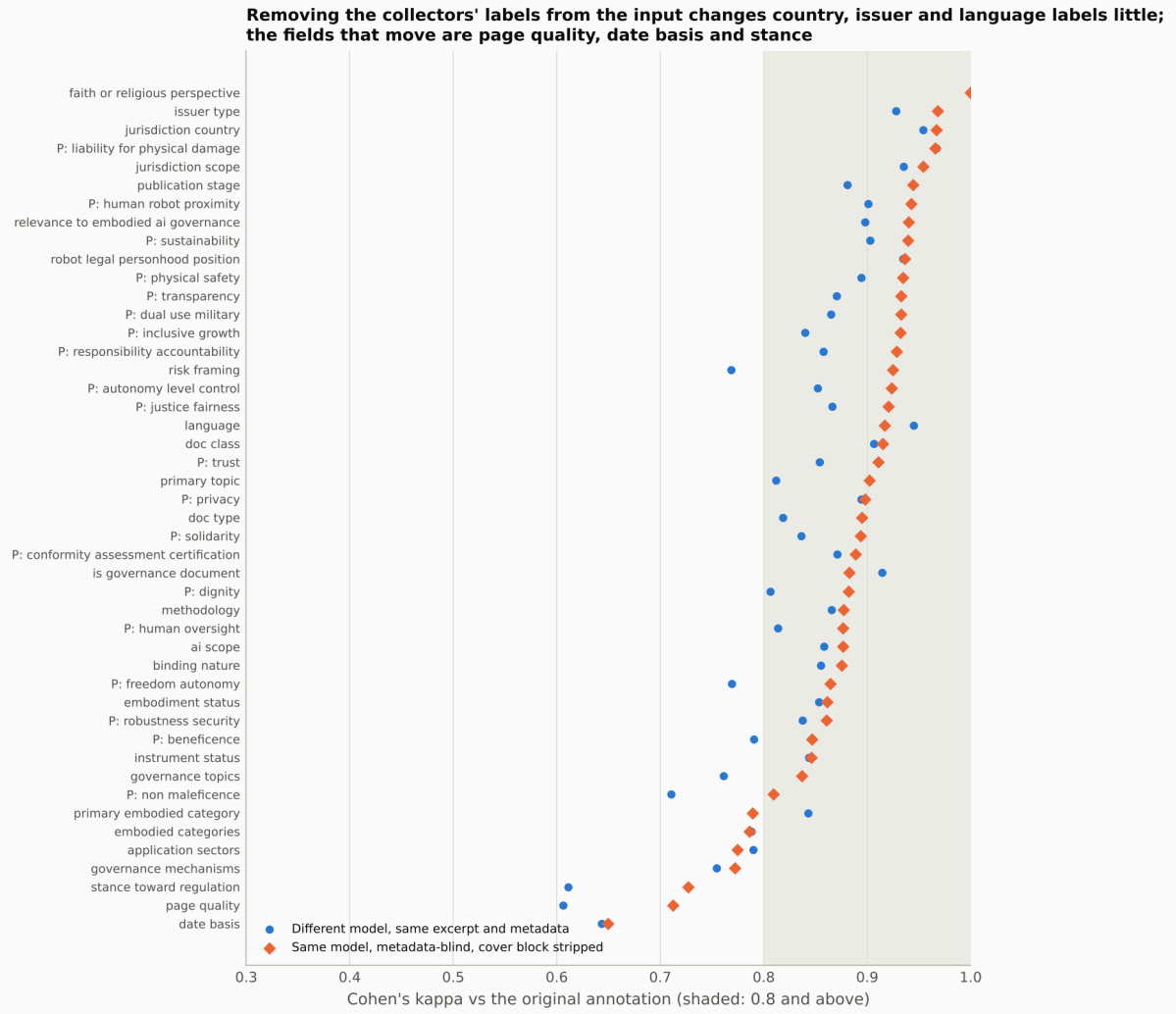


Figure 44. Per-field kappa for the metadata-blind, header-stripped re-annotation versus the original annotation (leakage check).

Table 26. Per-field agreement and kappa against the v1 final annotations: independent model on the same excerpt (160 docs); same model re-run (161 docs); same model on the full text (160 docs); same model metadata-blind with the collector cover block stripped (160 docs); and, for governance documents only, the v2 blind extended-excerpt annotation (up to 110,000 characters)

field	type	n	agreement_inter_model	kappa_inter_model	agreement_retest	kappa_retest	agreement_fulltext	kappa_fulltext	agre
issuer_type	single	160	0.944	0.928	0.981	0.976	0.962	0.951	
date_basis	single	160	0.869	0.643	0.875	0.657	0.881	0.714	
jurisdiction_country	single	160	0.956	0.954	0.981	0.98	0.988	0.987	
jurisdiction_scope	single	160	0.956	0.935	0.969	0.953	0.956	0.934	
doc_type	single	160	0.838	0.818	0.925	0.915	0.962	0.958	
doc_class	single	160	0.931	0.906	0.95	0.931	0.981	0.974	
is_governance_document	single	160	0.969	0.915	0.962	0.898	0.988	0.967	
instrument_status	single	160	0.938	0.843	0.956	0.891	0.975	0.939	
binding_nature	single	160	0.95	0.855	0.962	0.892	0.988	0.965	
publication_stage	single	160	0.931	0.881	0.962	0.934	0.981	0.966	
ai_scope	single	160	0.906	0.858	0.938	0.907	0.938	0.907	
embodiment_status	single	160	0.912	0.853	0.956	0.927	0.931	0.886	
primary_embodied_category	single	160	0.906	0.843	0.944	0.903	0.931	0.883	
primary_topic	single	160	0.831	0.812	0.944	0.937	0.919	0.909	
robot_legal_personhood_position	single	160	0.988	0.935	0.981	0.902	0.944	0.748	
risk_framing	single	160	0.806	0.768	0.875	0.85	0.894	0.872	
stance_toward_regulation	single	160	0.731	0.611	0.838	0.753	0.819	0.728	
methodology	single	160	0.925	0.865	0.956	0.921	0.925	0.865	
faith_or_religious_perspective	single	160	1	1	1	1	1	1	
page_quality	single	160	0.825	0.606	0.862	0.678	0.9	0.752	
language	single	160	0.975	0.945	0.981	0.956	0.975	0.94	
relevance_to_embodied_ai_governance	ordinal	160	0.794	0.898	0.906	0.956	0.881	0.946	
embodied_categories	multi-label	160	0.879	0.788	0.933	0.864	0.892	0.807	
application_sectors	multi-label	160	0.848	0.79	0.912	0.861	0.868	0.823	

field	type	n	agreement_inter_model	kappa_inter_model	agreement_retest	kappa_retest	agreement_fulltext	kappa_fulltext	agre
governance_topics	multi-label	160	0.714	0.761	0.814	0.85	0.79	0.795	
governance_mechanisms	multi-label	160	0.63	0.754	0.865	0.855	0.811	0.732	
principle:transparency	ordinal 0-2	160	0.844	0.87	0.894	0.912	0.844	0.869	
principle:justice_fairness	ordinal 0-2	160	0.862	0.866	0.9	0.905	0.85	0.862	
principle:non_maleficence	ordinal 0-2	160	0.794	0.71	0.888	0.805	0.838	0.763	
principle:responsibility_accountability	ordinal 0-2	160	0.844	0.858	0.9	0.905	0.862	0.895	
principle:privacy	ordinal 0-2	160	0.881	0.894	0.944	0.949	0.888	0.9	
principle:beneficence	ordinal 0-2	160	0.856	0.79	0.881	0.804	0.906	0.863	
principle:freedom_autonomy	ordinal 0-2	160	0.862	0.769	0.956	0.908	0.888	0.839	
principle:trust	ordinal 0-2	160	0.888	0.854	0.888	0.839	0.919	0.916	
principle:sustainability	ordinal 0-2	160	0.931	0.903	0.969	0.956	0.906	0.87	
principle:dignity	ordinal 0-2	160	0.894	0.806	0.944	0.906	0.85	0.772	
principle:solidarity	ordinal 0-2	160	0.919	0.836	0.975	0.945	0.906	0.824	
principle:human_oversight	ordinal 0-2	160	0.856	0.814	0.888	0.876	0.875	0.883	
principle:robustness_security	ordinal 0-2	160	0.869	0.837	0.9	0.871	0.862	0.858	
principle:inclusive_growth	ordinal 0-2	160	0.825	0.84	0.919	0.923	0.912	0.934	
principle:physical_safety	ordinal 0-2	160	0.919	0.894	0.956	0.94	0.912	0.885	
principle:human_robot_proximity	ordinal 0-2	160	0.906	0.901	0.938	0.935	0.956	0.955	
principle:liability_for_physical_damage	ordinal 0-2	160	0.975	0.967	0.969	0.958	0.944	0.904	
principle:autonomy_level_control	ordinal 0-2	160	0.925	0.852	0.956	0.89	0.938	0.891	
principle:conformity_assessment_certification	ordinal 0-2	160	0.956	0.871	0.975	0.873	0.919	0.788	
principle:dual_use_military	ordinal 0-2	160	0.975	0.865	0.994	0.965	0.956	0.822	

Table 27. Excerpt vs full-text annotation (same model, 160 documents): label prevalence, label counts and agreement

item	excerpt_pct	fulltext_pct	diff_pp	kappa
principle: Transparency	50.6	53.8	3.1	0.737
principle: Justice & fairness	43.1	48.8	5.6	0.837
principle: Non-maleficence	74.4	73.8	-0.6	0.723
principle: Responsibility & accountability	65	66.2	1.2	0.834
principle: Privacy	46.2	47.5	1.2	0.824
principle: Beneficence	69.4	69.4	0	0.706
principle: Freedom & autonomy	25.6	30	4.4	0.736
principle: Trust	43.8	43.8	0	0.822
principle: Sustainability	27.5	30.6	3.1	0.833
principle: Dignity	21.9	27.5	5.6	0.682
principle: Solidarity	23.1	26.9	3.7	0.8
principle: Human oversight	29.4	33.8	4.4	0.726
principle: Robustness & security	40	45	5	0.745
principle: Inclusive growth	57.5	59.4	1.9	0.859
principle: Physical safety	26.9	29.4	2.5	0.846
principle: Human-robot proximity	25.6	28.7	3.1	0.921
principle: Liability for physical damage	15	16.2	1.3	0.905
principle: Autonomy level & control	23.8	25.6	1.9	0.815
principle: Conformity assessment	11.2	15.6	4.4	0.706
principle: Dual use & military	7.5	11.2	3.8	0.78
labels: embodied categories (mean per document)	0.95	1.09	0.14	nan
labels: mechanisms (mean per document)	2.95	3.29	0.34	nan
labels: topics (mean per document)	4.78	5.18	0.41	nan
labels: sectors (mean per document)	2.24	2.41	0.18	nan

item	excerpt_pct	fulltext_pct	diff_pp	kappa
labels: named instruments (mean per document)	0.96	1.21	0.26	nan
embodied-set membership	45	46.2	1.3	0.924
governance document	24.4	25	0.6	0.916

Table 27b. Excerpt vs full-text agreement by whether the v1 excerpt truncated the document

subset	n	median_principle_kappa	doc_class_agreement	exact_category_set_agreement	governance_flag_agreement
truncated by the excerpt	97	0.749	0.979	0.722	0.99
fully contained in the excerpt	63	0.838	0.937	0.889	0.937

No person read or labelled any document; the checks below compare the model with itself or with another model. Five comparisons on round-1 samples (Table 26), computed against the annotations the analysis uses: a different model on the same input agrees at a median kappa of 0.85 over 46 fields (minimum 0.61 for page quality; 11 fields below 0.80, mainly the multi-label fields, stance, framing, date basis and three principle levels); the same model re-run agrees at 0.91 (minimum 0.66); the same model reading the full text instead of the excerpt agrees at 0.88 (minimum 0.71) and finds slightly more of everything (0.1 to 0.4 more labels per document), with lower agreement on documents the excerpt truncated (median principle kappa 0.75 vs 0.84; Table 27b); the same model with the collectors' cover block and metadata removed agrees at 0.90 (minimum 0.65 for date basis); and the extended blind re-annotation of governance documents agrees with the original at 0.83, with the largest movements in robot-type labels (0.60), stance (0.55) and publication stage (0.49).

Table 09b. Reliability variants: quadratic-weighted and unweighted kappa for ordinal fields, and agreement/kappa on the level-2 (treated in depth) threshold for principles, per comparison (round-1 and round-2 samples)

comparison	n	field	kappa_quadratic	kappa_unweighted	agreement_level2	kappa_level2
Opus, round-1 sample	160	relevance (0-3)	0.903	0.726	nan	nan
Opus, round-1 sample	160	principles (median over 20)	0.846	0.747	0.956	0.688
Sonnet re-run, round-1 sample	160	relevance (0-3)	0.956	0.871	nan	nan
Sonnet re-run, round-1 sample	160	principles (median over 20)	0.887	0.829	0.966	0.837
Sonnet full text, round-1 sample	160	relevance (0-3)	0.94	0.819	nan	nan
Sonnet full text, round-1 sample	160	principles (median over 20)	0.877	0.795	0.969	0.832
Sonnet blind, round-1 sample	160	relevance (0-3)	0.939	0.838	nan	nan
Sonnet blind, round-1 sample	160	principles (median over 20)	0.905	0.84	0.975	0.847
Opus, round-2 sample	160	relevance (0-3)	0.932	0.755	nan	nan
Opus, round-2 sample	160	principles (median over 20)	0.864	0.791	0.988	0.812
Sonnet re-run, round-2 sample	160	relevance (0-3)	0.952	0.856	nan	nan
Sonnet re-run, round-2 sample	160	principles (median over 20)	0.914	0.867	0.991	0.888

Two comparisons on a stratified round-2 sample of 160 documents (supplement S4.7) now sit beside them. Against a second model (Claude Opus 5.5) the median kappa over single-label and ordinal fields is 0.86 (0.81 over all 129 fields including the multi-label items); against a same-model re-run it is 0.91 (0.89). The weakest round-2 fields against the second model are page quality (0.53), stance toward regulation (0.65), publication stage (0.66), date basis (0.70), instrument status (0.75) and document type (0.79); the set-defining fields are at 0.83 (AI scope), 0.86 (document class, binding nature, embodiment status) and 0.93 (relevance, quadratic-weighted; 0.76 unweighted); 19 of 20 principle levels are at or above 0.79, solidarity at 0.72; among the multi-label items the weakest are the taxation topic (0.16), the creation of an oversight body (0.35), domestic robots (0.32) and self-regulation (0.43). The re-run reaches 0.84 for AI scope, 0.95 for relevance, 0.89 for instrument status and 0.90 for binding nature, with page quality (0.69) and publication stage (0.66) weakest. Table S09b gives the unweighted and level-2 variants for every comparison: on the "treated in depth" threshold the round-2 principle kappas are 0.81 (second model) and 0.89 (re-run) against 0.69 and 0.84 on the round-1 samples, so the depth scores that Table 9 rests on are at least as stable in round 2 as in round 1. Two limits: the second-model and re-run annotations were produced under the v2 excerpt policy before the character-aware cleaner, so for the 36 Chinese, Japanese and Korean documents in the sample the comparison also includes the cleaner change; and 27 of the 160 are landing pages whose labels agree because there is little to disagree about. The collector-metadata agreement check (Section 2) is a consistency check between model outputs, not an independent reference.

11.2 Precision of the instrument counts

The independent precision audit (Section 4; Table 62; supplement S13) read 100 randomly sampled v3 binding embodied texts: 67% distinct instruments, 97% binding, 56% with substantive embodied provisions, 67% with autonomy or AI content, 14% naming AI explicitly; 54% (44 to 63) distinct, binding and substantive together, implying about 161 (132 to 190) such instruments in the v3 set of 299. The failures were individual exemptions, approvals and directions captured as register pages (14), binding regimes with no embodied-specific provision (10), register, catalogue, index or table-of-contents captures (9), ancillary EU and UK acts (8), non-binding Chinese policy documents (3) and definitions-only sections (2); 60 of the 100 belonged to a family with another member in the population. On the same 100 documents the v4 rules keep 61% (51 to 70), agreeing with the auditor on 77% (kappa 0.53). Precision and coverage pull in opposite directions and are reported separately: the v4 counts are upper bounds on distinct, binding, substantive instruments and lower bounds on what exists.

11.3 Model-free check: full-text lexicons

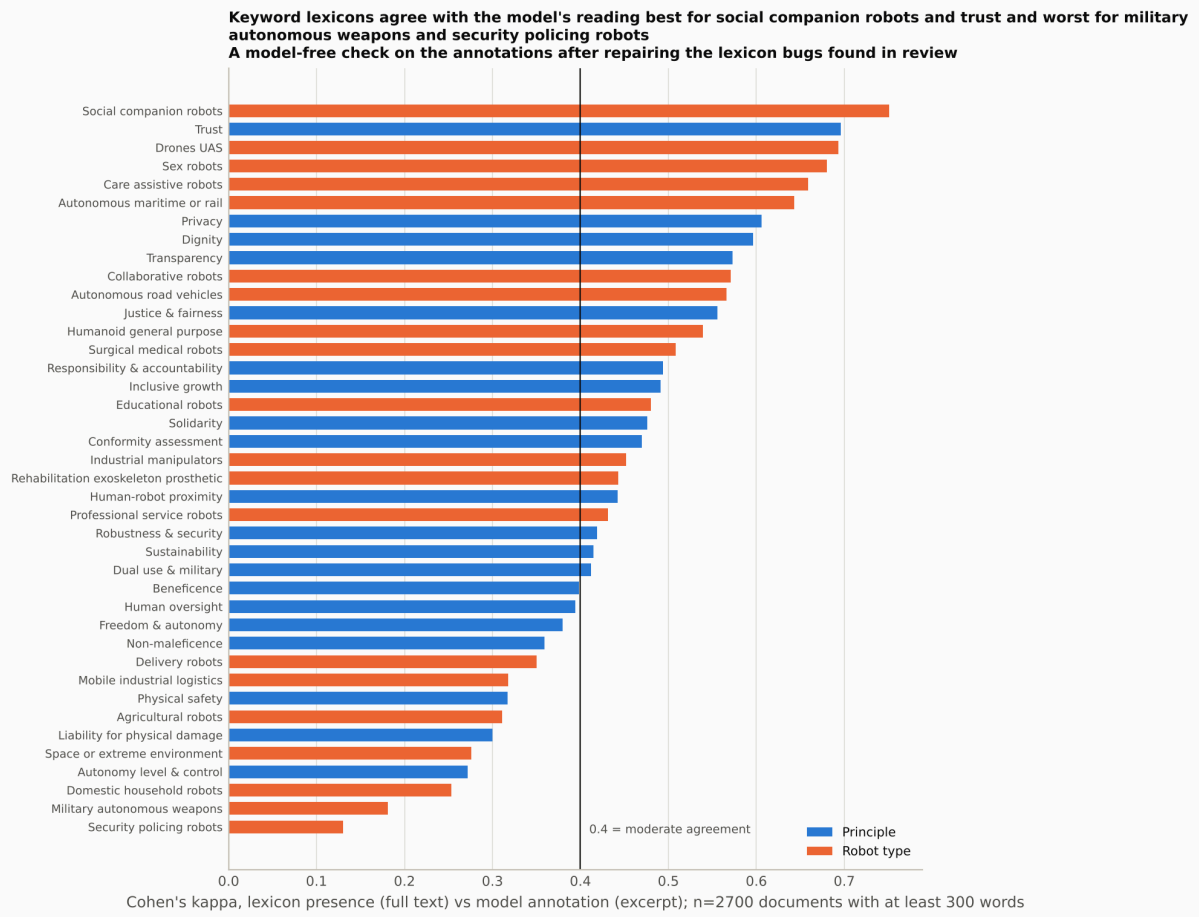


Figure 36. Lexicon-based measures from full text versus model annotation of the excerpt: agreement and kappa per principle and category.

Table 18. Agreement between full-text lexicon presence and excerpt-based annotation (documents with >= 300 words, n=2700); 'density05' = at least 0.5 hits per 1,000 words

item	label	n	annotated_pct	lexicon_any_pct	agreement_any	kappa_any	agreement_density05	kappa_density05
principle:transparency	Transparency	2700	46.7	36.9	0.79	0.573	0.731	0.443
principle:justice_fairness	Justice & fairness	2700	37.7	43.4	0.785	0.556	0.802	0.554
principle:non_maleficence	Non-maleficence	2700	73.6	74.1	0.753	0.359	0.727	0.371
principle:responsibility_accountability	Responsibility & accountability	2700	61.3	64.1	0.763	0.494	0.727	0.452
principle:privacy	Privacy	2700	35.3	46.8	0.807	0.606	0.793	0.55
principle:beneficence	Beneficence	2700	57.3	47.1	0.697	0.398	0.648	0.335
principle:freedom_autonomy	Freedom & autonomy	2700	21.6	29.1	0.764	0.38	0.792	0.269
principle:trust	Trust	2700	36	35.8	0.86	0.696	0.821	0.576
principle:sustainability	Sustainability	2700	24.9	47.2	0.716	0.415	0.8	0.5
principle:dignity	Dignity	2700	18.9	11.8	0.894	0.596	0.844	0.259
principle:solidarity	Solidarity	2700	22.8	28	0.801	0.476	0.834	0.431
principle:human_oversight	Human oversight	2700	35.6	14.3	0.76	0.394	0.693	0.168
principle:robustness_security	Robustness & security	2700	48.1	43.8	0.711	0.419	0.679	0.347
principle:inclusive_growth	Inclusive growth	2700	48.1	52.5	0.745	0.491	0.74	0.477
principle:physical_safety	Physical safety	2700	40	20.9	0.698	0.317	0.669	0.204
principle:human_robot_proximity	Human-robot proximity	2700	24	14.5	0.824	0.442	0.82	0.346
principle:liability_for_physical_damage	Liability for physical damage	2700	12.9	14.5	0.834	0.3	0.878	0.226
principle:autonomy_level_control	Autonomy level & control	2700	27.7	9.5	0.768	0.272	0.745	0.111

item	label	n	annotated_pct	lexicon_any_pct	agreement_any	kappa_any	agreement_density05	kappa_density05
principle:conformity_assessment_certification	Conformity assessment	2700	31.3	31.9	0.771	0.47	0.796	0.469
principle:dual_use_military	Dual use & military	2700	12	32.1	0.786	0.412	0.906	0.605
category:humanoid_general_purpose	Humanoid general purpose	2700	5.7	8.5	0.939	0.539	0.964	0.61
category:collaborative_robots	Collaborative robots	2700	3.8	3.3	0.971	0.571	0.973	0.489
category:industrial_manipulators	Industrial manipulators	2700	7.3	6.9	0.928	0.452	0.939	0.361
category:mobile_industrial_logistics	Mobile industrial logistics	2700	3.4	4.6	0.948	0.318	0.97	0.366
category:delivery_robots	Delivery robots	2700	1.8	1.1	0.981	0.35	0.986	0.335
category:professional_service_robots	Professional service robots	2700	6.8	5.6	0.934	0.431	0.944	0.367
category:domestic_household_robots	Domestic household robots	2700	2.5	1.9	0.968	0.253	0.975	0.054
category:social_companion_robots	Social companion robots	2700	8.7	9.7	0.959	0.751	0.969	0.775
category:care_assistive_robots	Care assistive robots	2700	7.6	6.4	0.956	0.659	0.958	0.61
category:educational_robots	Educational robots	2700	1.8	0.8	0.987	0.48	0.985	0.306
category:sex_robots	Sex robots	2700	1	1.8	0.991	0.68	0.996	0.813
category:surgical_medical_robots	Surgical medical robots	2700	5	4.7	0.954	0.508	0.964	0.472
category:rehabilitation_exoskeleton_prosthetic	Rehabilitation exoskeleton prosthetic	2700	1.9	3.7	0.969	0.443	0.984	0.515
category:autonomous_road_vehicles	Autonomous road vehicles	2700	13.3	18.1	0.885	0.566	0.926	0.631
category:drones_uas	Drones UAS	2700	11.6	16.4	0.926	0.693	0.951	0.741
category:autonomous_maritime_or_rail	Autonomous maritime or rail	2700	1.4	1.1	0.991	0.643	0.99	0.515
category:agricultural_robots	Agricultural robots	2700	2.6	0.6	0.978	0.311	0.975	0.104
category:space_or_extreme_environment	Space or extreme environment	2700	1.4	1.2	0.981	0.276	0.986	0.199
category:military_autonomous_weapons	Military autonomous weapons	2700	6.8	31.3	0.723	0.181	0.862	0.313
category:security_policing_robots	Security policing robots	2700	2	0.7	0.977	0.13	0.98	0.099

Lexicon presence agrees with annotated presence at 70 to 89% for principles and 72 to 99% for robot types (n = 2,700 documents of at least 300 words; median kappa 0.40 to 0.43, moderate), best for distinctive vocabularies (social and care robots; trust, privacy and dignity) and worst for diffuse concepts (domestic robots, autonomy-level control, liability for physical damage, security robots). The Spearman correlation between annotated and lexicon-based embodied-minus-general differences across the 20 principles is 0.88.

11.4 Sensitivity

Table 19. Headline statistics under alternative document subsets

subset	n	embodied_share_pct	governance_share_pct	binding_share_of_gov_pct	research_share_embodied_pct	accountability_any_embodied_pct	physical_s
All usable documents	3165	50.6	54	41	31.8	59.3	
Collectors' sprint set only	877	72.7	12.3	5.6	79.5	65.5	
Collectors' database set only	662	5.6	35.6	13.1	2.7	75.7	
Substantive captures only	2256	41.5	56	46.8	31.8	74.6	
Confidence >= 0.5 only	3140	50.5	54.4	40.9	31.2	59.5	
Collector tier Core only	255	19.2	45.1	8.7	49	79.6	
Collector tier Supporting/Context only	909	57.2	11.9	6.5	80	64.8	

subset	n	embodied_share_pct	governance_share_pct	binding_share_of_gov_pct	research_share_embodied_pct	accountability_any_embodied_pct	physical_s
Excluding Europe	1833	52.9	60.2	33.6	21.1	54.5	
English only	2373	53.2	49.4	44.5	36.9	59.7	
Non-English only	792	42.9	67.7	33.2	12.6	57.6	
All annotated incl. unusable captures	3295	49.5	52.7	40.6	32.4	58.8	

Table 32. Headline quantities under alternative definitions of the embodied-AI set

definition	n	research_share_pct	governance_documents	binding	physical_safety_any_pct	proximity_any_pct	accountability_any_pct	social_robots_research_pct	socia
Embodied set (relevance >= 2 or any category)	1603	31.8	822	195	67.3	42.9	59.3	43	
Strict embodied set (relevance >= 2 only)	1374	32.3	708	168	71.3	46.2	60.5	43.7	
Embodied set, substantive captures only	936	31.8	453	175	70.7	45.8	74.6	38.6	

Headline quantities are stable across subsets (Table 19) and across definitions of the embodied set (Table 32): under the strict definition (relevance only, 1,374 documents) the research share is 32%, governance documents fall to 708 (168 principal binding instruments) and the social-robot gap is unchanged (44% of research, 16 instruments); restricting to substantive captures (936 documents) leaves 453 governance documents (175 binding) and the same ordering of robot types.

11.5 Coverage and known gaps

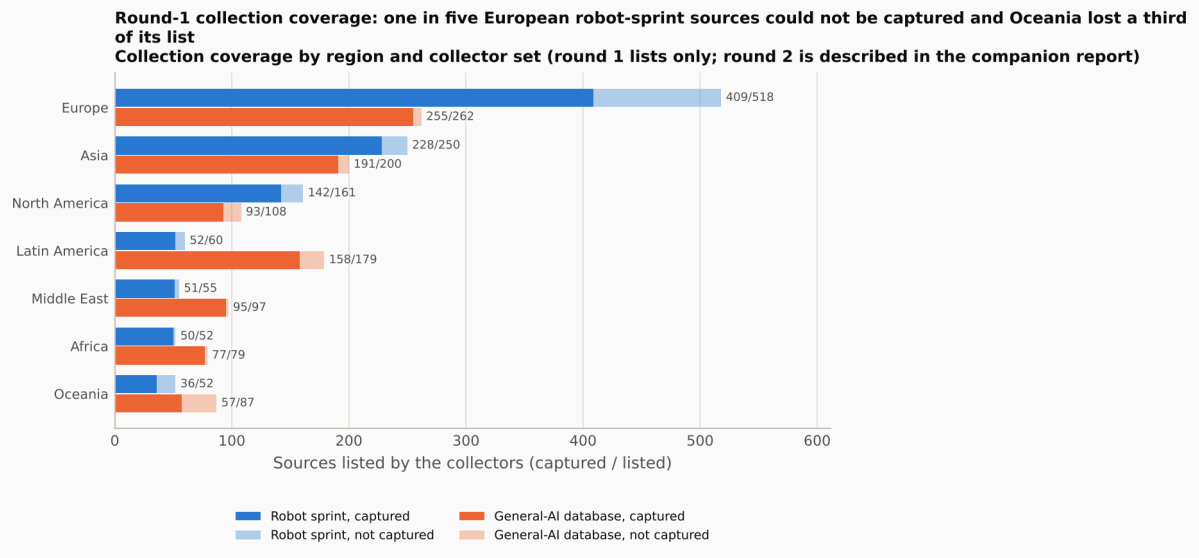


Figure 37. Collection coverage by region: sources captured versus listed, by collector set.

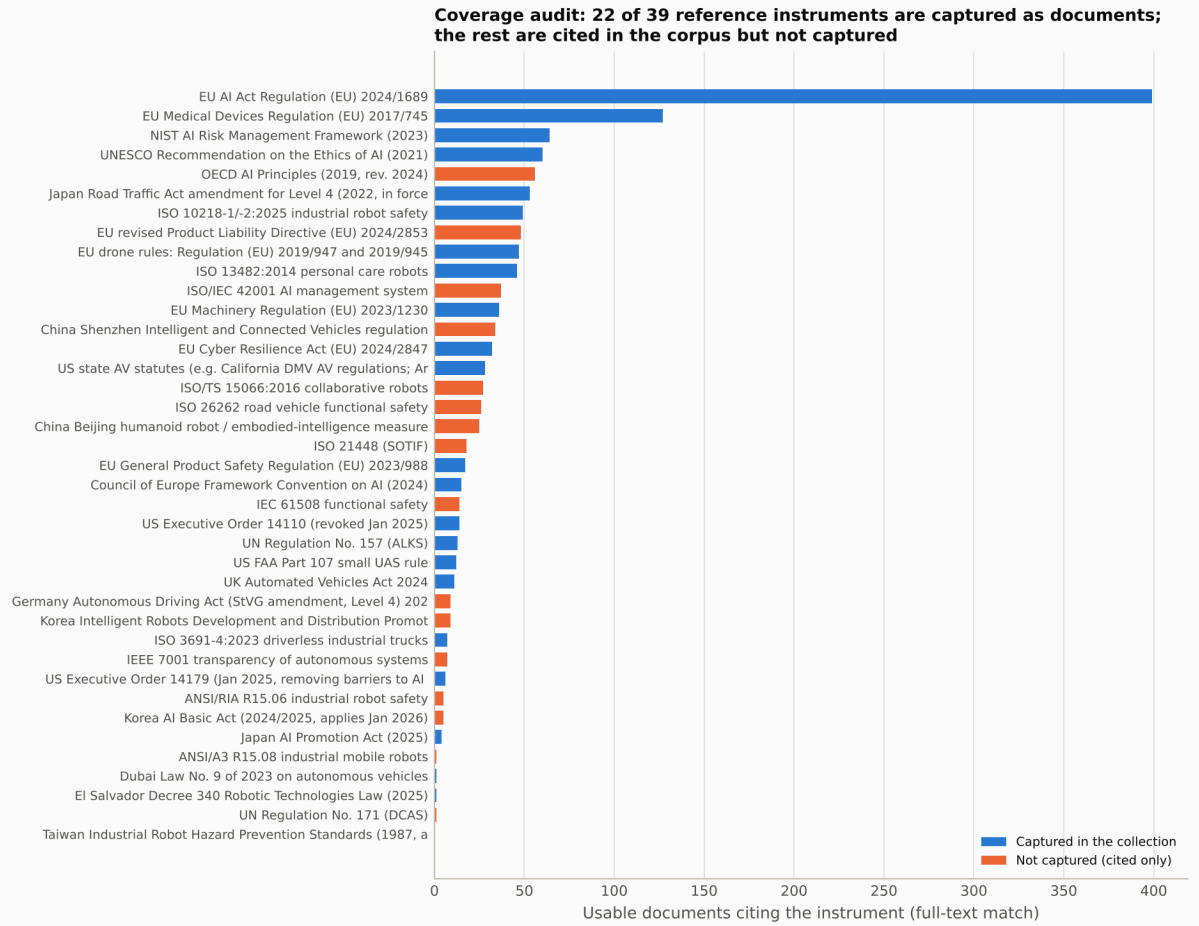


Figure 41. Coverage audit: documents citing each reference instrument, coloured by whether the instrument itself was captured as a principal instrument with a content-analysed capture.

Table 25. Coverage audit: reference list of binding instruments and standards relevant to embodied AI, whether each is captured in the collection as a principal instrument with a content-analysed capture (title match) and how many usable documents cite it (full-text regex)

instrument	jurisdiction	year	nature	status	captured_as_document	n_title_matches	n_documents_citing	pct_citing
UK Automated Vehicles Act 2024	GB	2024	binding	in_force	True	2	11	0.3
Germany Autonomous Driving Act (StVG amendment, Level 4) 2021	DE	2021	binding	in_force	False	0	9	0.3
Japan Road Traffic Act amendment for Level 4 (2022, in force 2023)	JP	2023	binding	in_force	True	6	53	1.7
Korea Intelligent Robots Development and Distribution Promotion Act (2008)	KR	2008	binding	in_force	False	0	9	0.3
EU Machinery Regulation (EU) 2023/1230	EU	2023	binding	adopted_applies_2027	True	4	36	1.1
EU revised Product Liability Directive (EU) 2024/2853	EU	2024	binding	adopted_transposition_2026	False	0	48	1.5
EU drone rules: Regulation (EU) 2019/947 and 2019/945	EU	2019	binding	in_force	True	14	47	1.5
EU Cyber Resilience Act (EU) 2024/2847	EU	2024	binding	adopted_applies_2027	True	1	32	1
EU Medical Devices Regulation (EU) 2017/745	EU	2017	binding	in_force	True	17	127	4
EU General Product Safety Regulation (EU) 2023/988	EU	2023	binding	in_force	True	3	17	0.5
UN Regulation No. 157 (ALKS)	INT	2021	binding	in_force	True	2	13	0.4
US FAA Part 107 small UAS rule	US	2016	binding	in_force	True	2	12	0.4
US state AV statutes (e.g. California DMV AV regulations; Arizona EO)	US	2018	binding	in_force	True	15	28	0.9

instrument	jurisdiction	year	nature	status	captured_as_document	n_title_matches	n_documents_citing	pct_citing
China Shenzhen Intelligent and Connected Vehicles regulation (2022)	CN	2022	binding	in_force	False	0	34	1.1
China Beijing humanoid robot / embodied-intelligence measures (2024-2025)	CN	2024	soft	published	False	0	25	0.8
ISO 10218-1/-2:2025 industrial robot safety	INT	2025	standard	published	True	3	49	1.5
ISO/TS 15066:2016 collaborative robots	INT	2016	standard	published	False	0	27	0.9
ISO 13482:2014 personal care robots	INT	2014	standard	published	True	3	46	1.5
ISO 3691-4:2023 driverless industrial trucks	INT	2023	standard	published	True	1	7	0.2
IEC 61508 functional safety	INT	2010	standard	published	False	0	14	0.4
ISO 26262 road vehicle functional safety	INT	2018	standard	published	False	0	26	0.8
ISO 21448 (SOTIF)	INT	2022	standard	published	False	0	18	0.6
ISO/IEC 42001 AI management system	INT	2023	standard	published	False	0	37	1.2
IEEE 7001 transparency of autonomous systems	INT	2021	standard	published	False	0	7	0.2
ANSI/RIA R15.06 industrial robot safety	US	2012	standard	published	False	0	5	0.2
ANSI/A3 R15.08 industrial mobile robots	US	2020	standard	published	False	0	1	0
EU AI Act Regulation (EU) 2024/1689	EU	2024	binding	in_force_phased	True	25	399	12.6
US Executive Order 14110 (revoked Jan 2025)	US	2023	binding	revoked	True	1	14	0.4
US Executive Order 14179 (Jan 2025, removing barriers to AI leadership)	US	2025	binding	in_force	True	1	6	0.2
Japan AI Promotion Act (2025)	JP	2025	binding	in_force	True	1	4	0.1
Korea AI Basic Act (2024/2025, applies Jan 2026)	KR	2025	binding	adopted	False	0	5	0.2
Council of Europe Framework Convention on AI (2024)	INT	2024	binding_treaty	open_for_signature	True	5	15	0.5
UNESCO Recommendation on the Ethics of AI (2021)	INT	2021	soft	adopted	True	1	60	1.9
OECD AI Principles (2019, rev. 2024)	INT	2019	soft	adopted	False	0	56	1.8
Dubai Law No. 9 of 2023 on autonomous vehicles	AE	2023	binding	in_force	True	1	1	0
El Salvador Decree 340 Robotic Technologies Law (2025)	SV	2025	binding	in_force	True	1	1	0
Taiwan Industrial Robot Hazard Prevention Standards (1987, amended 2018)	TW	2018	binding	in_force	False	0	0	0
NIST AI Risk Management Framework (2023)	US	2023	soft	published	True	2	64	2
UN Regulation No. 171 (DCAS)	INT	2024	binding	in_force	False	0	1	0

In round 1 the collectors could not capture 266 of 2,160 listed sources, 109 of them in the European robot sprint; 34 captured files were oversized or restricted, 27 recovered. In round 2, 355 of 2,075 candidates yielded no text, most on hosts that serve bot challenges or sit outside the sandbox network policy (eur-lex.europa.eu catalogue pages 59, the Chinese national law database 27, the UNODA library 17, Légifrance 17, the Korean law portal 17, the Taiwanese law portal 11; Table 52); a retry pass after the review of v3 recovered four texts (a Tanzanian RPAS amendment regulation, a GB/T catalogue page and two Korean ministry press pages); the Product Liability Directive 2024/2853, the Toy Safety Regulation 2025/2509 and the Toy Safety Directive 2009/48/EC still returned no text. The coverage audit (Table 25, Figure 41) checks 39 reference instruments and standards (R157 and R171 now separate rows; Taiwan dated 2018; a stricter pattern for US state automated-vehicle statutes) and counts an instrument as captured only when a principal instrument with a content-analysed capture matches its title: 22 are captured as documents in the collection, 16 are cited by documents in the corpus but were not captured, and 1 are neither captured nor cited. On this strict criterion 22 of 39 are captured (11 in v2); on the looser title-family match used in v3, 31 (12 in v2). The 17 not captured are the German Autonomous Driving Act (StVG amendment, 2021), Korea's Intelligent Robots Act (2008) and AI Basic Act, the revised Product Liability Directive (EU) 2024/2853, the Shenzhen intelligent-connected-vehicle regulation (2022), Beijing's humanoid and embodied-intelligence measures, ISO/TS 15066, IEC 61508, ISO 26262, ISO 21448, ISO/IEC 42001, IEEE 7001, ANSI/RIA R15.06, ANSI/A3 R15.08, the OECD AI Principles, Taiwan's industrial-robot hazard prevention standards and UN Regulation No. 171.

Beyond the reference list, the round-2 collectors located and verified instruments that the environment could not read, which are the first targets of a round 3: the Hangzhou regulations on promoting the embodied intelligent robot industry (2026); the Shenzhen regulations on intelligent connected vehicles (2022), on the AI industry (2022) and on the low-altitude economy (2024); Hong Kong's Road Traffic (Autonomous Vehicles) Regulation, Cap. 374AF (2024); Korea's Intelligent Robots Development and Distribution Promotion Act and Enforcement Decree, and the enforcement decree and rule of its Autonomous Vehicles Act; US DoD Directive 3000.09 on autonomy in weapon systems (2023) and California SB 243 on companion chatbots (2025); India's Drone Rules 2021; the EU Product Liability Directive 2024/2853, the Toy Safety Regulation (EU) 2025/2509 and the Toy Safety Directive 2009/48/EC; France's ordonnance 2021-443 and décret 2021-873 on criminal liability for automated vehicles; Switzerland's DETEC ordinance on aircraft of special categories (2022); ANSI/A3 R15.06-2025 and GB/T 48317.1-2026 (humanoid robots, part 1). The Korean, Chinese NPC, Hong Kong, Légifrance, defense.gov and Indian hosts block the environment or serve script-only pages. Their absence, with the jurisdictions round 2 could not reach, is why every instrument count in this report is a lower bound on what exists, even as the precision audit makes it an upper bound on what the collection holds.

12. Limitations

The corpus is a purposive multilingual collection, not a census, and its composition drives several ratios: Europe is 42% of usable documents, English 75% and Chinese 9%, the last almost entirely one round-2 harvest of the State Council policy library; the research-to-instrument ratio reflects the sprint, database and round-2 designs; and jurisdiction coverage follows the availability of open legal databases and of hosts the environment could reach (round 2 added 251 embodied documents in Europe and 223 in Asia but 20 in Latin America, 15 in Africa and 10 in the Middle East, Table 54; France, Singapore, Korea, Taiwan, Hong Kong, Brazil and Chile are among the jurisdictions whose portals could not be read), so the regional instrument counts measure API availability as much as law.

Precision. The rule-based counting unit removes most but not all of the inflation that the v3 unit carried: on the audited sample the rules keep 61% where the auditor keeps 54%, and the audit's interval (44 to 63%) is the honest statement of how many of the v3 binding texts were distinct, binding and substantive. The 195 and 168 binding counts are upper bounds. Instrument families are grouped by a normalised title key; 86 near-duplicate title pairs noted in review are only partly merged because distinct parts of one regime are legitimately distinct documents. Status is one field: "in force" and "applicable" are not separated, no applicability date is recorded, the Machinery Directive, which governs machinery until January 2027, was coded superseded by a mechanically applied audit code until that value was corrected to in force, and court decisions and soft law carry lifecycle labels designed for legislation.

Labels. Every label was assigned by a language model; no human validation exists. The seven reliability checks compare model with model and are consistent, not certified; the fields the analysis depends on most are among the weaker ones in round 2 (instrument status 0.75, document type 0.79 against the second model), and the multi-label fields that define mechanisms and robot types have per-label kappas as low as 0.16. The excerpt covered the whole text for only half of the usable documents (median fraction seen 0.99; 21% with under a quarter seen), and the longest statutes, whose robot articles lie beyond the window, are labelled from what the model could see and, in the cases the reviewers found, from what it knew; a full-text pass for texts over 110,000 characters was not done. The excerpt cleaner deleted most of the Chinese, Japanese and Korean text before the model in v3; the v4 re-annotation corrects this for every such document, but the earlier failure shows that the pipeline's checks did not catch a defect that changed 57 of 422 set memberships. The relevance field that defines the sets was produced by the same model that produced the outcomes. The model's confidence field is a page-quality proxy and is not used.

Dating. Consolidated statutes harvested from legal databases carry the enactment year or the version year as the annotation found it; the two are not separated, "first appearance" years for sector law are not reported, time analyses are restricted to 2014 to 2026 and 2026 is partial. Dates are inferred from content for 16% of usable documents; 2,526 of the 2,618 set members have a known year and enter the adjusted models (Table 20).

Design. Collection round is confounded with genre, text length, capture quality and language; the adjusted models carry round as a covariate and the stratified tables report within-round contrasts, but no within-round table exists for mechanisms, robot-type trends (Figure 42) or topic change (Table 28), and the within-genre regional-divergence tables (29b) keep only regions with at least 20 documents of the genre, which leaves four regions in the research panel and five in the instrument panel. Technical standards are annotated on their public scope text only, never on the normative text; 16 of those scope texts were obtained through a committee mirror or written as abstracts by a collector agent (Table 51), and the copyright status of mirrored scope pages was not checked. Statistics about depth therefore describe scope pages for standards. The six embodied-specific items are considerations scored like principles, not principles in the ethical sense.

People and agents. The round-2 collectors, the six adversarial reviewers of v3, the precision auditor and the writers of these documents were AI agents (Claude) run in one Claude Code session; the regional teams of round 1 were people, but their labels were not used as a reference. Nothing in this report has been checked by a lawyer or a domain coder.

Not done, with reasons. Human validation of labels (no human coders in this setting; the protocol is in supplement S4.5). Version dating of consolidated texts (deferred to round 3). Reading the verified-but-unread instruments named in Section 11.5 (hosts block the environment). Stricter title merging of near-duplicate pairs (would merge distinct parts of one regime). A full-text pass for the longest statutes (cost). Measurement-error correction of the headline shares and odds ratios (requires a human-coded sample). Small groups (Africa, the Middle East, Latin America, humanoid and social-robot instruments) are reported as counts, and conclusions about them are correspondingly tentative.

13. Conclusions

Within this collection, embodied AI is governed by the law of the machine classes it is deployed in, by technical standards and voluntary guidance, and by the enumeration of robots inside general-AI law. On the headline unit, 505 principal instruments reach robots and 195 are binding, 182 of them in force, almost all vehicle, aviation, workplace, machinery and medical-device rules; 140 of the 195 contain any provision about autonomy or automation, and 15 name AI. How many instruments there are depends on how one counts (810 legal texts, 505 principal instruments, 161 by an independent reading of the v3 set), and how different robot rules look from AI rules depends on which collection one reads: the robot literature's own considerations (physical safety, autonomy-level control, conformity assessment, physical liability) and its tools (standards, licensing, conformity assessment, insurance) distinguish robot instruments from AI instruments in both rounds and after every adjustment, whereas the thinner ethical vocabulary of robot rules is a property of the

database-harvested sector law of round 2 and does not appear among the collector-found instruments of round 1. Specification lags mention: binding law and standards' scope pages set out physical safety in roughly six cases in ten, autonomy-level control in one in four. Research and rules point at different robots, with companion and social robots the clearest case of attention without instruments and drones and vehicles of instruments without research. Regions differ in the mix of documents they contribute and in their tools; the claim that they differ on values is not supported once genre is held constant. The frontier is moving toward industrial policy, humanoids and the vocabulary of "physical AI", and away from personhood; compliance machinery grows within the original collection. For governance practice: the operative texts for robots are sector regulation, standards and product law, so attention to transport, aviation and occupational-safety regimes, to ISO/TC 299 and IEC, and to the AI Act's harmonised-standards mandate buys more than new principles; robots that live with people have almost no instruments of their own and are governed by product, toy, data-protection and consumer law that this collection does not capture as robot documents; and autonomy-level control needs the specification that three binding instruments in four still leave open. For method: a study of this kind should state its counting unit, report its construct tiers, and model its own collection design, because the collection strategy changed the headline counts by an order of magnitude and would otherwise have reversed the conclusions.

Glossary

- **Embodied AI, physical AI.** AI that controls a machine with a body: robots, self-driving vehicles, drones, surgical and factory machines. The object of study here is the governance of these machine classes, whether or not an instrument mentions AI.
- **Governance instrument, rule or policy; legal text.** A law, regulation, decree, bill, strategy, guideline, code, standard or resolution meant to steer behaviour. Research papers and news are not instruments. "Legal text" means every captured document of instrument class, the comparable unit across versions.
- **Principal instrument.** The headline counting unit: a legal text that is not a commencement order, levy or fee instrument, individual exemption, direction, approval or permit, definitions-only part, table of contents, EEA incorporation decision, language corrigendum, duplicate text, catalogue or landing page or stub under 150 words. Amendments count.
- **Instrument family.** A principal instrument with its yearly renewals, successive versions and parts within one jurisdiction, counted once.
- **Robot-relevant; with embodied provisions.** Rated at least moderately relevant to embodied-AI governance (2 or 3 of 3); rated 0 or 1 but carrying a robot-type label.
- **AI-content tiers.** Whether the full text contains autonomy or AI provisions, and whether it names AI explicitly (two or more mentions).
- **Binding law vs voluntary guidance (soft law).** Binding: breaking it has legal consequences. Voluntary: guidance, principles, strategies and codes that cannot be enforced on their own.
- **Technical standard.** A detailed engineering specification (ISO 10218, ISO 13482) that is voluntary unless a law makes it binding by reference or a national standard is declared mandatory (Chinese GB standards). Annotated here on its public scope page only.
- **Conformity assessment.** The pre-market check (testing, certification, CE marking) that a product meets safety rules.
- **Mentioned vs specified in depth.** A principle or consideration is mentioned when named; specified in depth when the document says what must be done, by whom and how it is checked. Depth is measured only on content-analysed captures of at least 150 words.
- **Content-analysed capture; catalogue page; stub.** A substantive or partial capture of a document's text; a catalogue, register, committee or landing page captured in place of the text; a capture under 150 words.
- **Collection round.** Round 1: the regional teams' web collection of research and official texts. Round 2: the harvest of legal databases and collector-agent search for instruments. Round is a covariate in the adjusted models and a stratum in Tables 20b and 30.
- **Governance-gap index.** Share of research papers about a robot type minus the share of principal instruments; positive means studied more than governed.
- **Precision audit.** An independent reading of a random sample of counted instruments against stated criteria, giving the share that genuinely meets the definition, with a sampling interval.
- **Cohen's kappa.** Agreement between two labellers after discounting chance; 0.8 and above is conventionally "almost perfect", 0.6 "substantial", 0.4 "moderate".
- **Holm correction.** Raises the bar for significance when many comparisons are made at once; applied within one table or one stratum.
- **Cramér's V.** Strength of association from 0 to 1; 0.1 small, 0.3 medium; bias-corrected versions remove the inflation small samples produce.
- **Odds ratio.** How much more likely an outcome is in one group than another; 1 means no difference. Adjusted odds ratios allow for other differences between the groups (here document class, region, collection round and year).
- **Wilson interval, bootstrap interval.** The range within which a percentage or difference would most likely fall if the sample were drawn again.

Appendix. Figure and table index

Figures 1 to 44 (9, 15 and 32 were merged into neighbours) and the round-2 figures 50 to 52 are in `figures/` as PNG and SVG in light and dark variants with their data in `figures/data/`; Figures 23, 31, 39, 40 and 41 were rebuilt in v4 (capture-tier base, dendrogram order and region set, single axis, computed title, strict coverage criterion). Tables T01 to T35 with T20b, the round-2 tables T50 to T62 with T53b and S01 to S13 with S09b are in `tables/`; every statistic is in `data/results.json`, every annotation run in `data/run_manifest.json`. The supplement holds the codebook, taxonomies, reliability detail including the round-2 sample (S4.7), instrument lists, coverage audit, quality-assurance rules and the override file (S8), version history (S10), datasheet and the precision audit (S13); the companion report holds the round-2 inventory, funnel and collector-metadata checks.